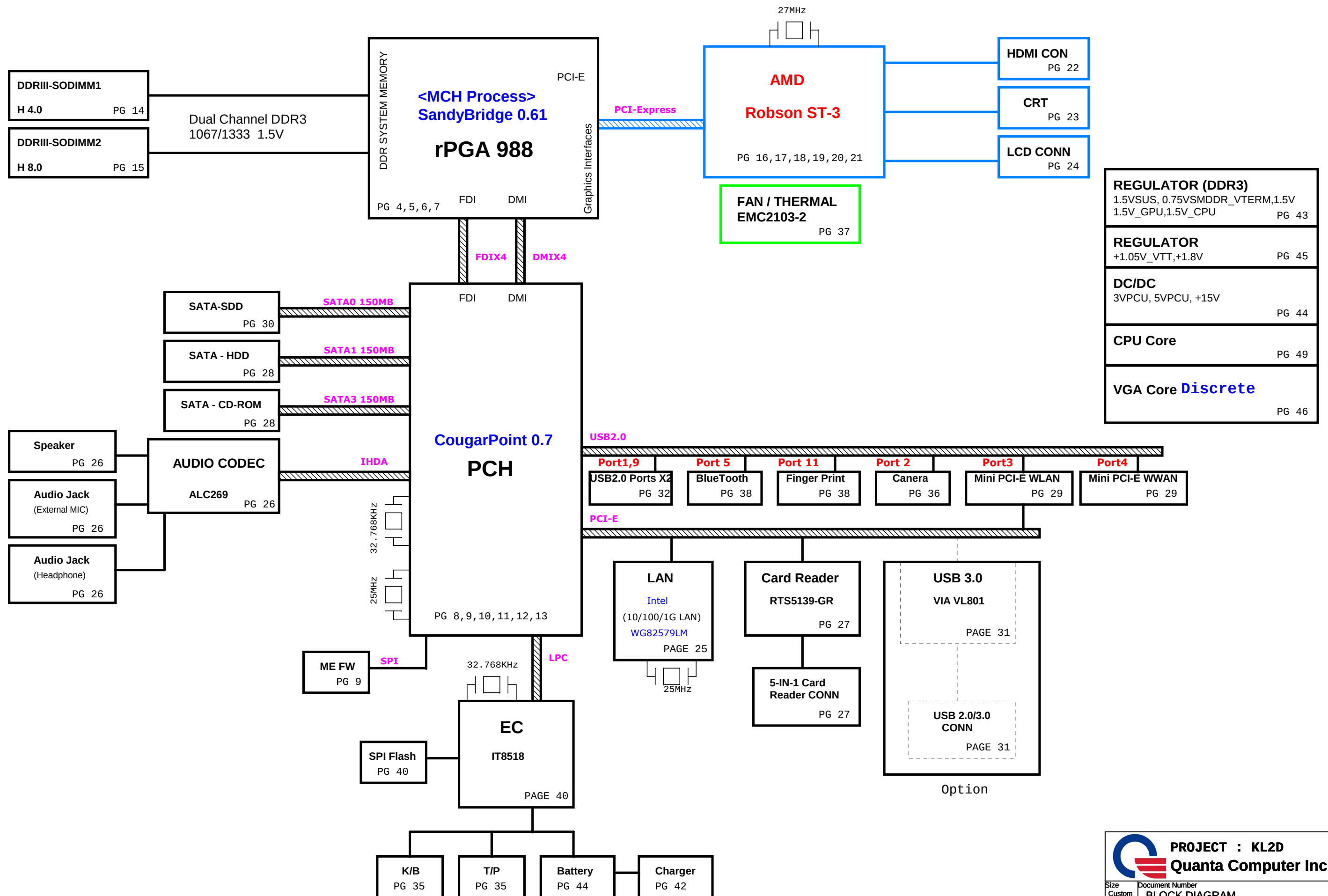
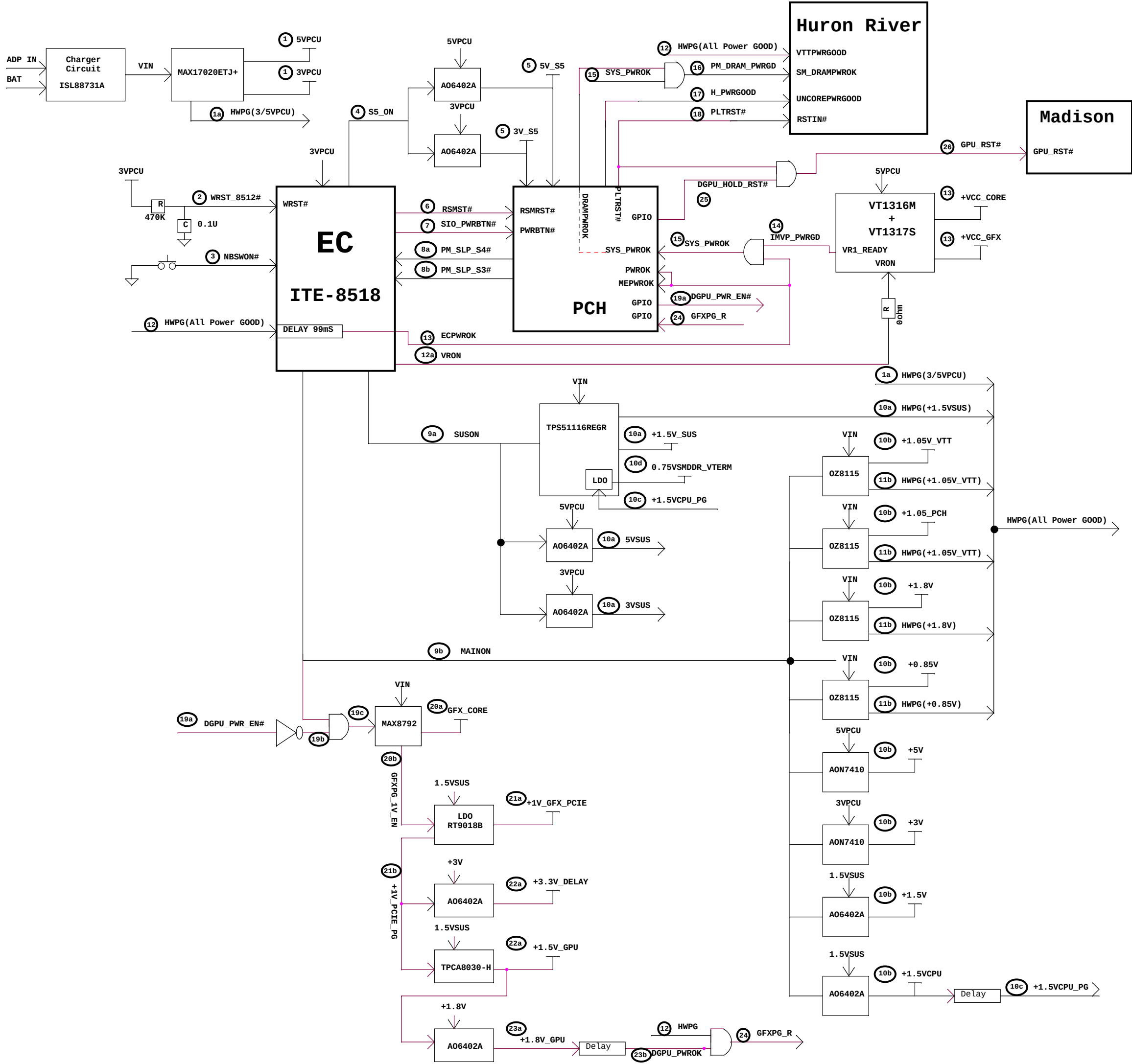


## KL9A Intel Huron River Platform with AMD Discrete GFX





02/20 DEL for Pre-ES1

CPU\_CLK select(CLK)

02/20 DEL for Pre-ES1

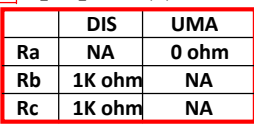
|         |                            |               |
|---------|----------------------------|---------------|
|         | 0                          | 1             |
| CPU_SEL | CPU0/1=133MHz<br>(default) | CPU0/1=100MHz |



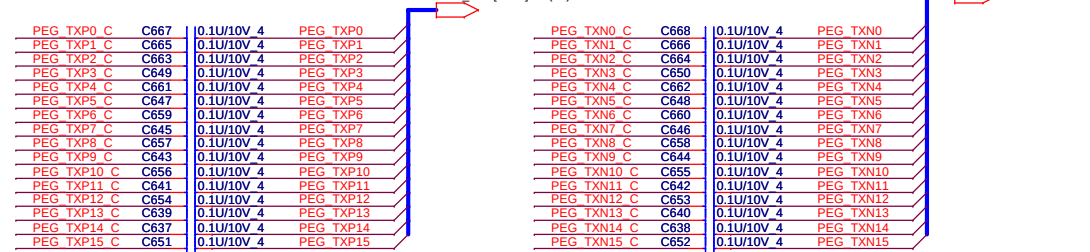
PROJECT : KL2D  
Quanta Computer Inc.

|       |                             |  |       |   |       |
|-------|-----------------------------|--|-------|---|-------|
| Size  | Document Number             |  |       |   | Rev   |
|       | <b>Clock Generator</b>      |  |       |   | 1A    |
| Date: | Thursday, November 04, 2010 |  | Sheet | 3 | of 53 |

## Sandy Bridge Processor (CLK,MISC,JTAG)

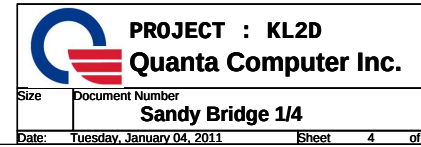
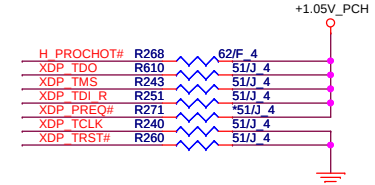


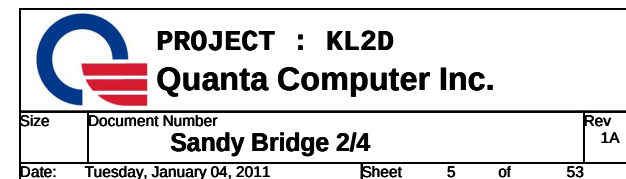
## PEG x16 (UMA Non-stuff)



### Processor pull-up(CPU)

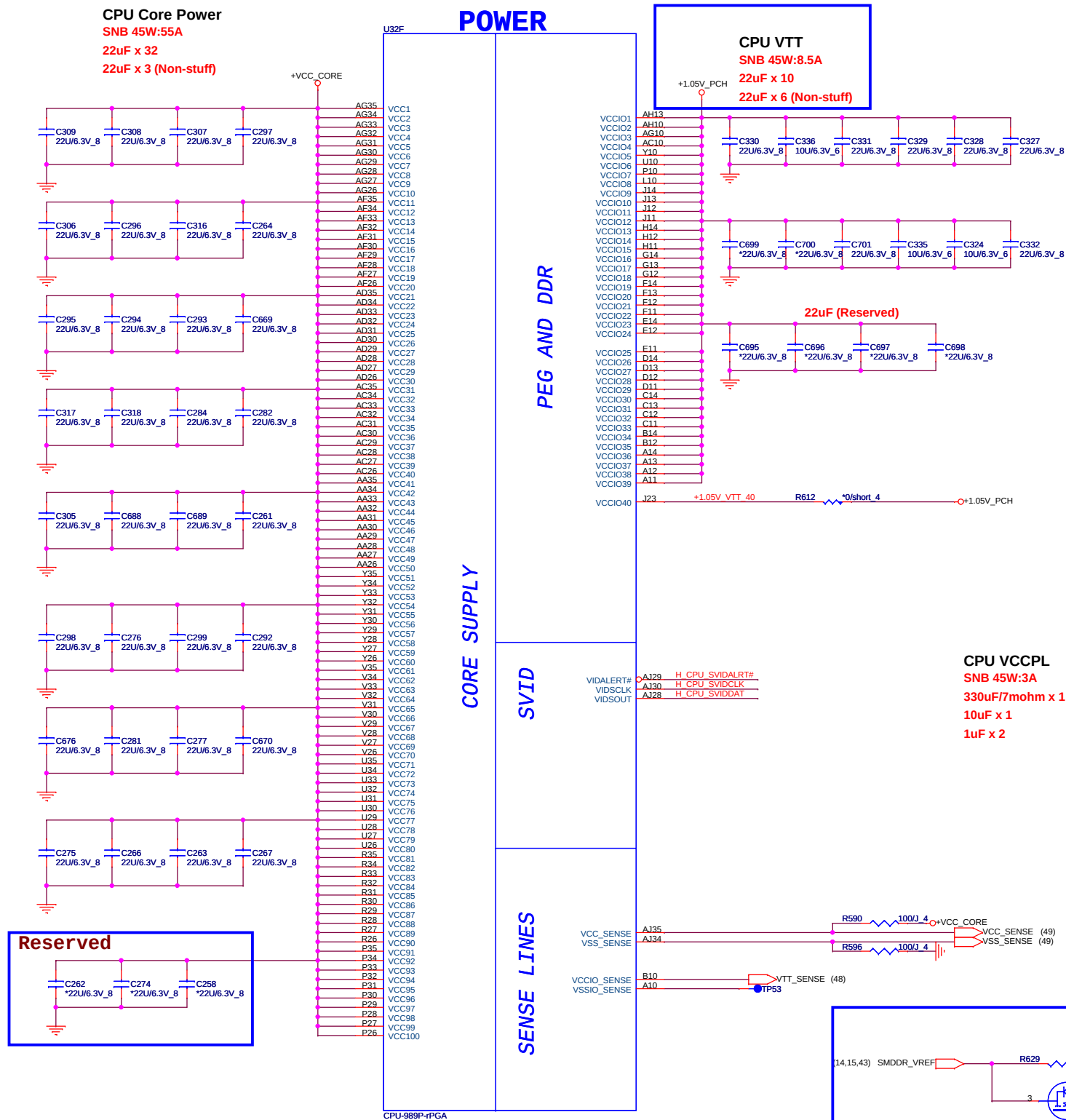
eDP\_COMPIO and ICOMPO signals should be shorted near balls and routed with typical impedance <25 mohms



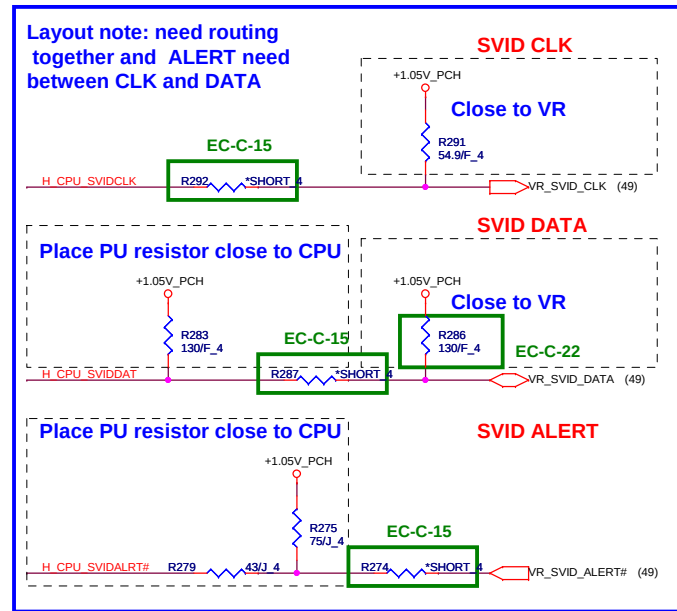
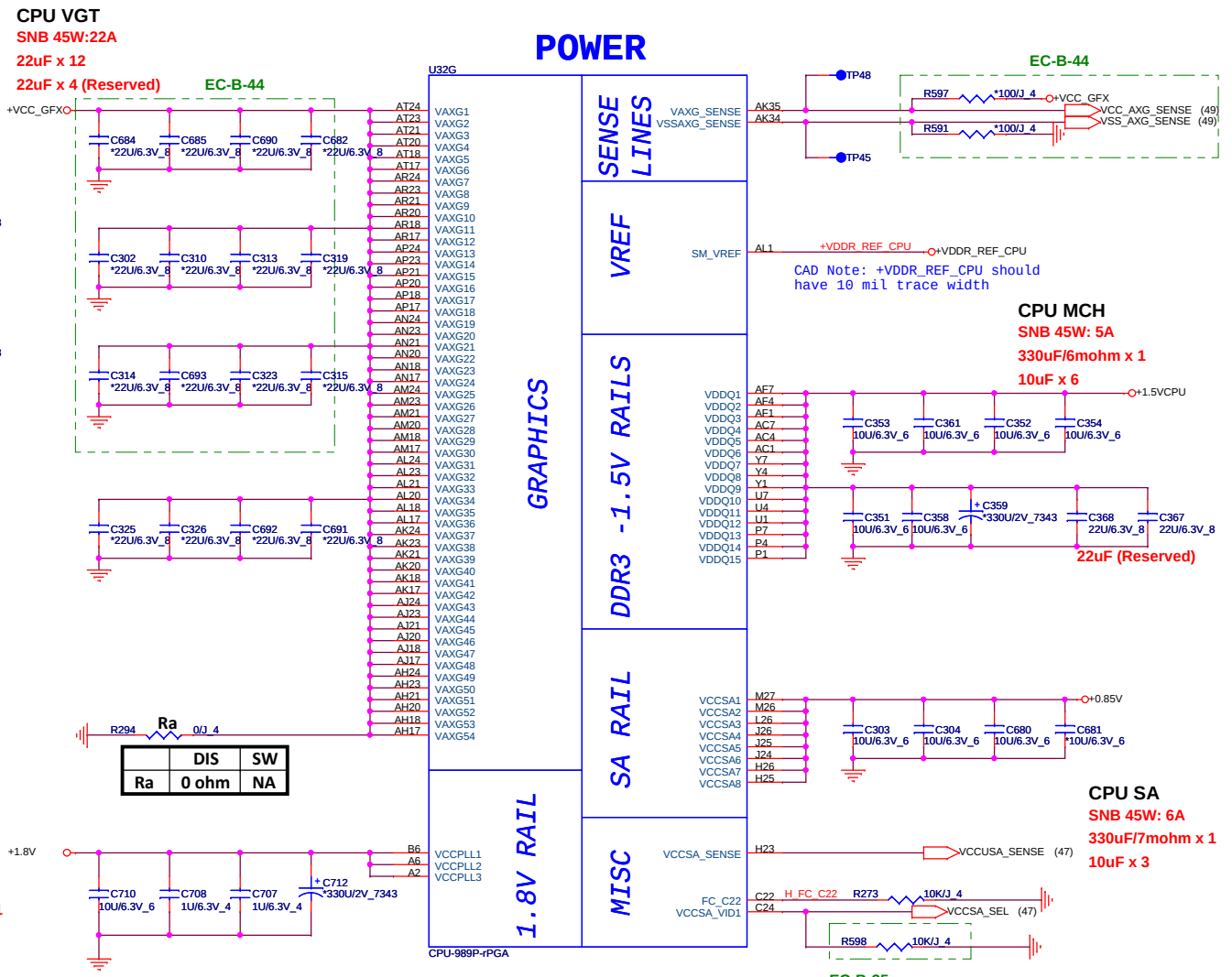


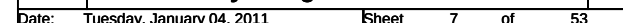


## Sandy Bridge Processor (POWER)



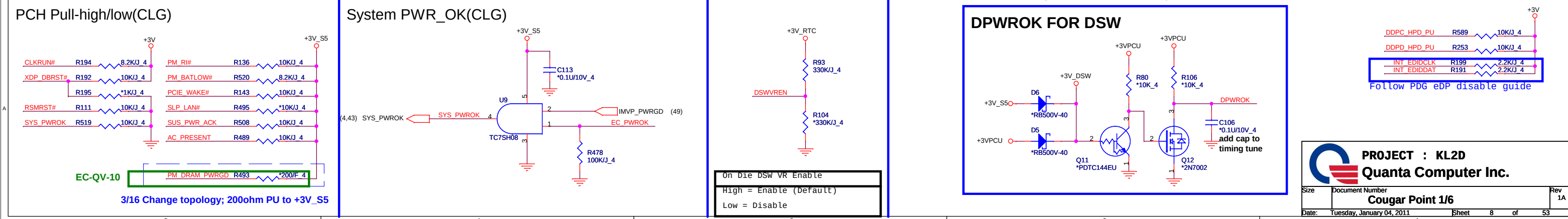
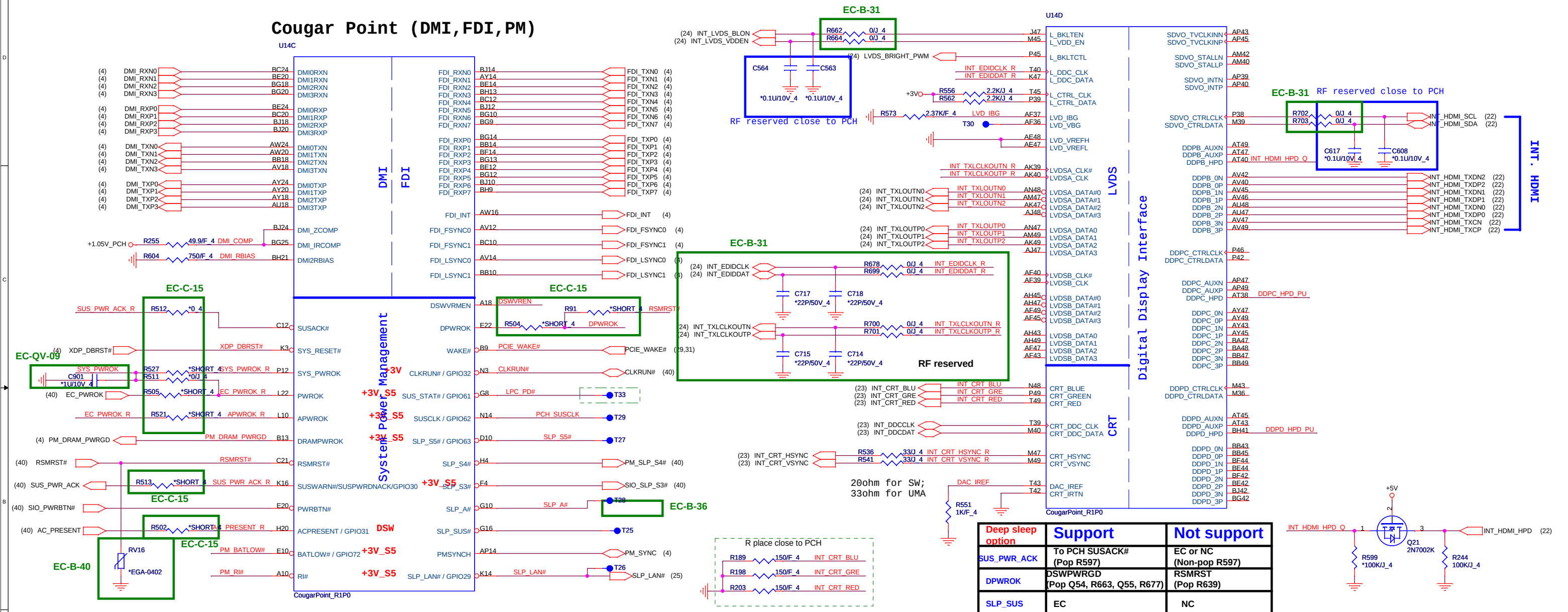
## Sandy Bridge Processor (GRAPHIC POWER)





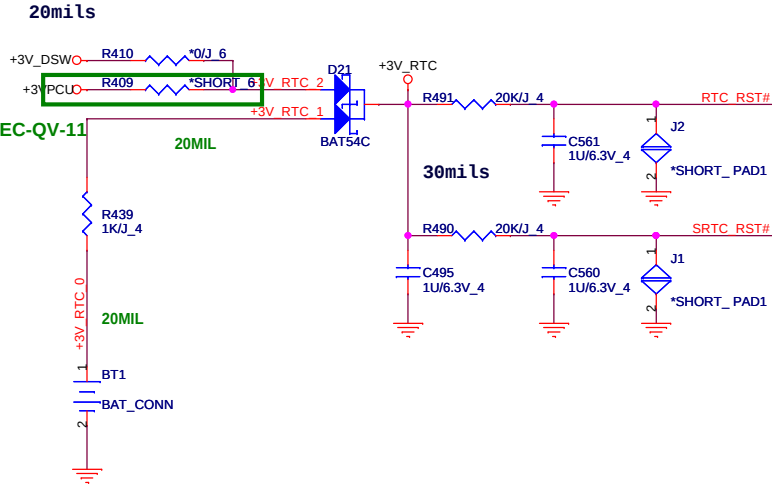
## Cougar Point (LVDS, DDI)

## Cougar Point (DMI, FDI, PM)

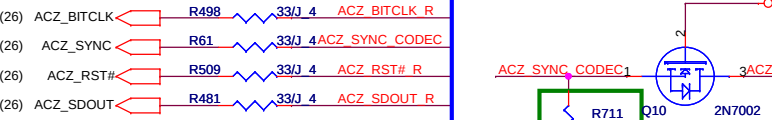




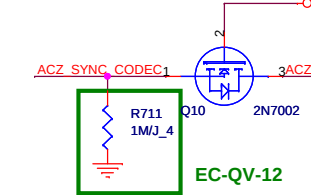
RTC Circuitry(RTC)



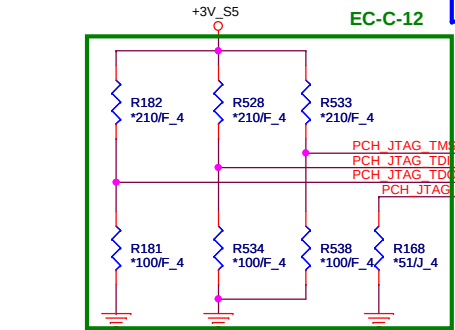
HDA Bus(CLG)



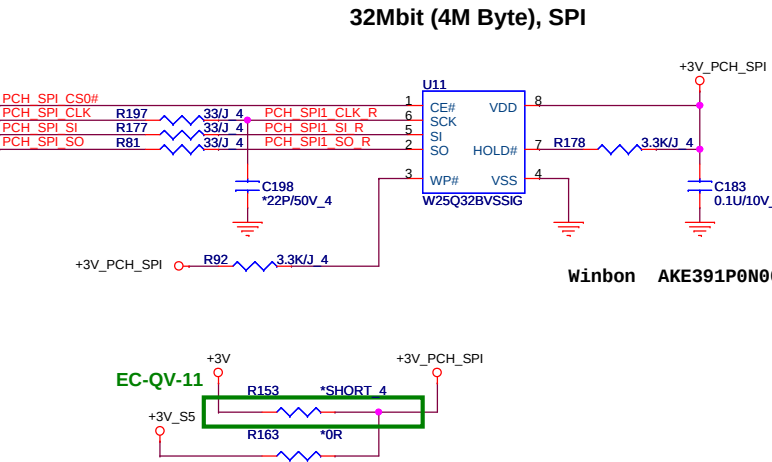
To Separate Codec Sync by PD3



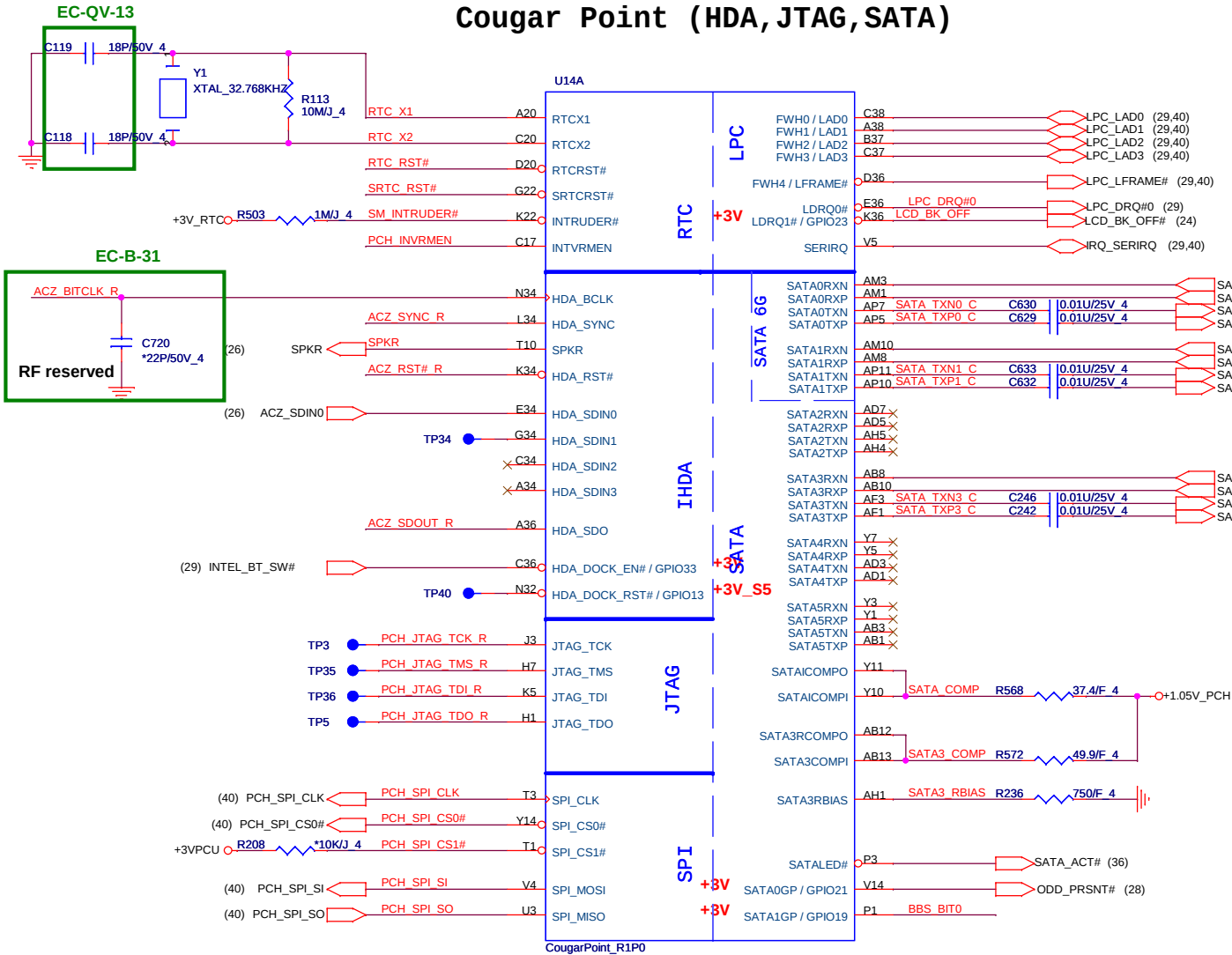
PCH JTAG Debug (CLG)



PCH Dual SPI (CLG)



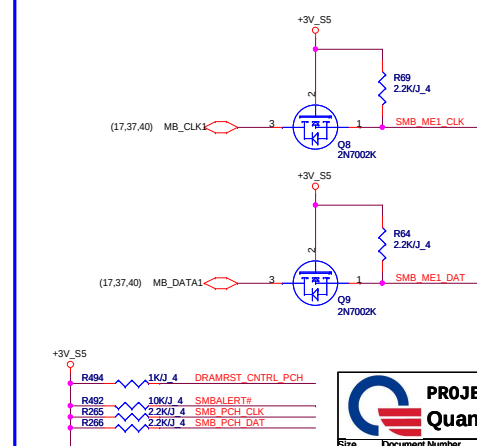
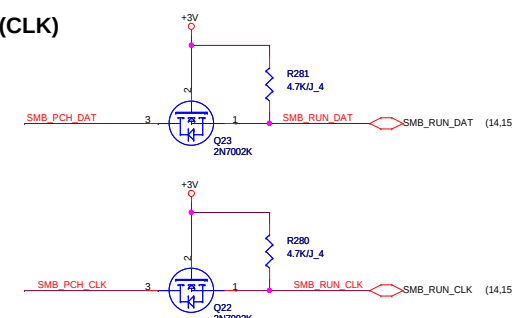
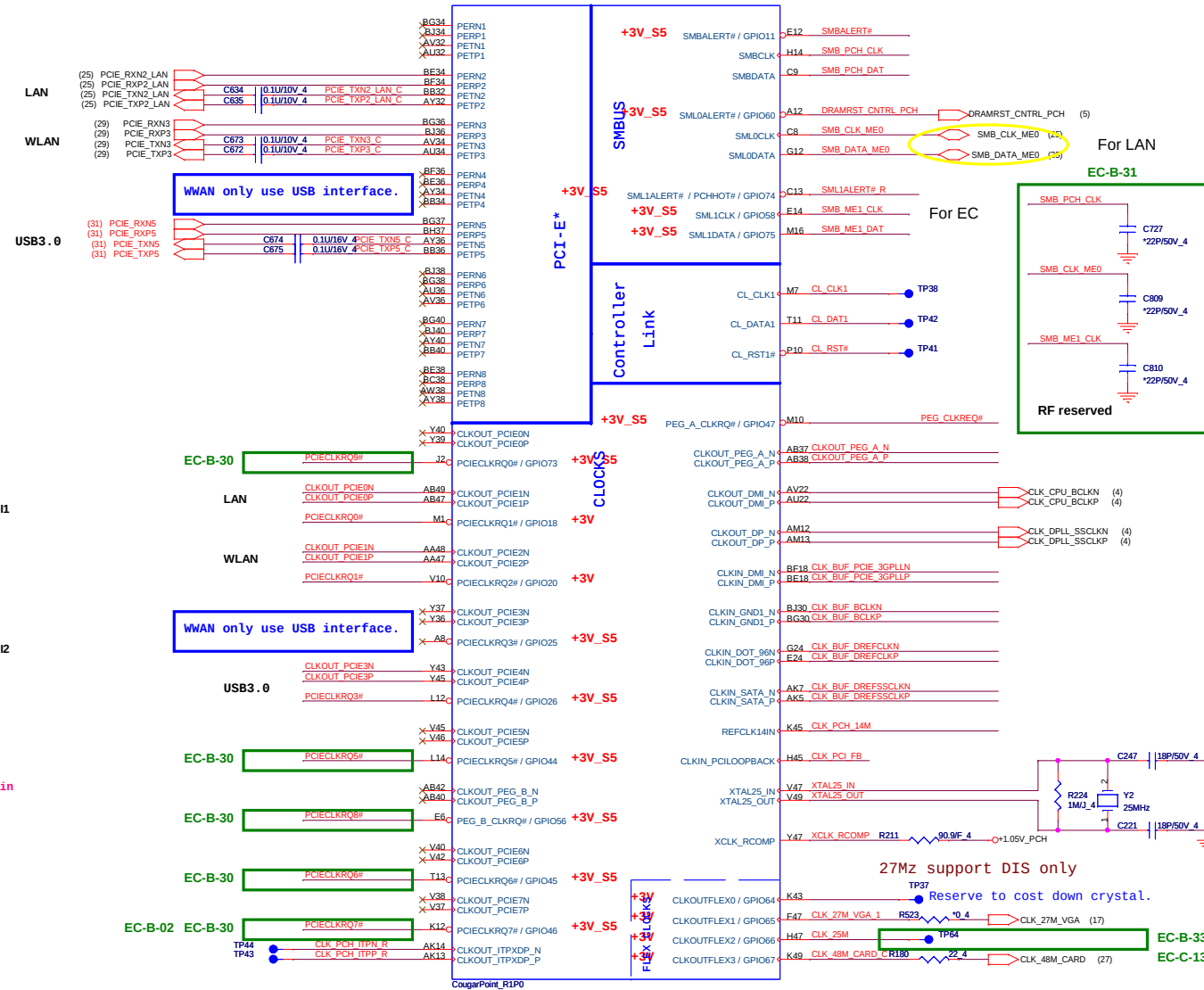
PCH2 (CLG)



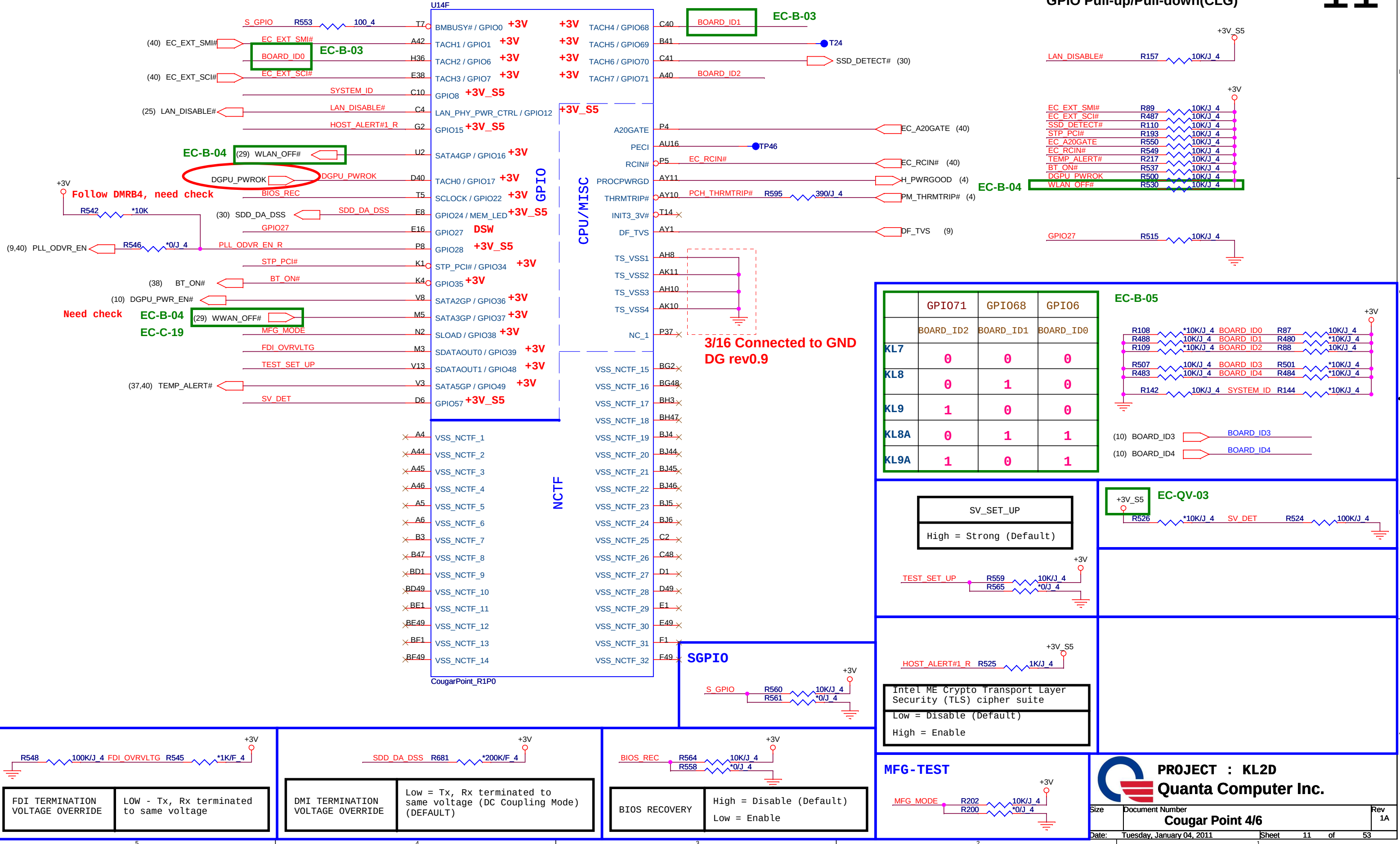
PCH Strap Table

| Pin Name       | Strap description               | Sampled       | Configuration                                                                                                                                                                                                                                                                             |       |       |               |   |   |       |   |   |     |  |
|----------------|---------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|---------------|---|---|-------|---|---|-----|--|
| SPKR           | No reboot mode setting          | PWROK         | 0 = Default (weak pull-down 20K)<br>1 = Setting to No-Reboot mode                                                                                                                                                                                                                         |       |       |               |   |   |       |   |   |     |  |
| GNT3# / GPIO55 | Top-Block Swap Override         | PWROK         | 0 = "top-block swap" mode<br>1 = Default (weak pull-up 20K)                                                                                                                                                                                                                               |       |       |               |   |   |       |   |   |     |  |
| INTVRMEN       | Integrated 1.05V VRM enable     | ALWAYS        | Should be always pull-up                                                                                                                                                                                                                                                                  |       |       |               |   |   |       |   |   |     |  |
| GNT1# / GPIO51 | Boot BIOS Selection 1 [bit-1]   | PWROK         | <table border="1"><thead><tr><th>GNT1#</th><th>GNT0#</th><th>Boot Location</th></tr></thead><tbody><tr><td>1</td><td>1</td><td>SPI *</td></tr><tr><td>0</td><td>0</td><td>LPC</td></tr></tbody></table> <b>Default weak pull-up on GNT0/1#<br/>[Need external pull-down for LPC BIOS]</b> | GNT1# | GNT0# | Boot Location | 1 | 1 | SPI * | 0 | 0 | LPC |  |
| GNT1#          | GNT0#                           | Boot Location |                                                                                                                                                                                                                                                                                           |       |       |               |   |   |       |   |   |     |  |
| 1              | 1                               | SPI *         |                                                                                                                                                                                                                                                                                           |       |       |               |   |   |       |   |   |     |  |
| 0              | 0                               | LPC           |                                                                                                                                                                                                                                                                                           |       |       |               |   |   |       |   |   |     |  |
| GPIO19         | Boot BIOS Selection 0 [bit-0]   | PWROK         |                                                                                                                                                                                                                                                                                           |       |       |               |   |   |       |   |   |     |  |
| HDA_SDO        | Flash Descriptor Security       | RSMRST        | 0 = Override<br>1 = Default (weak pull-up 20K)                                                                                                                                                                                                                                            |       |       |               |   |   |       |   |   |     |  |
| DF_TVS         | DMI/FDI Termination voltage     | PWROK         | 0 = Set to Vss<br>1 = Set to Vcc (weak pull-down 20K)                                                                                                                                                                                                                                     |       |       |               |   |   |       |   |   |     |  |
| GPIO28         | On-die PLL Voltage Regulator    | RSMRST#       | 0 = Disable<br>1 = Enable (Default)                                                                                                                                                                                                                                                       |       |       |               |   |   |       |   |   |     |  |
| HDA_SYNC       | On-Die PLL VR Voltage Select    | RSMRST        | 0 = Support by 1.8V (weak pull-down)<br>1 = Support by 1.5V                                                                                                                                                                                                                               |       |       |               |   |   |       |   |   |     |  |
| GPIO8          | Integrated Clock Chip Enable    | RSMRST#       | Should be pull-down<br>(weak pull-up 20K)                                                                                                                                                                                                                                                 |       |       |               |   |   |       |   |   |     |  |
| SPI_MOSI       | iTPM function Disable           | APWROK        | 0 = Default (weak pull-down 20K)<br>1 = Enable                                                                                                                                                                                                                                            |       |       |               |   |   |       |   |   |     |  |
| NV_ALE         | Intel Anti-Theft HDD protection | PWROK         | 0 = Disable (Internal pull-down 20kohm)                                                                                                                                                                                                                                                   |       |       |               |   |   |       |   |   |     |  |

## U14B

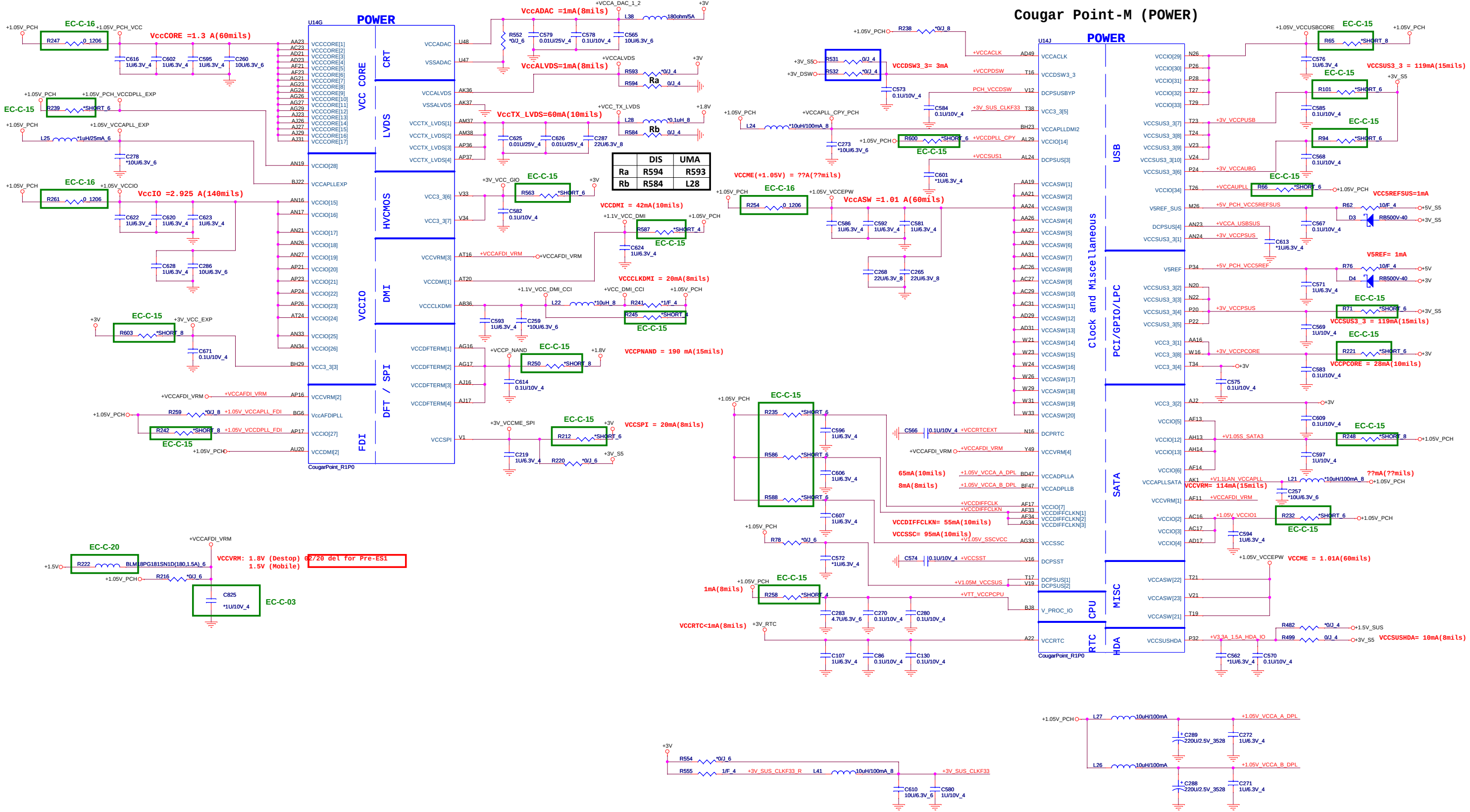


Cougar Point (GPIO,VSS\_NCTF,RSVD)



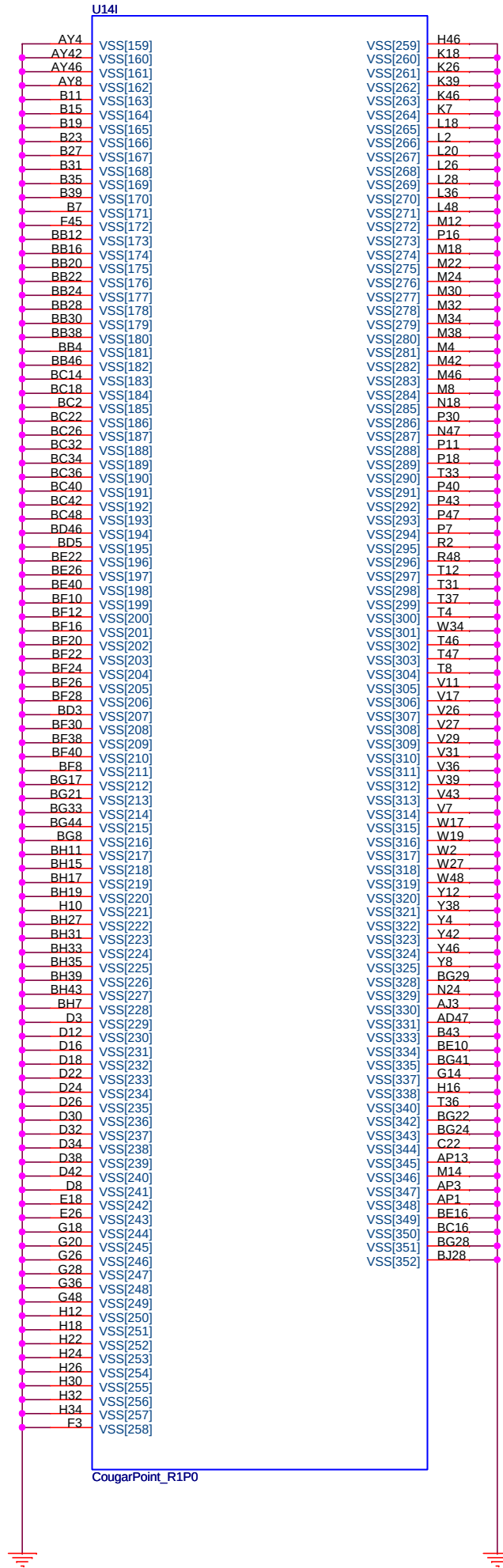
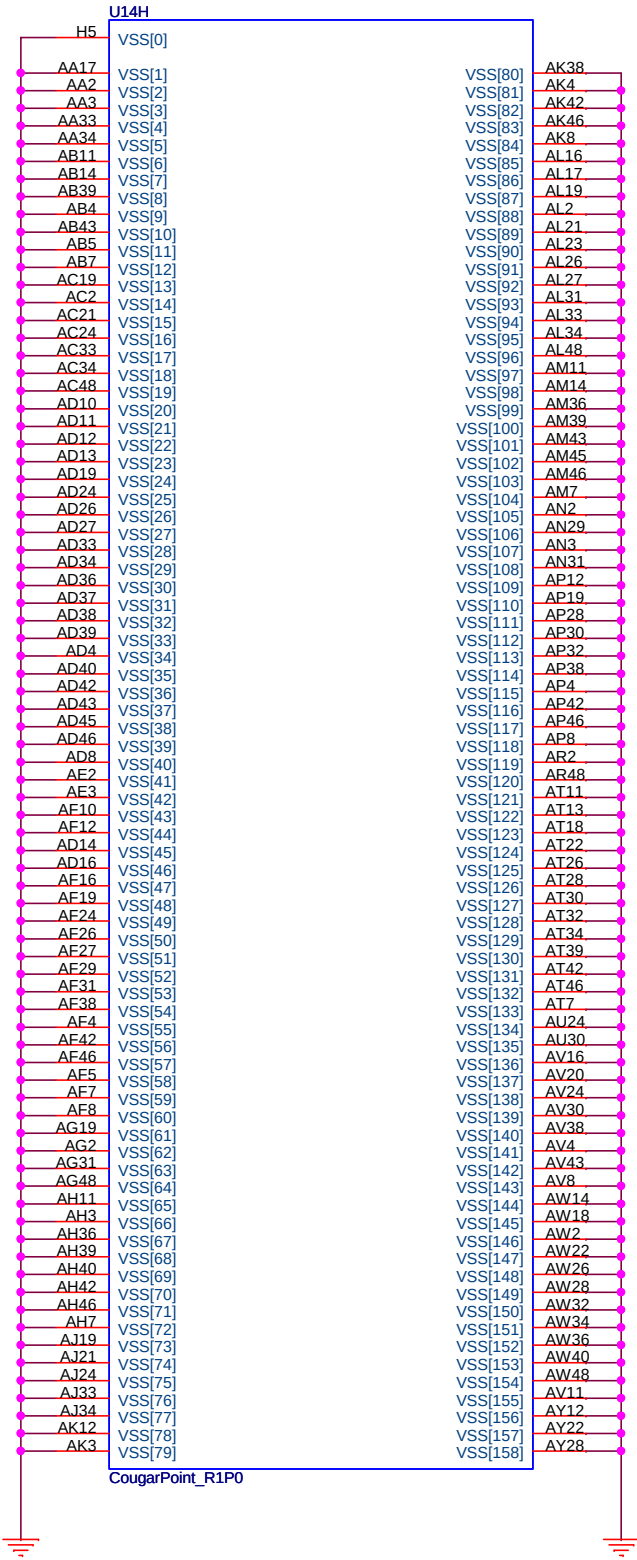
COUGAR POINT (POWER)

Cougar Point-M (POWER)





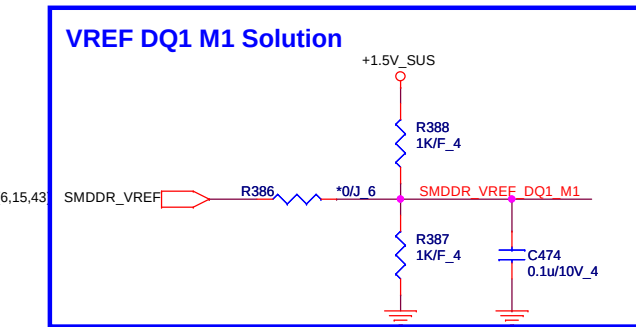
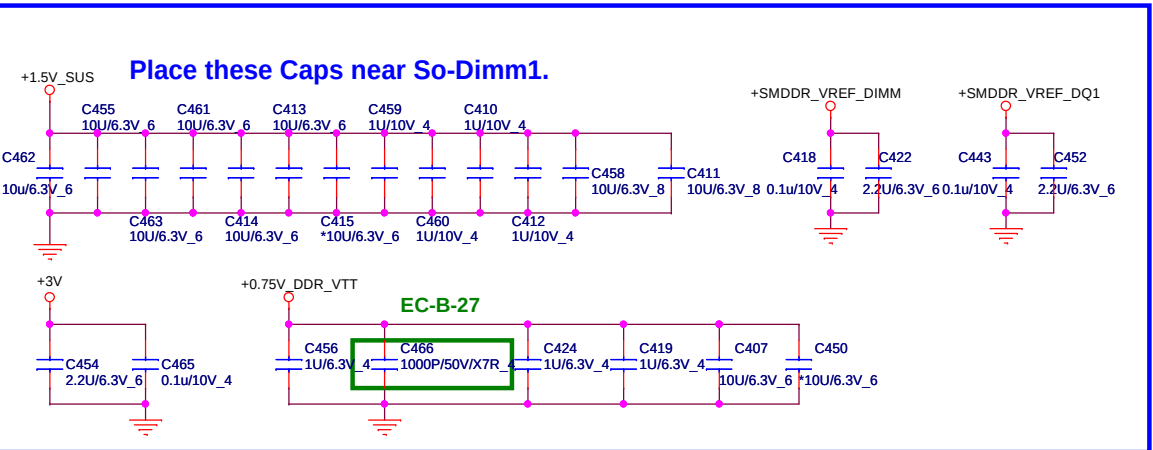
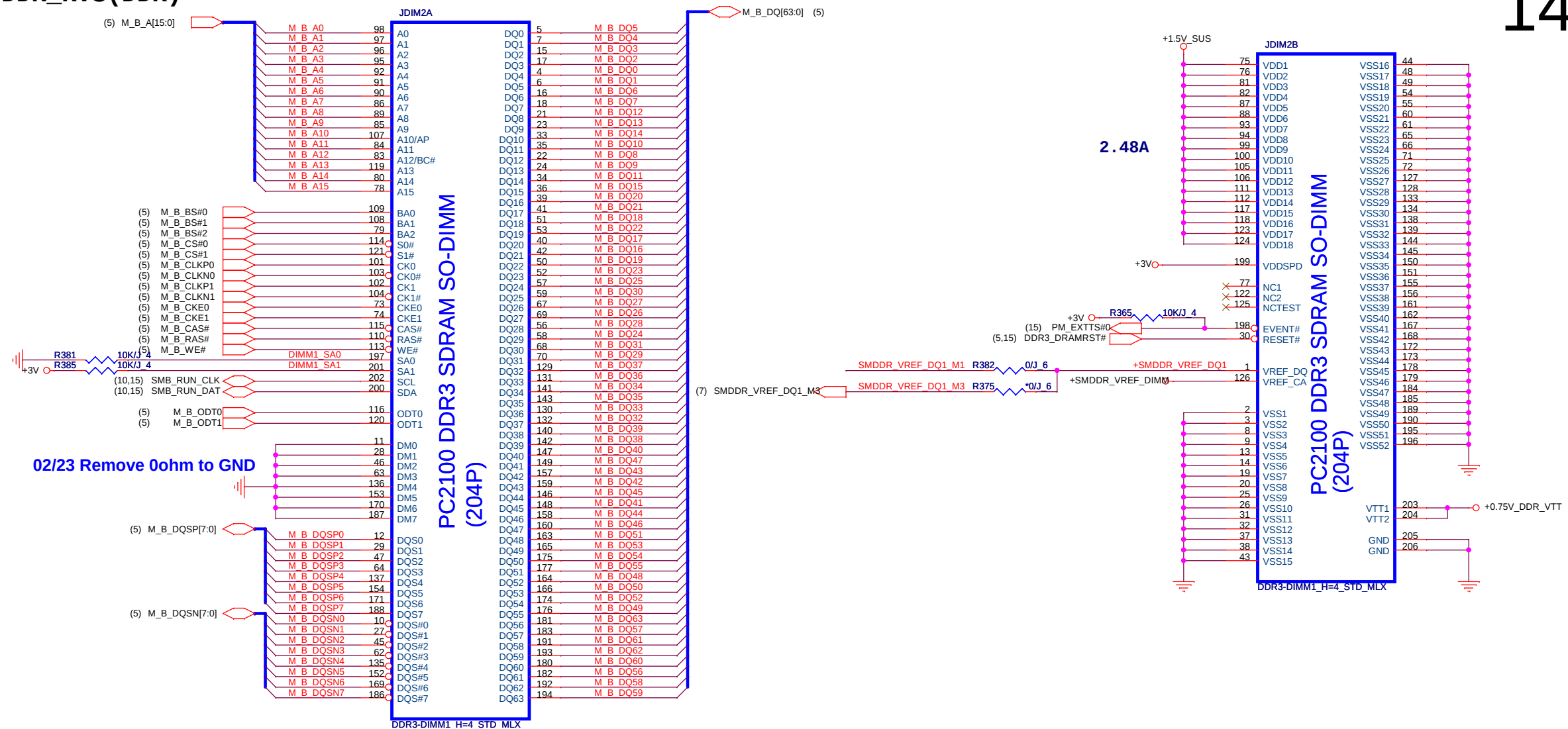
IBEX PEAK-M (GND)





DDR\_RVS(DDR)

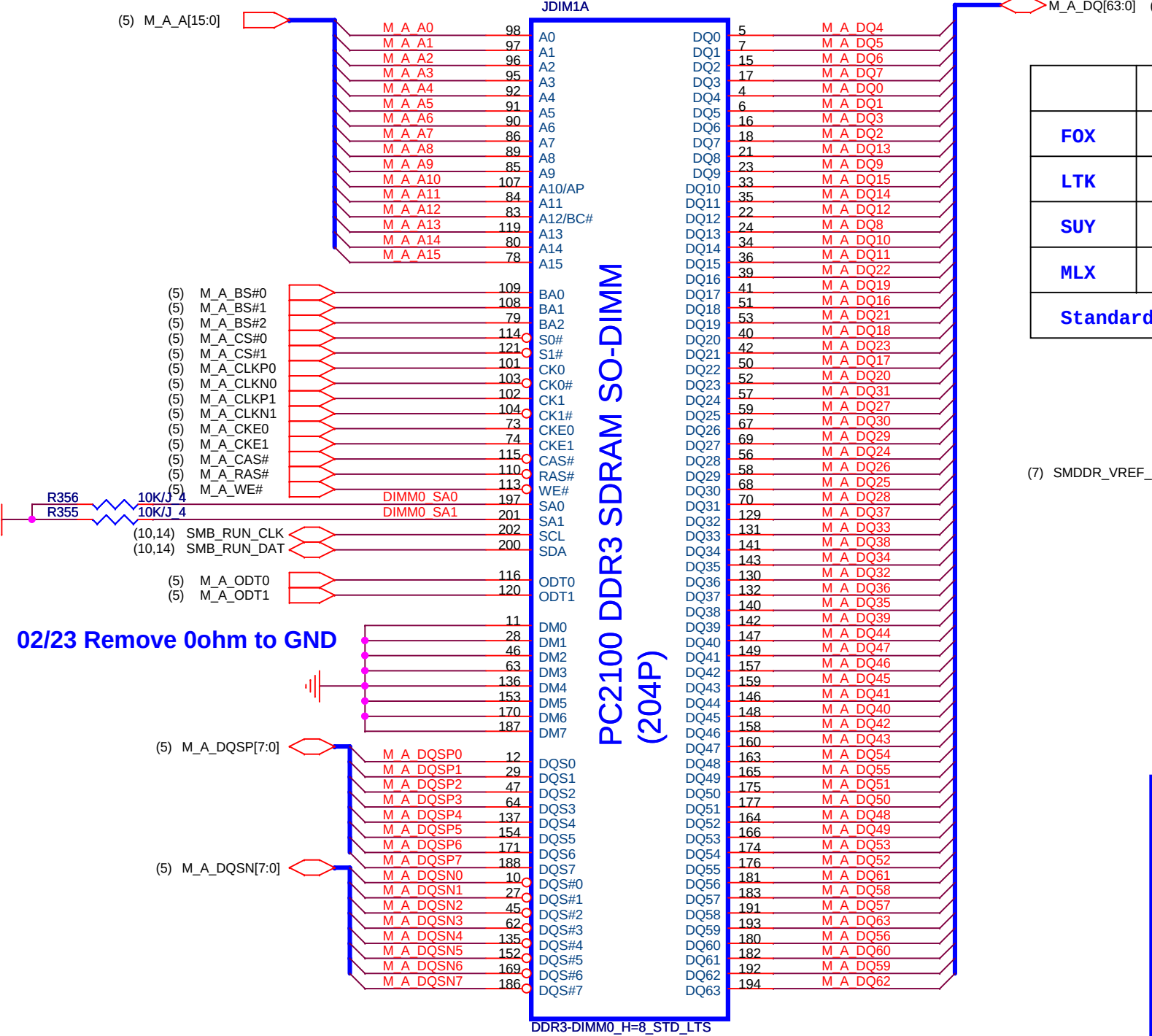
14



|                                       | STD 4H      | STD 8H      |
|---------------------------------------|-------------|-------------|
| FOX                                   |             |             |
| LTK                                   | DGMK4000004 | DGMK4000097 |
| SUY                                   |             |             |
| MLX                                   | DGMK4000011 | DGMK4000080 |
| Standard 8H type:DDR-C-2013310-204p-1 |             |             |

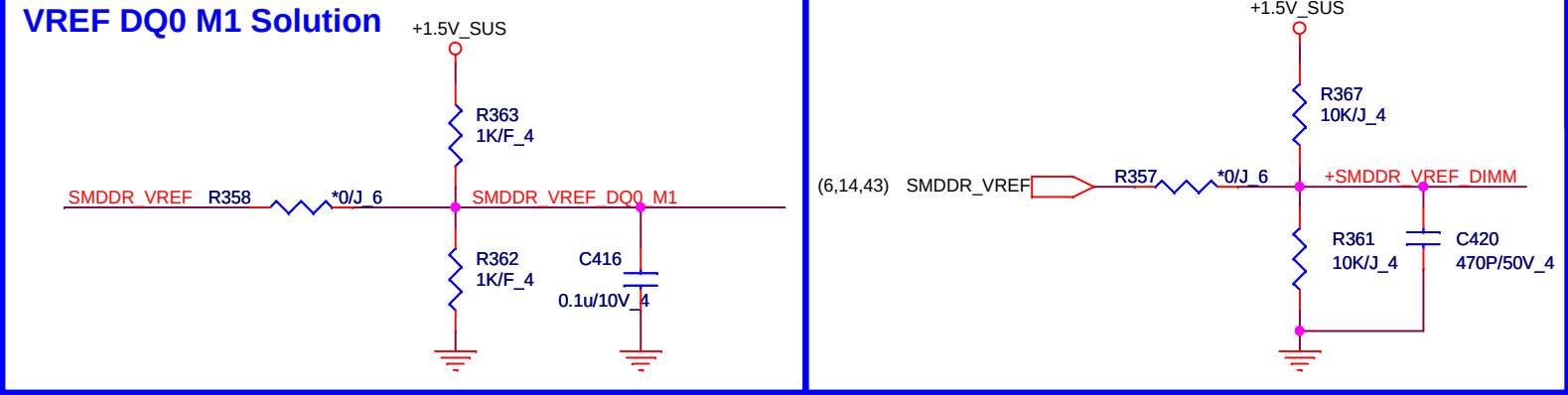


DDR\_RVS(DDR)

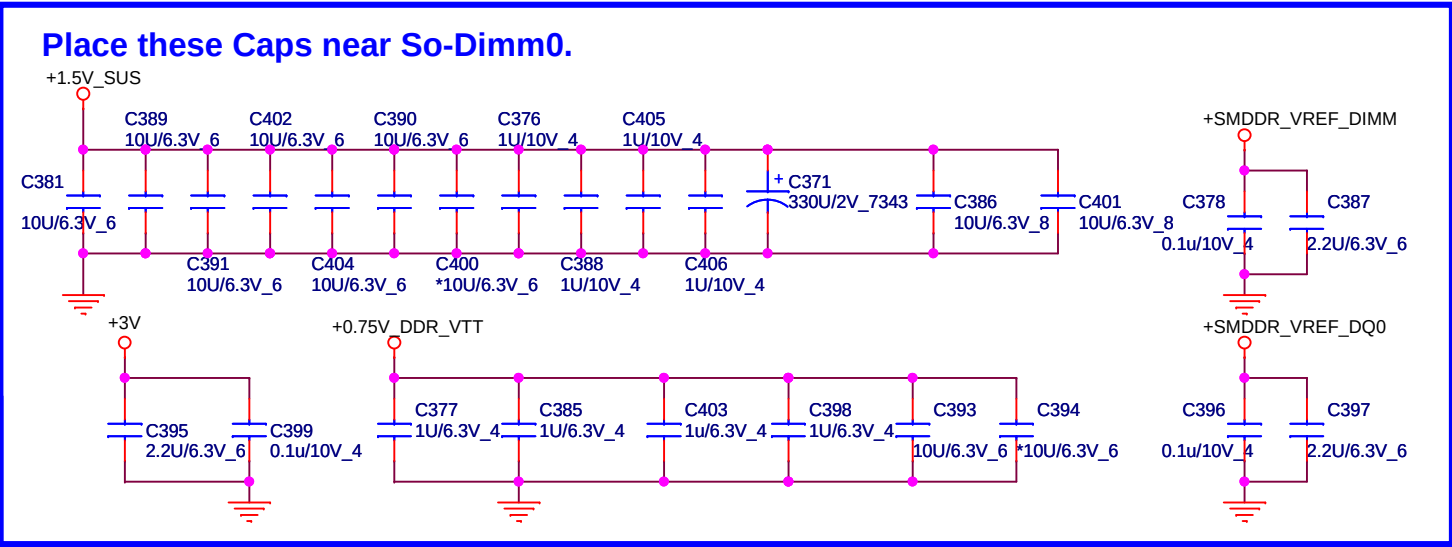


02/23 Remove 0ohm to GND

VREF DQ0 M1 Solution



|                                     | STD 4H      | STD 8H      |
|-------------------------------------|-------------|-------------|
| FOX                                 |             |             |
| LTK                                 | DGMK4000004 | DGMK4000097 |
| SUY                                 |             |             |
| MLX                                 | DGMK4000011 | DGMK4000080 |
| Standard 4H type:DDR-C-2013289-204p |             |             |





**PROJECT KL5A**  
**Quanta Computer Inc.**

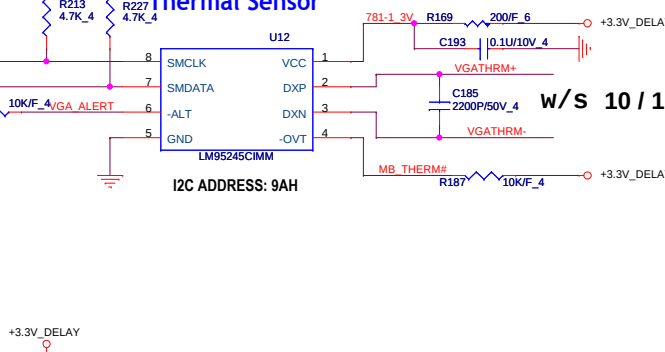
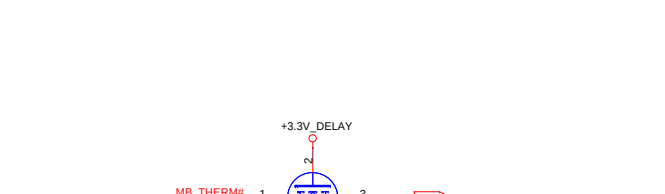
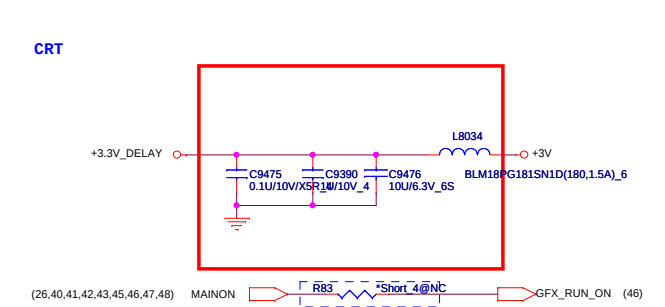
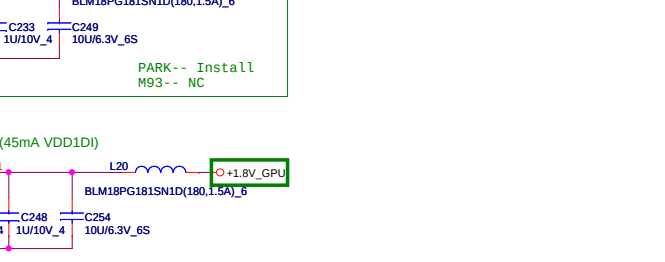
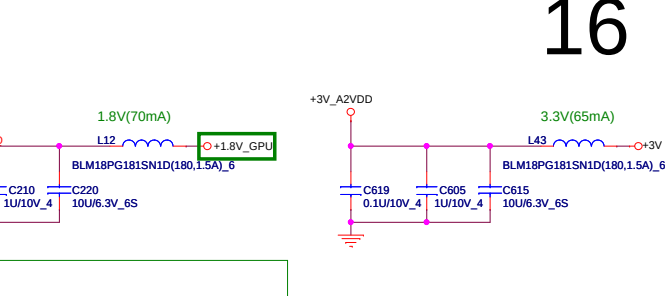
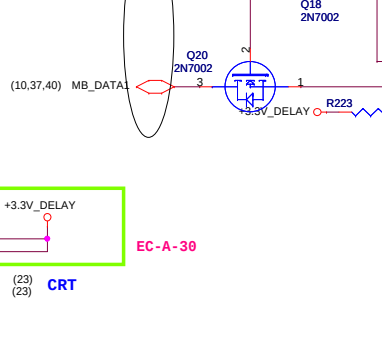
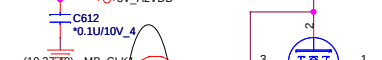
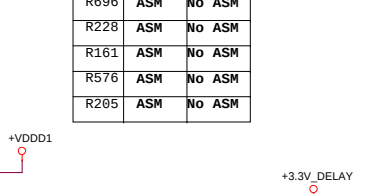
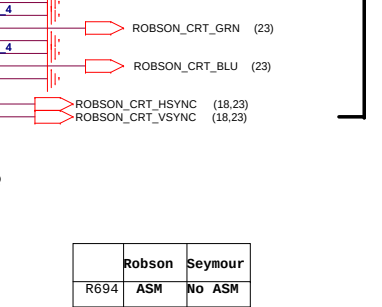
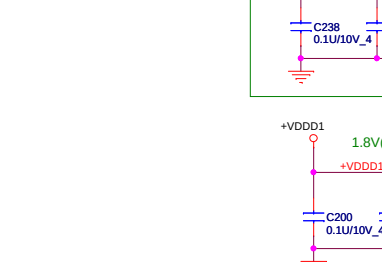
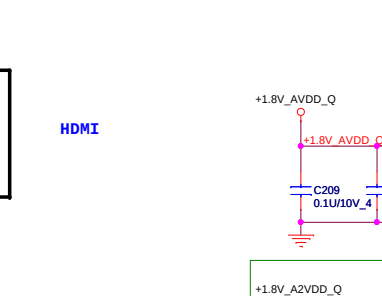
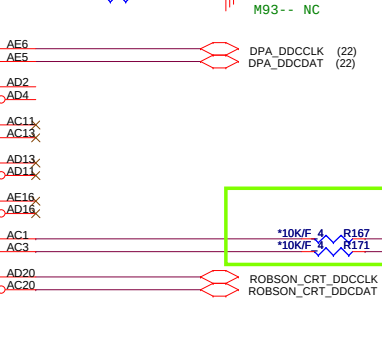
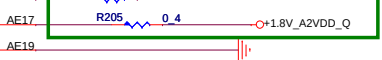
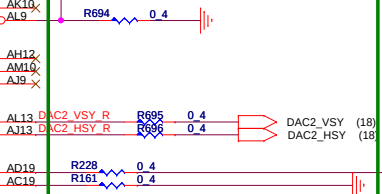
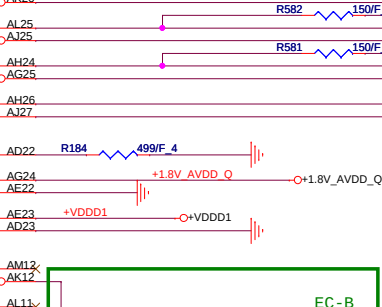
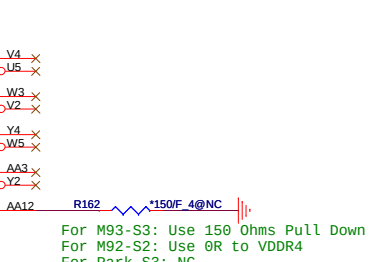
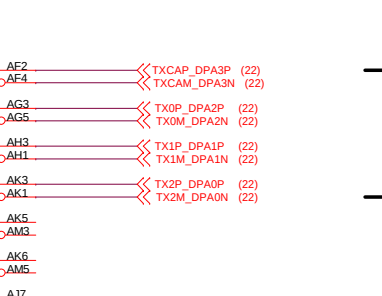
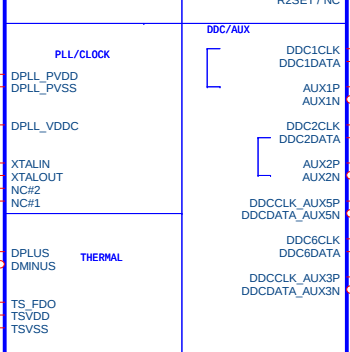
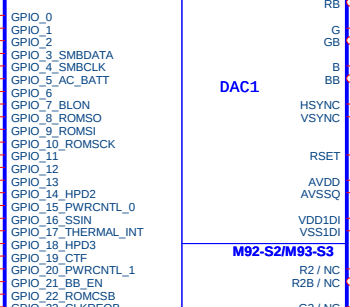
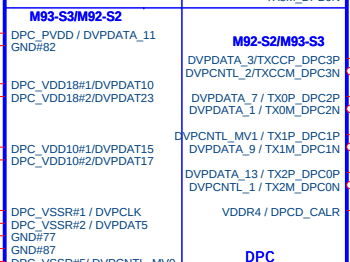
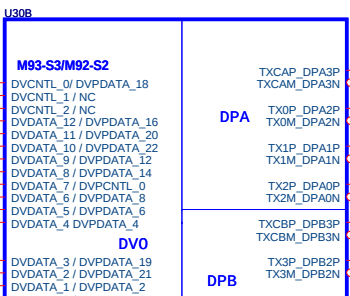
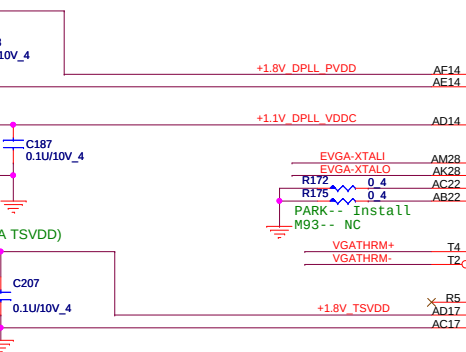
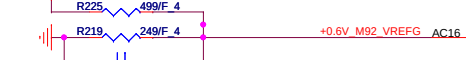
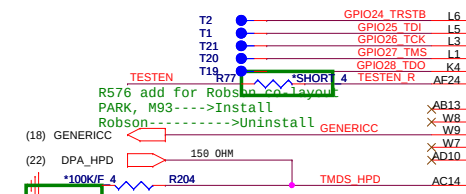
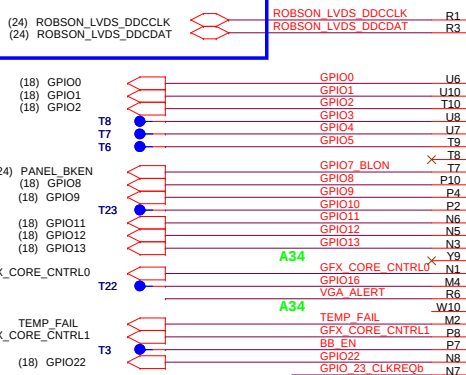
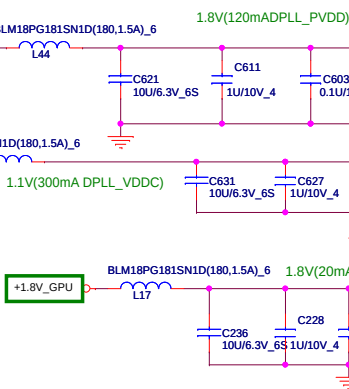
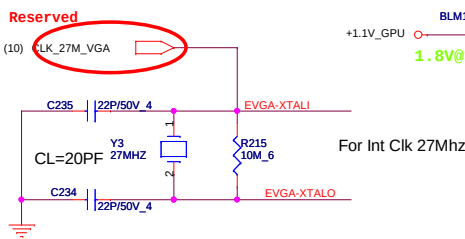
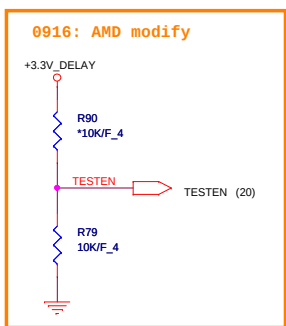
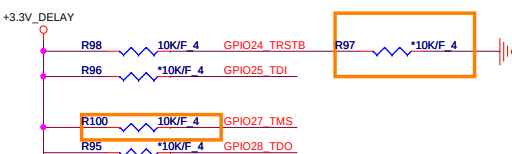
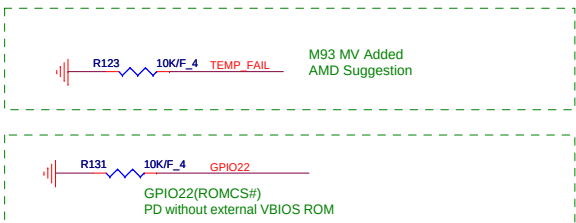
|                                 |                                            |        |
|---------------------------------|--------------------------------------------|--------|
| Size B                          | Document Number<br><b>DDRIII SO-DIMM-1</b> | Rev 1A |
| Date: Tuesday, January 04, 2011 | Sheet 15 of 53                             |        |



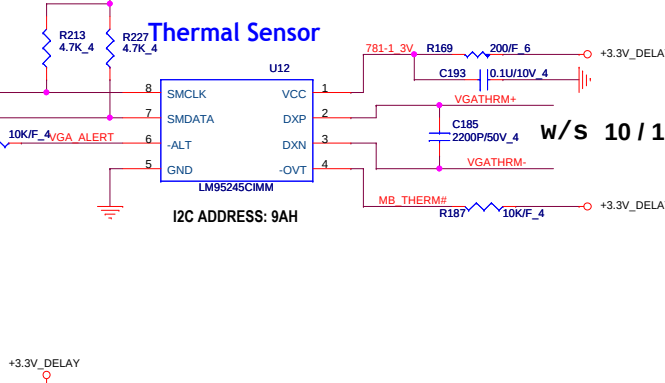
| MEM_ID[3:0] |      |      |      | Vendor  | Type             | Vendor P/N      | Project   |
|-------------|------|------|------|---------|------------------|-----------------|-----------|
| R160        | R152 | R146 | R151 |         |                  |                 |           |
| 0           | 0    | 0    | 0    | Samsung | 128MB*16-800MHZ  | K4W2G1646C-HC12 | KL8A/KL9A |
| 0           | 0    | 0    | 1    | Hynix   | 128MB*16 -800MHZ | H5TQ2G63BFR-12C | KL8A/KL9A |
| 0           | 0    | 1    | 0    | Samsung | 64x16 800MHZ     | K4W1G1646E-HC12 | KL9A only |
| 0           | 0    | 1    | 1    | Hynix   | 64x16 800MHZ     | H5TQ1G63DFR-12C | KL9A only |

**Table : VDDC\_OPT VID**

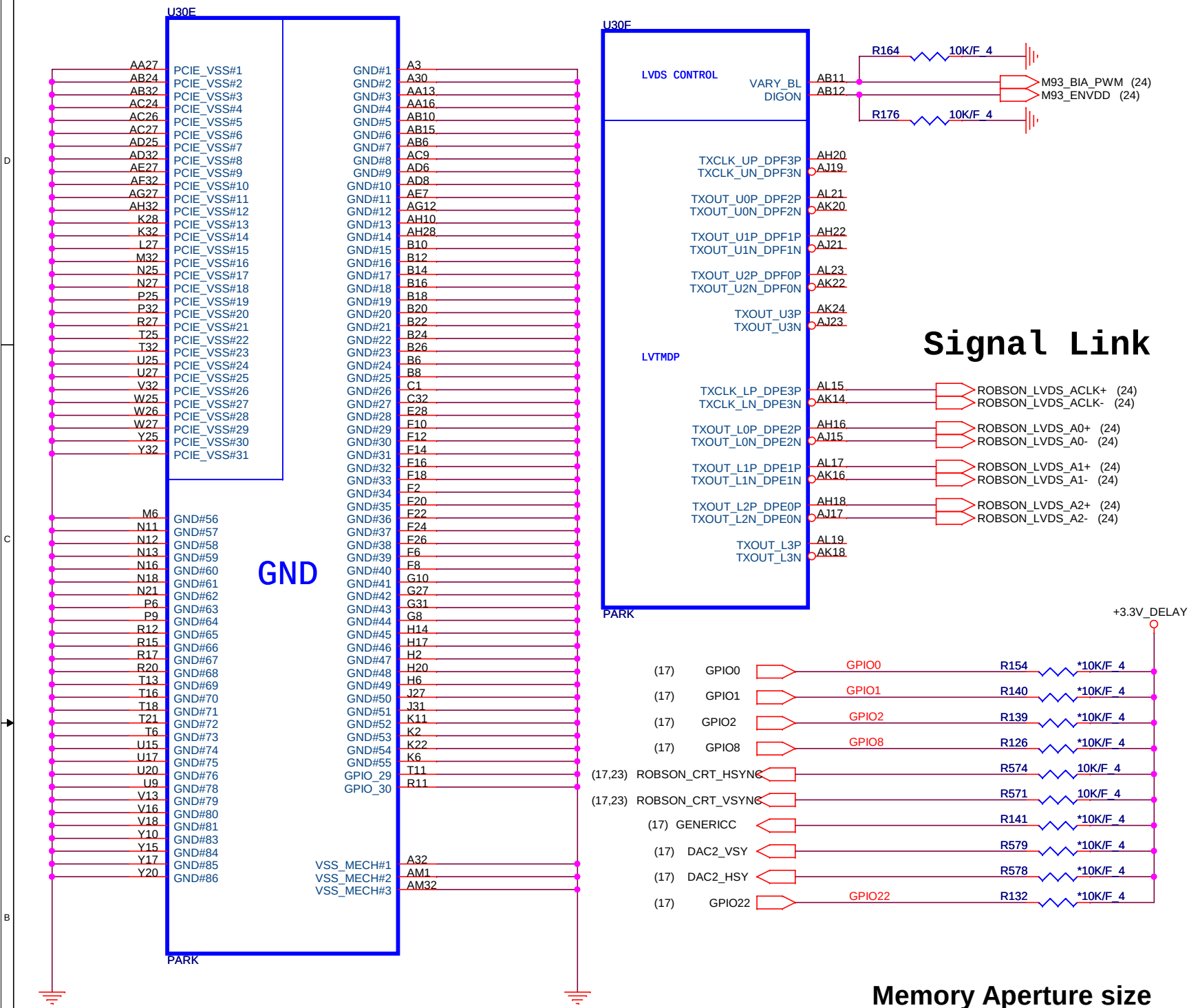
| GFX_CORE_CNTRL1 | GFX_CORE_CNTRL0 | +VCC_GFX_CORE |
|-----------------|-----------------|---------------|
| 1               | 1               | 1.12V         |
| 1               | 0               | 0.95V         |
| 0               | 1               | 1.05V         |
| 0               | 0               | 0.9V          |



|      | Robson | Seymour |
|------|--------|---------|
| R694 | ASM    | No ASM  |
| R695 | ASM    | No ASM  |
| R696 | ASM    | No ASM  |
| R228 | ASM    | No ASM  |
| R161 | ASM    | No ASM  |
| R576 | ASM    | No ASM  |
| R205 | ASM    | No ASM  |







| CONFIGURATION STRAPS                                                                                    |                            |                                                                                                                                                                                 | RECOMMENDED SETTINGS<br>0= DO NOT INSTALL RESISTOR<br>1 = INSTALL 10K RESISTOR<br>X = DESIGN DEPENDANT<br>NA = NOT APPLICABLE |
|---------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| ALLOW FOR PULLUP PADS FOR THESE STRAPS AND IF THESE GPIOs ARE USED, THEY MUST NOT CONFLICT DURING RESET |                            |                                                                                                                                                                                 |                                                                                                                               |
| STRAPS                                                                                                  | PIN                        | DESCRIPTION OF DEFAULT SETTINGS                                                                                                                                                 |                                                                                                                               |
| TX_PWRS_ENB                                                                                             | GPIO0                      | <b>Transmitter Power Savings Enable</b><br>0: 50% Tx output swing for mobile mode<br>1: full Tx output swing (Default setting for Desktop)                                      | 1                                                                                                                             |
| TX_DEEMPH_EN                                                                                            | GPIO1                      | <b>PCI Express Transmitter De-emphasis Enable</b><br>0: Tx de-emphasis disabled for mobile mode<br>1: Tx de-emphasis enabled (Default setting for Desktop)                      | 1                                                                                                                             |
| BIF_GEN2_EN_A                                                                                           | GPIO2                      | Enable CLKREQ# Power Management<br>0 - CLKREQ# power management capability is disabled<br>1 - CLKREQ# power management capability is enabled                                    | 0                                                                                                                             |
| RSVD<br>BIF_VGA_DIS<br>RSVD                                                                             | GPIO8<br>GPIO9<br>GPIO21   | VGA ENABLED                                                                                                                                                                     | 0<br>0<br>0                                                                                                                   |
| BIOS_ROM_EN                                                                                             | GPIO_22_ROMCSB             | ENABLE EXTERNAL BIOS ROM                                                                                                                                                        | 0                                                                                                                             |
| ROMIDCFG(2:0)                                                                                           | GPIO[13:11]                | SERIAL ROM TYPE OR MEMORY APERTURE SIZE SELECT                                                                                                                                  | 0 0 1                                                                                                                         |
| VIP_DEVICE_STRAP_ENA                                                                                    | V2SYNC                     | IGNORE VIP DEVICE STRAPS                                                                                                                                                        | 0                                                                                                                             |
| RSVD<br>AUD[1]<br>AUD[0]                                                                                | GENERICC<br>HSYNC<br>VSYNC | AUD[1] AUD[0]<br>0 0 No audio function<br>0 1 Audio for DisplayPort and HDMI if dongle is detected<br>1 0 Audio for DisplayPort only<br>1 1 Audio for both DisplayPort and HDMI | 0<br>0<br>11                                                                                                                  |

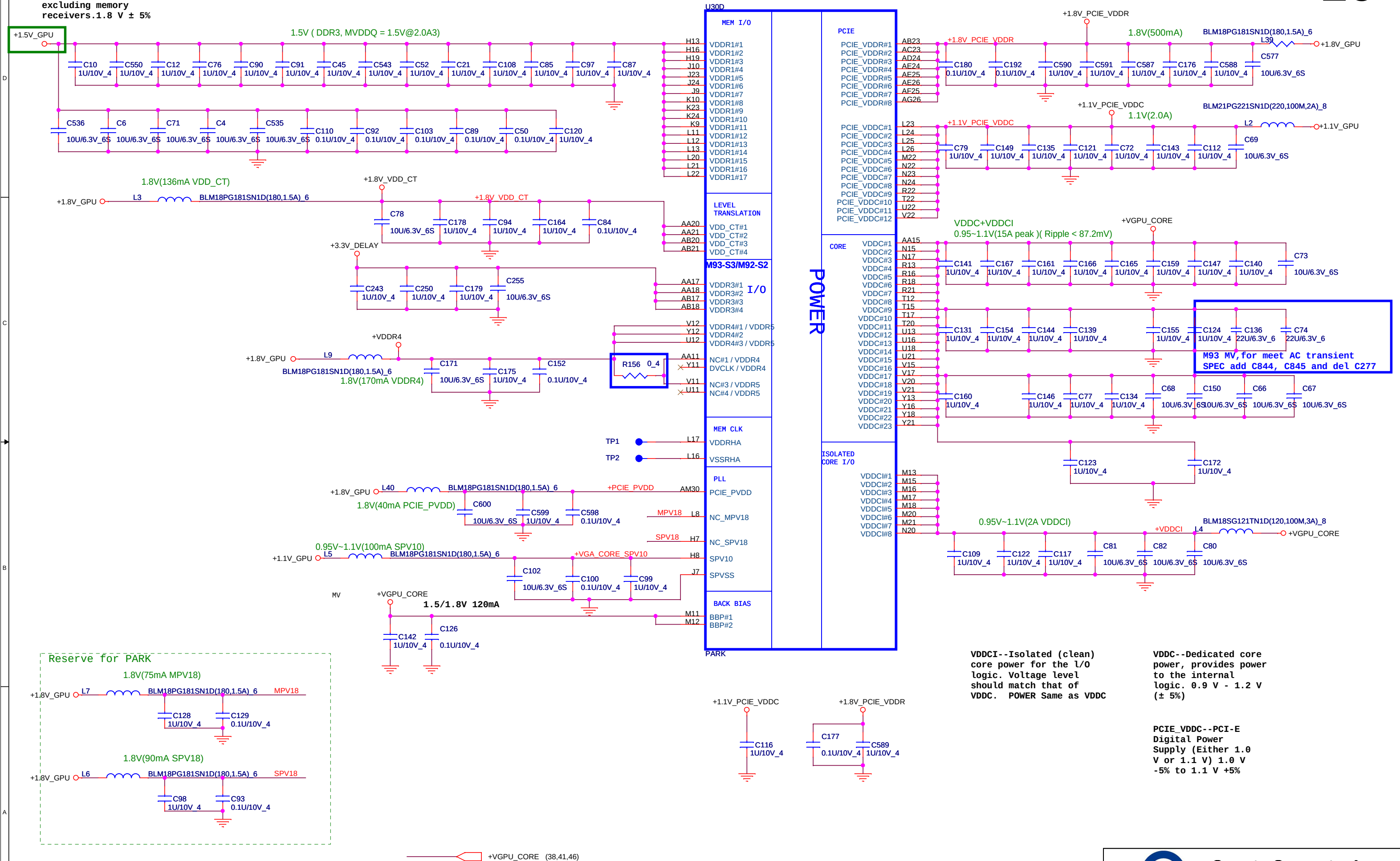
| AMD RESERVED CONFIGURATION STRAPS                                                                              |          |
|----------------------------------------------------------------------------------------------------------------|----------|
| ALLOW FOR PULLUP PADS FOR THESE STRAPS AND IF THESE GPIOs ARE USED, THEY MUST NOT CONFLICT DURING RESET        |          |
| H2SYNC                                                                                                         | GENERICC |
| PULLUP PADS ARE NOT REQUIRED FOR THESE STRAPS BUT IF THESE GPIOs ARE USED, THEY MUST NOT CONFLICT DURING RESET |          |
| GPIO21_BB_EN                                                                                                   |          |

| GPIO9<br>BIOSROM |      | GPIO13<br>ROMIDCFG2 | GPIO12<br>ROMIDCFG1 | GPIO11<br>ROMIDCFG0 |
|------------------|------|---------------------|---------------------|---------------------|
| 0                | 128M | 0                   | 0                   | 0                   |
| 0                | 256M | 0                   | 0                   | 1                   |
| 0                | 64M  | 0                   | 1                   | 0                   |
| 0                | 32M  | 0                   | 1                   | 1                   |
| 0                | 512M | 1                   | 0                   | 0                   |
| 0                | 1G   | 1                   | 0                   | 1                   |
| 0                | 2G   | 1                   | 1                   | 0                   |
| 0                | 4G   | 1                   | 1                   | 1                   |

It is a shared pin strap with CONFIG[2:0] if BIOS\_ROM\_EN is set to 0.

VDD\_CT -- Level translation between core and I/O, excluding memory receivers. 1.8 V ± 5%

PCIE\_VDDR--PCI-E I/O power. 1.8 V ± 5%



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**PROJECT : DM4/RB4**

MEMORY INTERFACE

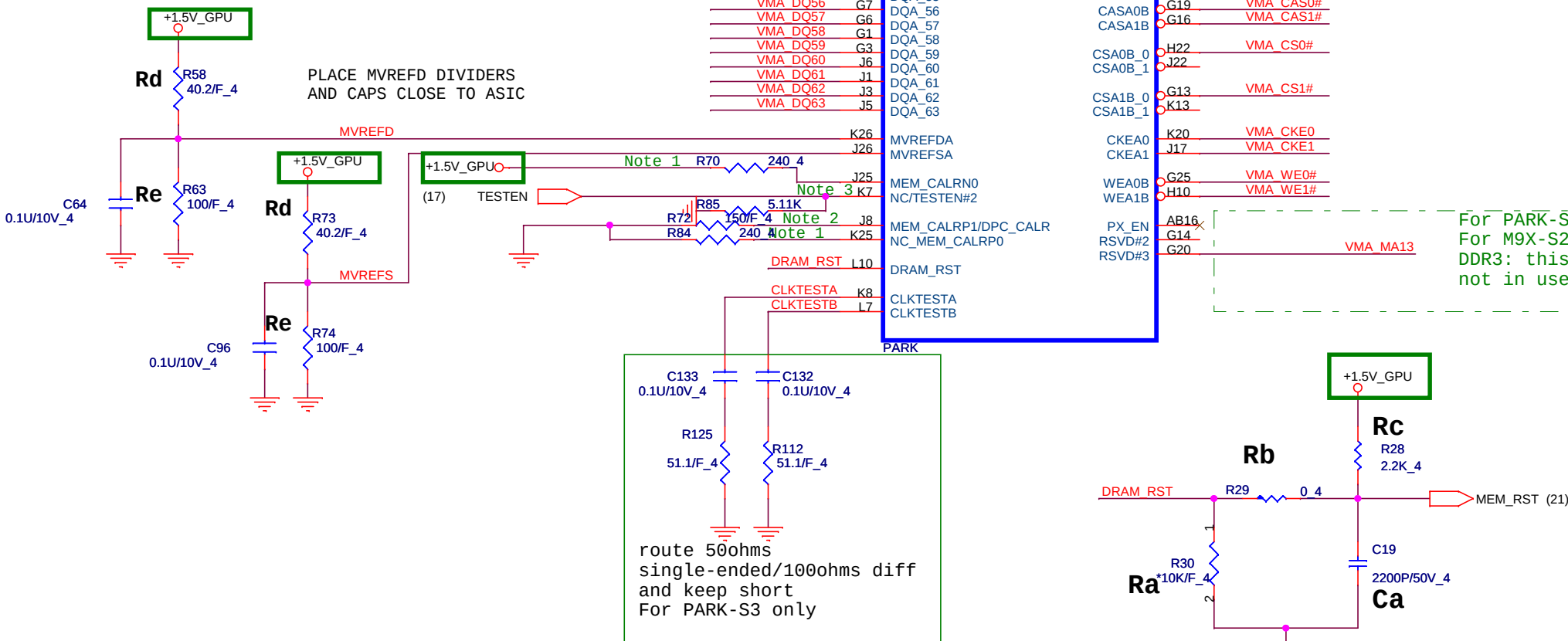
U30C

|            |      |           |
|------------|------|-----------|
| MAA_0      | K17  | VMA MA0   |
| MAA_1      | J20  | VMA MA1   |
| MAA_2      | H23  | VMA MA2   |
| MAA_3      | G23  | VMA MA3   |
| MAA_4      | G24  | VMA MA4   |
| MAA_5      | H24  | VMA MA5   |
| MAA_6      | J19  | VMA MA6   |
| MAA_7      | K19  | VMA MA7   |
| MAA_8      | J14  | VMA MA8   |
| MAA_9      | K14  | VMA MA9   |
| MAA_10     | J11  | VMA MA10  |
| MAA_11     | J13  | VMA MA11  |
| MAA_12     | H11  | VMA MA12  |
| MAA_13/BA2 | G11  | VMA BA2   |
| MAA_14/BA0 | J16  | VMA BA0   |
| MAA_15/BA1 | L15  | VMA BA1   |
|            |      |           |
| DQMA_0     | E32  | VMA DM0   |
| DQMA_1     | E30  | VMA DM1   |
| DQMA_2     | A21  | VMA DM2   |
| DQMA_3     | C21  | VMA DM3   |
| DQMA_4     | E13  | VMA DM4   |
| DQMA_5     | D12  | VMA DM5   |
| DQMA_6     | E3   | VMA DM6   |
| DQMA_7     | F4   | VMA DM7   |
|            |      |           |
| RDQSA_0    | H28  | VMA RDQS0 |
| RDQSA_1    | C27  | VMA RDQS1 |
| RDQSA_2    | A23  | VMA RDQS2 |
| RDQSA_3    | E19  | VMA RDQS3 |
| RDQSA_4    | E15  | VMA RDQS4 |
| RDQSA_5    | D10  | VMA RDQS5 |
| RDQSA_6    | D6   | VMA RDQS6 |
| RDQSA_7    | G5   | VMA RDQS7 |
|            |      |           |
| WDQSA_0    | H27  | VMA WDQS0 |
| WDQSA_1    | A27  | VMA WDQS1 |
| WDQSA_2    | C23  | VMA WDQS2 |
| WDQSA_3    | C19  | VMA WDQS3 |
| WDQSA_4    | C15  | VMA WDQS4 |
| WDQSA_5    | E9   | VMA WDQS5 |
| WDQSA_6    | C5   | VMA WDQS6 |
| WDQSA_7    | H4   | VMA WDQS7 |
|            |      |           |
| ODTA0      | L18  | VMA ODT0  |
| ODTA1      | K16  | VMA ODT1  |
|            |      |           |
| CLKA0      | H26  | VMA CLK0  |
| CLKA0B     | H25  | VMA CLK0# |
|            |      |           |
| CLKA1      | G9   | VMA CLK1  |
| CLKA1B     | H9   | VMA CLK1# |
|            |      |           |
| RASA0B     | G22  | VMA RAS0# |
| RASA1B     | G17  | VMA RAS1# |
|            |      |           |
| CASA0B     | G19  | VMA CAS0# |
| CASA1B     | G16  | VMA CAS1# |
|            |      |           |
| CSA0B_0    | H22  | VMA CS0#  |
| CSA0B_1    | J22  |           |
|            |      |           |
| CSA1B_0    | G13  | VMA CS1#  |
| CSA1B_1    | K13  |           |
|            |      |           |
| CKEA0      | K20  | VMA CKE0  |
| CKEA1      | J17  | VMA CKE1  |
|            |      |           |
| WEA0B      | G25  | VMA WE0#  |
| WEA1B      | H10  | VMA WE1#  |
|            |      |           |
| PX_EN      | AB16 |           |
| RSVD#2     | G14  |           |
| RSVD#3     | G20  | VMA MA13  |

|          |     |        |
|----------|-----|--------|
| VMA DQ0  | K27 | DQA_0  |
| VMA DQ1  | J29 | DQA_1  |
| VMA DQ2  | H30 | DQA_2  |
| VMA DQ3  | H32 | DQA_3  |
| VMA DQ4  | G29 | DQA_4  |
| VMA DQ5  | E28 | DQA_5  |
| VMA DQ6  | F32 | DQA_6  |
| VMA DQ7  | F30 | DQA_7  |
| VMA DQ8  | C30 | DQA_8  |
| VMA DQ9  | F27 | DQA_9  |
| VMA DQ10 | A28 | DQA_10 |
| VMA DQ11 | C28 | DQA_11 |
| VMA DQ12 | E27 | DQA_12 |
| VMA DQ13 | G26 | DQA_13 |
| VMA DQ14 | D26 | DQA_14 |
| VMA DQ15 | E25 | DQA_15 |
| VMA DQ16 | A25 | DQA_16 |
| VMA DQ17 | C25 | DQA_17 |
| VMA DQ18 | E25 | DQA_18 |
| VMA DQ19 | D24 | DQA_19 |
| VMA DQ20 | E23 | DQA_20 |
| VMA DQ21 | E23 | DQA_21 |
| VMA DQ22 | D22 | DQA_22 |
| VMA DQ23 | F21 | DQA_23 |
| VMA DQ24 | E21 | DQA_24 |
| VMA DQ25 | D20 | DQA_25 |
| VMA DQ26 | F19 | DQA_26 |
| VMA DQ27 | A19 | DQA_27 |
| VMA DQ28 | D18 | DQA_28 |
| VMA DQ29 | F17 | DQA_29 |
| VMA DQ30 | A17 | DQA_30 |
| VMA DQ31 | C17 | DQA_31 |
| VMA DQ32 | E17 | DQA_32 |
| VMA DQ33 | D16 | DQA_33 |
| VMA DQ34 | F15 | DQA_34 |
| VMA DQ35 | A15 | DQA_35 |
| VMA DQ36 | D14 | DQA_36 |
| VMA DQ37 | F13 | DQA_37 |
| VMA DQ38 | A13 | DQA_38 |
| VMA DQ39 | C13 | DQA_39 |
| VMA DQ40 | F11 | DQA_40 |
| VMA DQ41 | A11 | DQA_41 |
| VMA DQ42 | C11 | DQA_42 |
| VMA DQ43 | F11 | DQA_43 |
| VMA DQ44 | A9  | DQA_44 |
| VMA DQ45 | C9  | DQA_45 |
| VMA DQ46 | F9  | DQA_46 |
| VMA DQ47 | D8  | DQA_47 |
| VMA DQ48 | E7  | DQA_48 |
| VMA DQ49 | A7  | DQA_49 |
| VMA DQ50 | C7  | DQA_50 |
| VMA DQ51 | F7  | DQA_51 |
| VMA DQ52 | A5  | DQA_52 |
| VMA DQ53 | E5  | DQA_53 |
| VMA DQ54 | C3  | DQA_54 |
| VMA DQ55 | F1  | DQA_55 |
| VMA DQ56 | G7  | DQA_56 |
| VMA DQ57 | G6  | DQA_57 |
| VMA DQ58 | G1  | DQA_58 |
| VMA DQ59 | G3  | DQA_59 |
| VMA DQ60 | J6  | DQA_60 |
| VMA DQ61 | J1  | DQA_61 |
| VMA DQ62 | J3  | DQA_62 |
| VMA DQ63 | J5  | DQA_63 |

support 16bit  
VRAM ( 64M X 16 )

| DIVIDER RESISTORS  | ROBSON |
|--------------------|--------|
| MVREF TO 1.8V (Rd) | 40.2R  |
| MVREF TO GND (Re)  | 100R   |



Ra Rb Rc Rd

Place all these components very close to GPU (Within 25mm) and keep all component close to each Other (within 5mm) except Rser2

This basic topology should be used for DRAM\_RST for DDR3/GDDR5. These Capacitors and Resistor values are an example only. The Series R and || Cap values will depend on the DRAM load and will have to be calculated for different Memory ,DRAM Load and board to pass Reset Signal Spec.

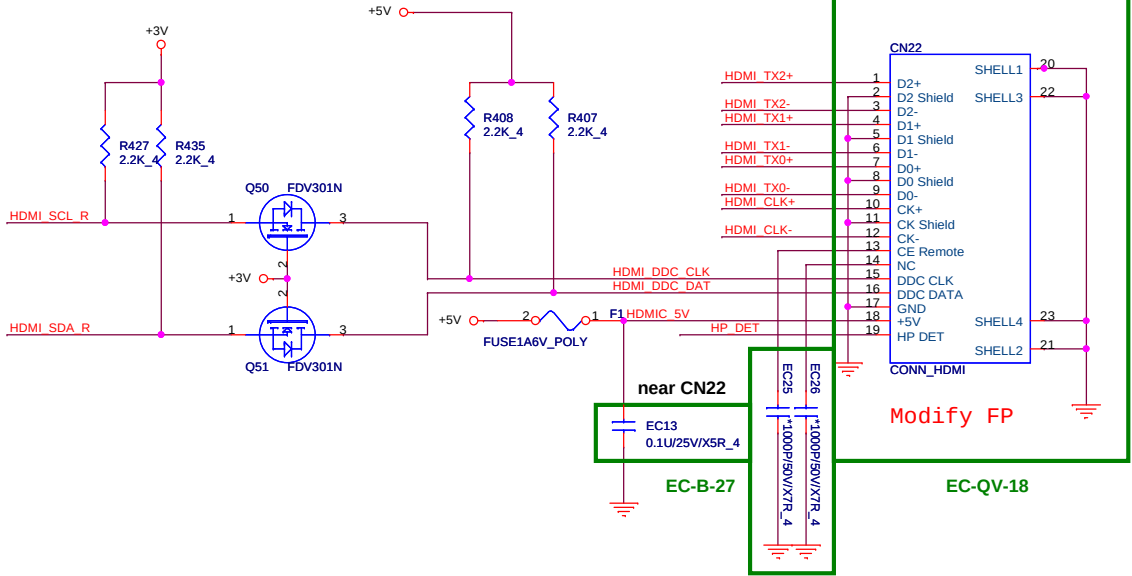
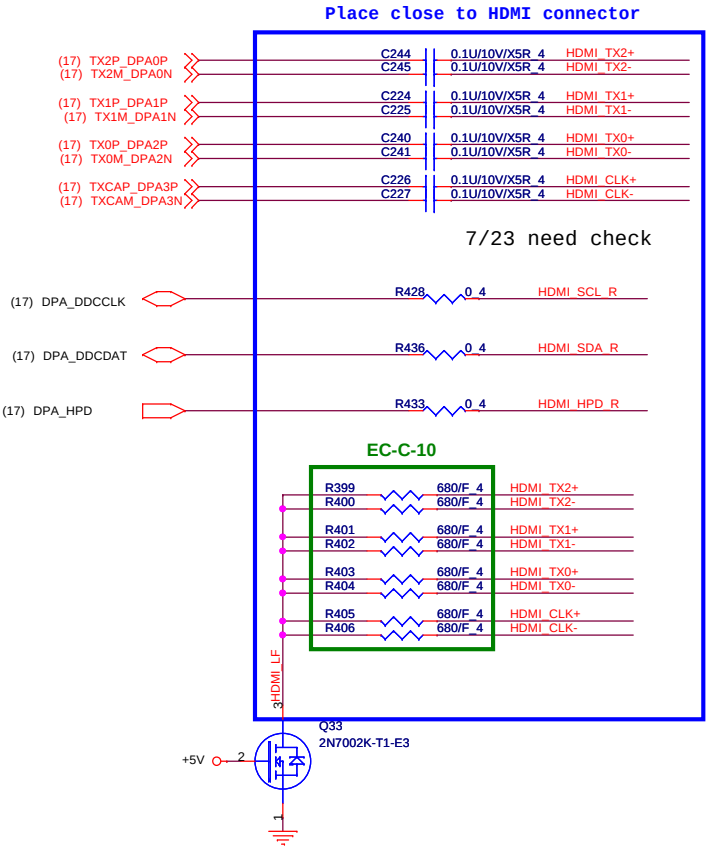


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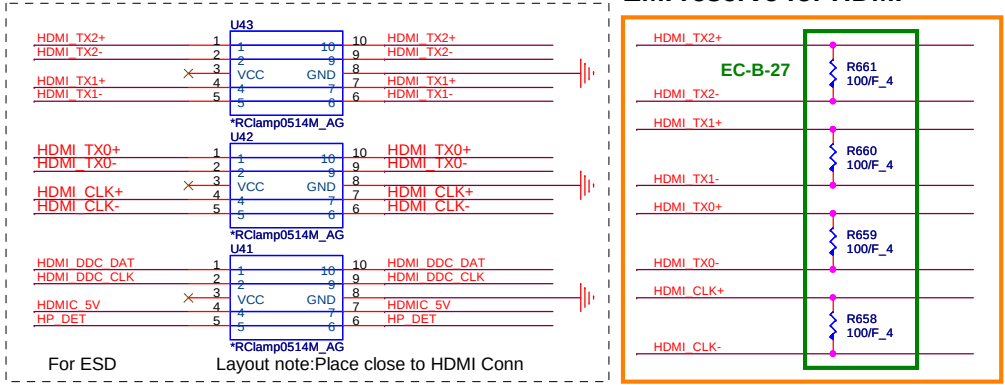
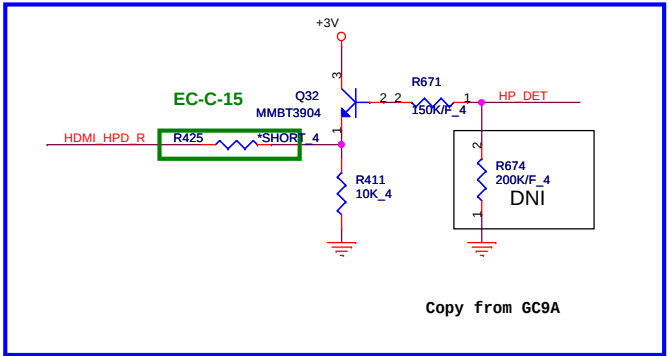
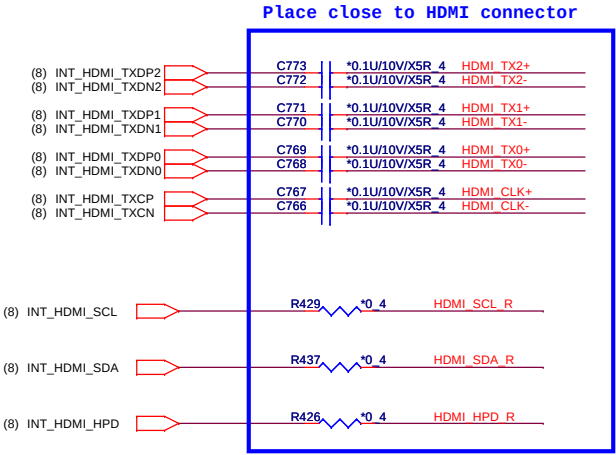
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|-------|---------------------------|----------------|
|       | Robson_MEM_Interface      | 1A             |
| Date: | Tuesday, January 04, 2011 | Sheet 20 of 53 |

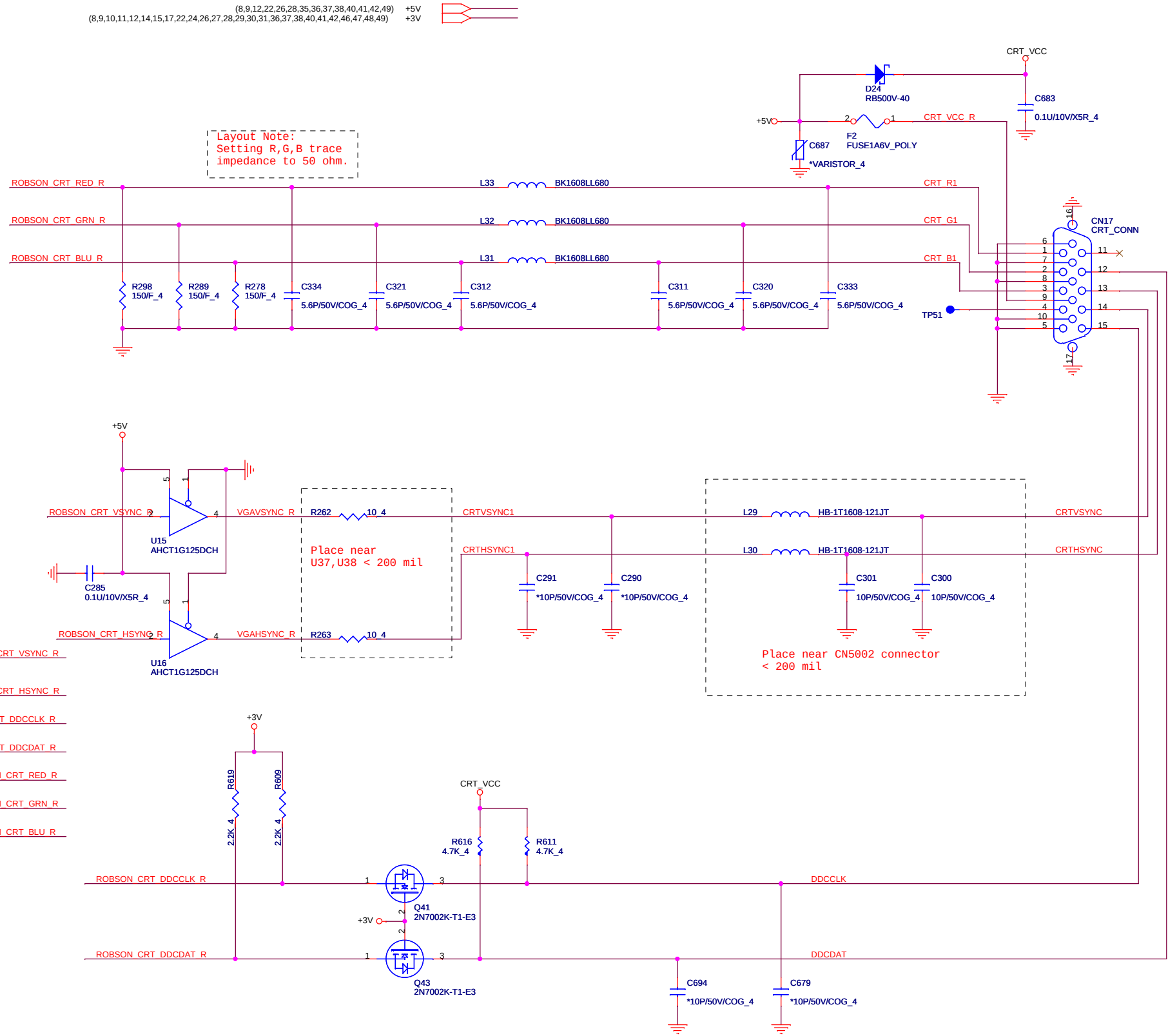
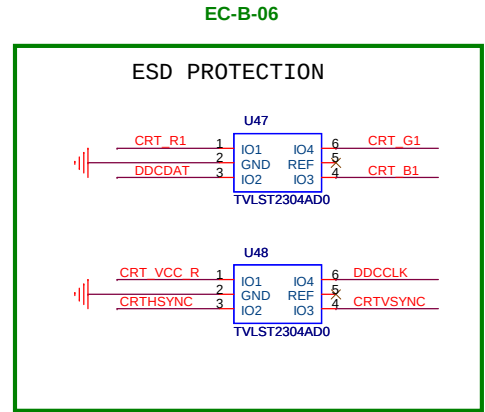




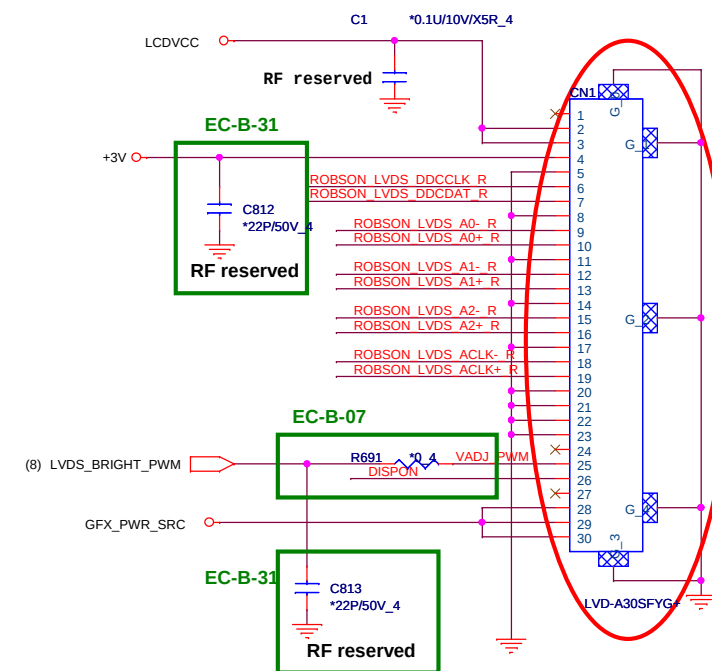
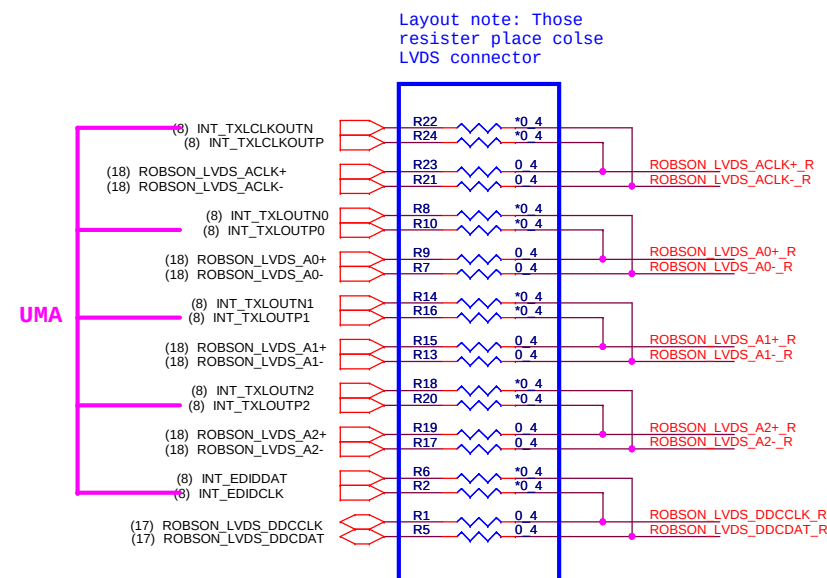
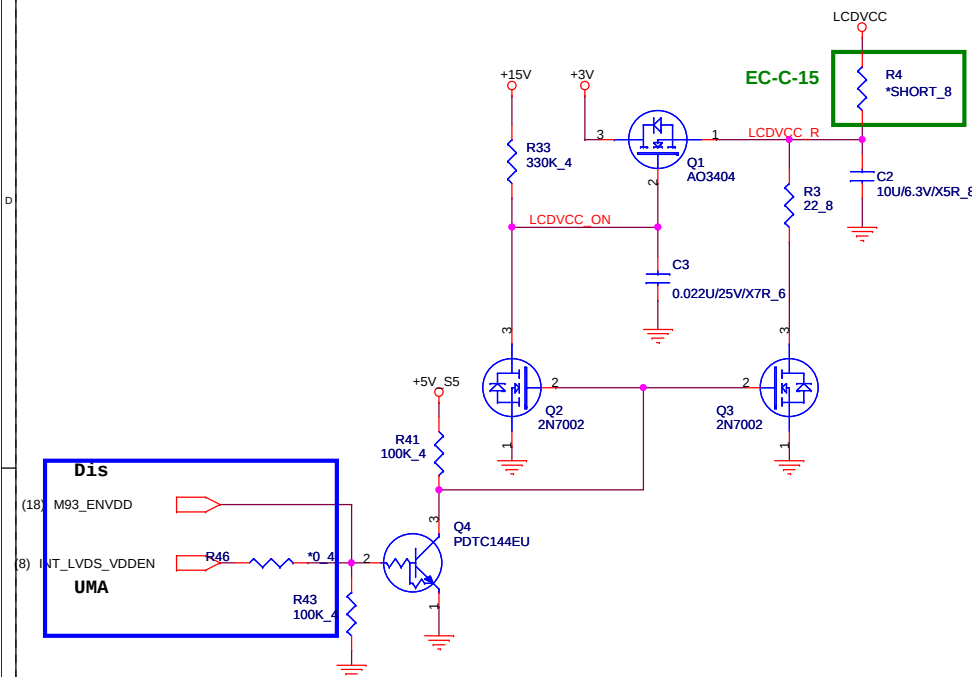


UMA Only

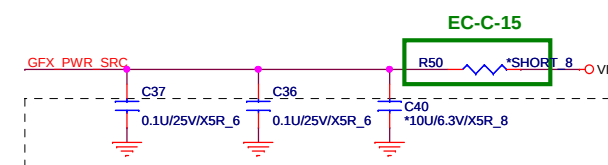
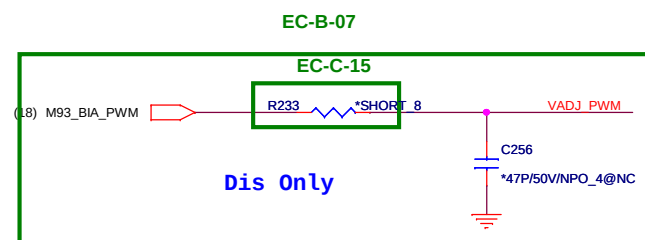
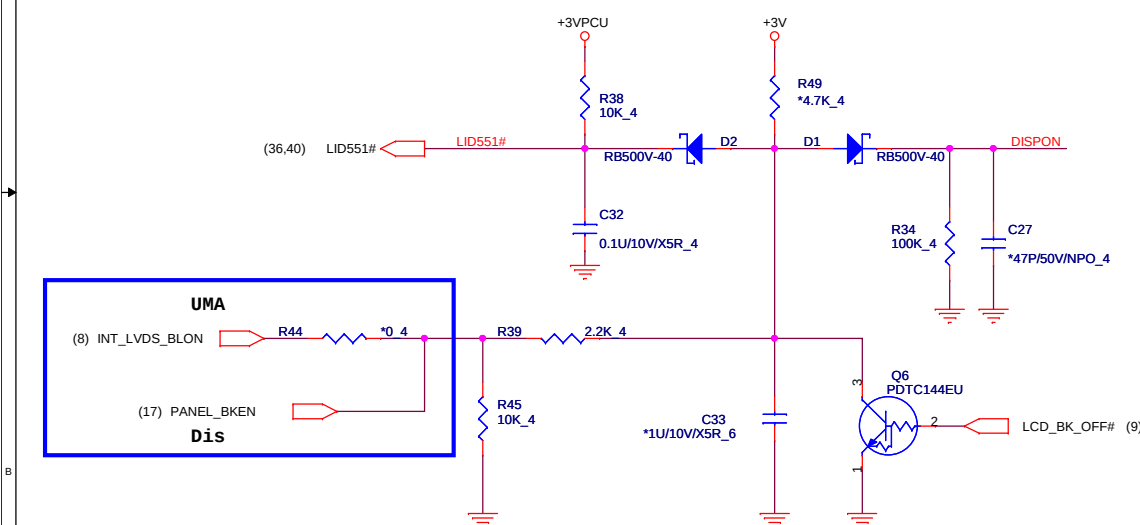


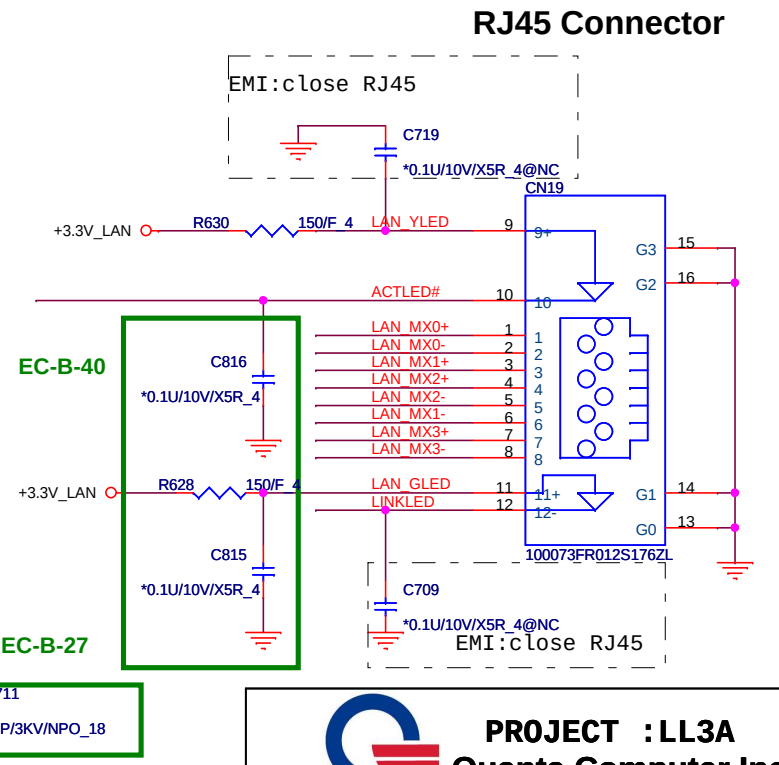
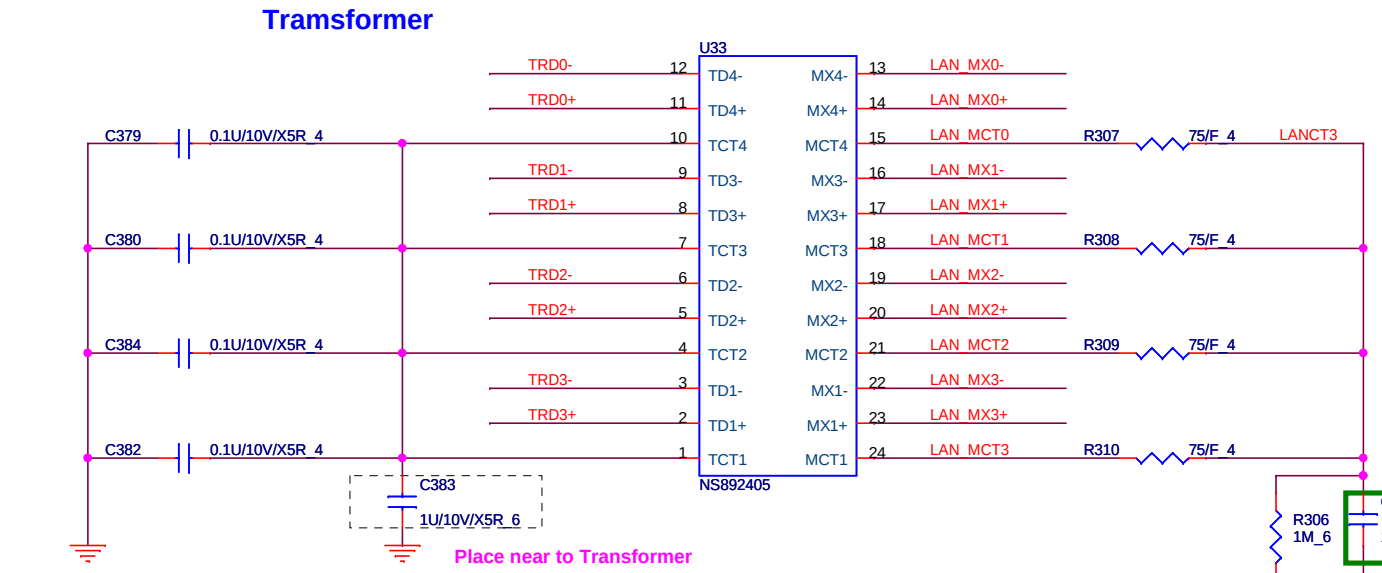
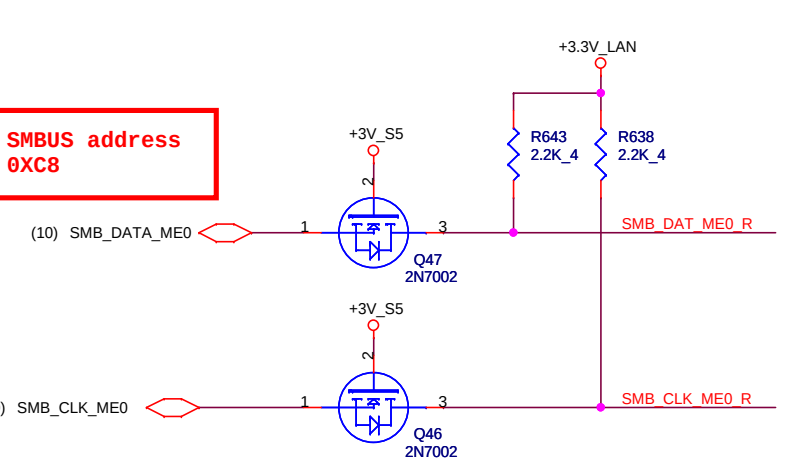
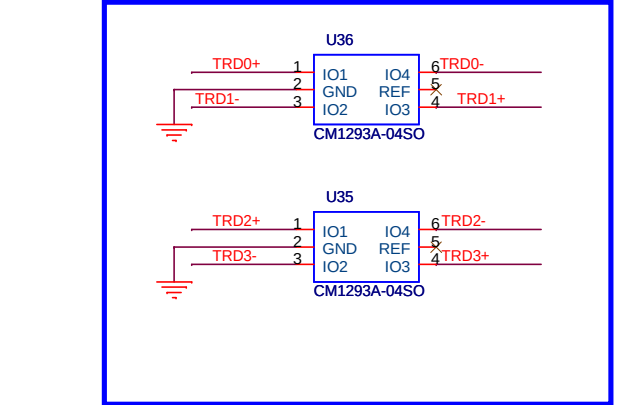
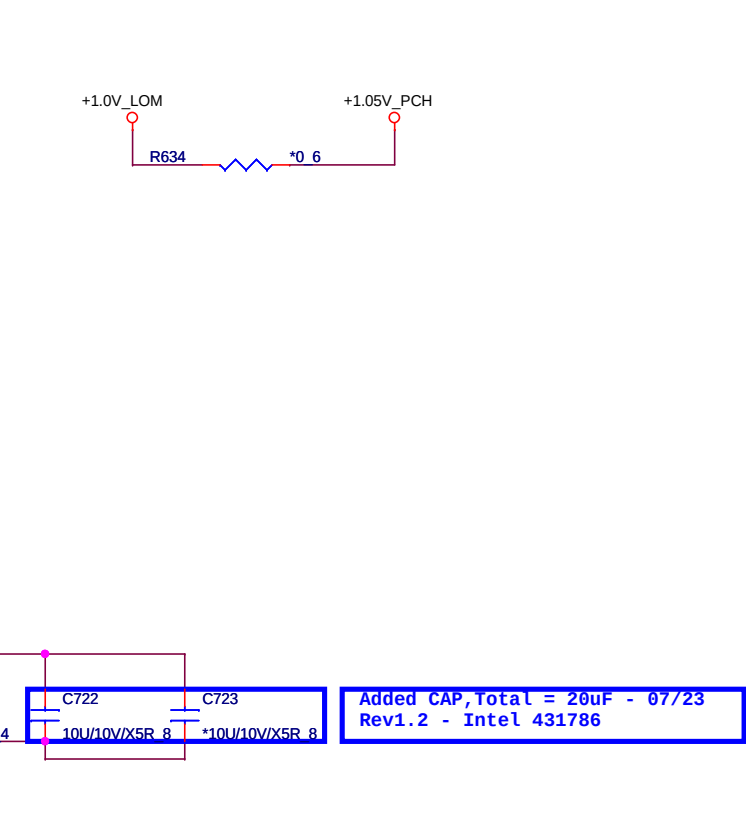
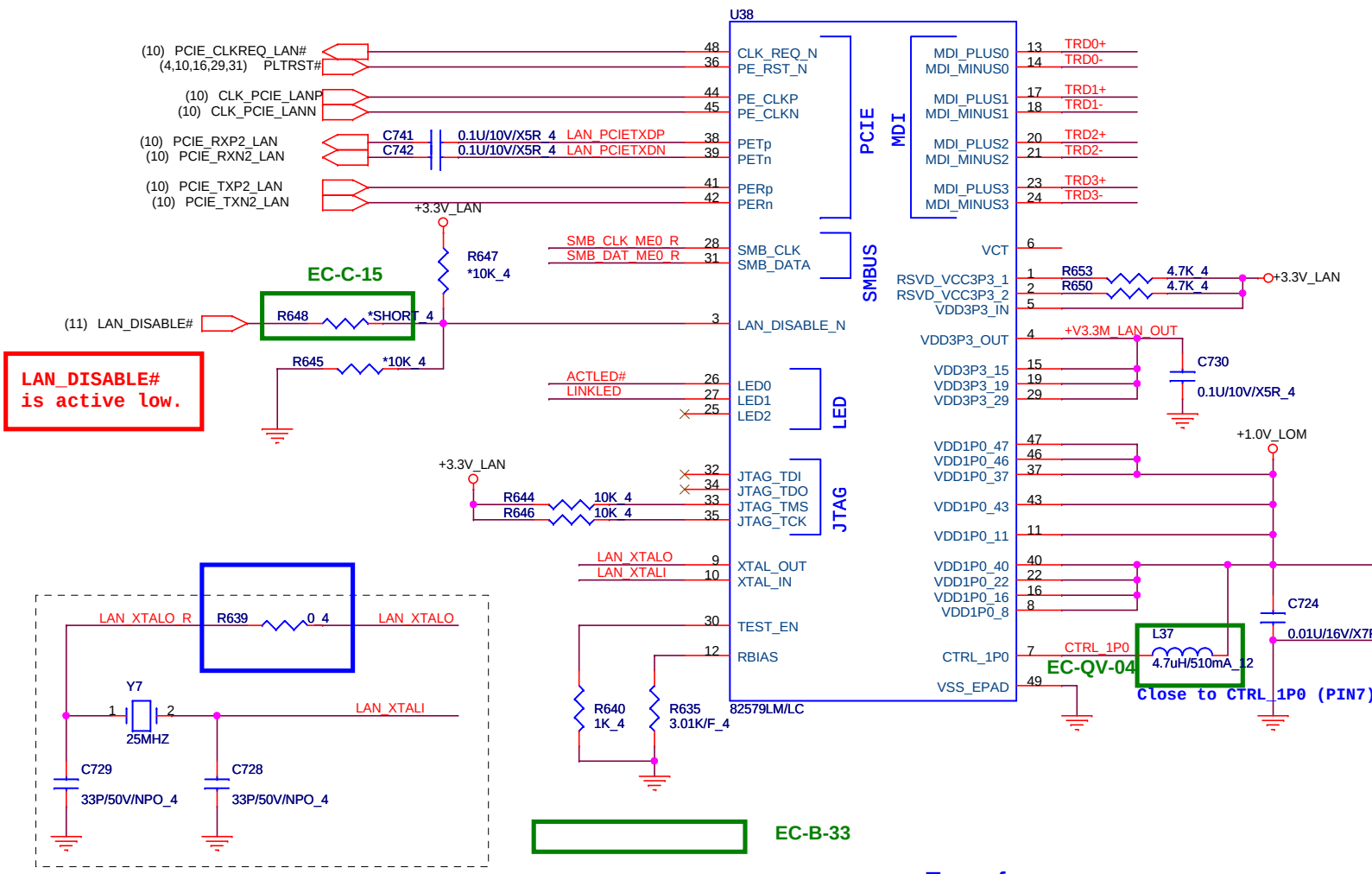
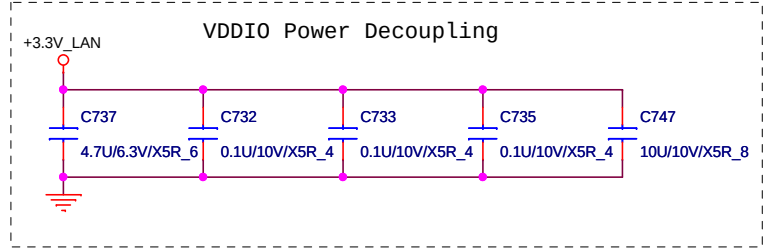
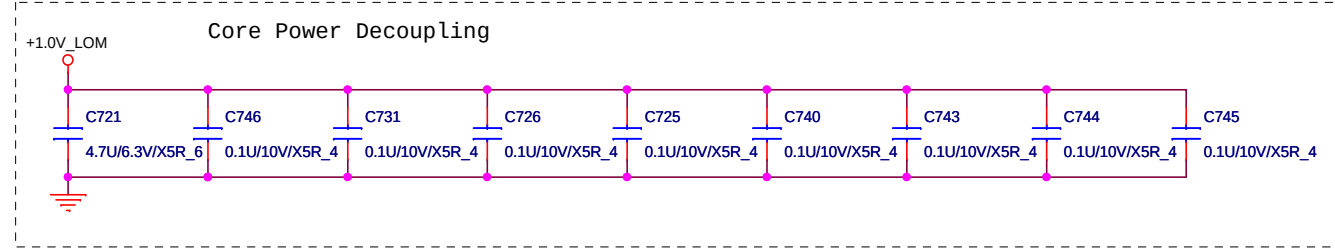
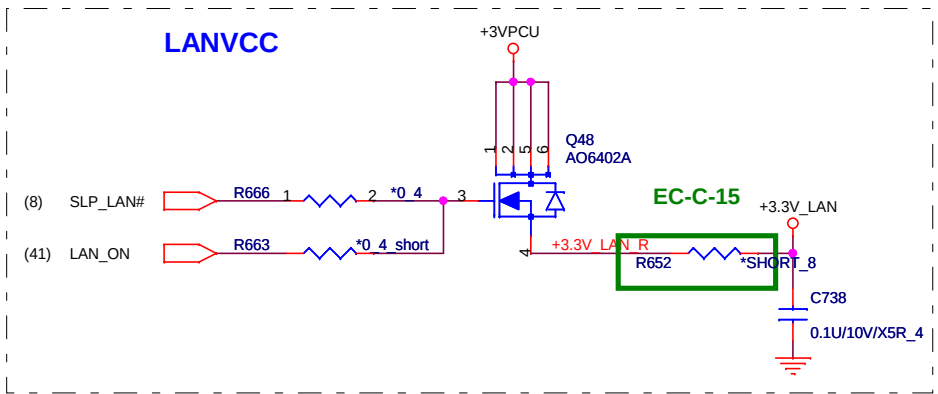


**LCDVCC**



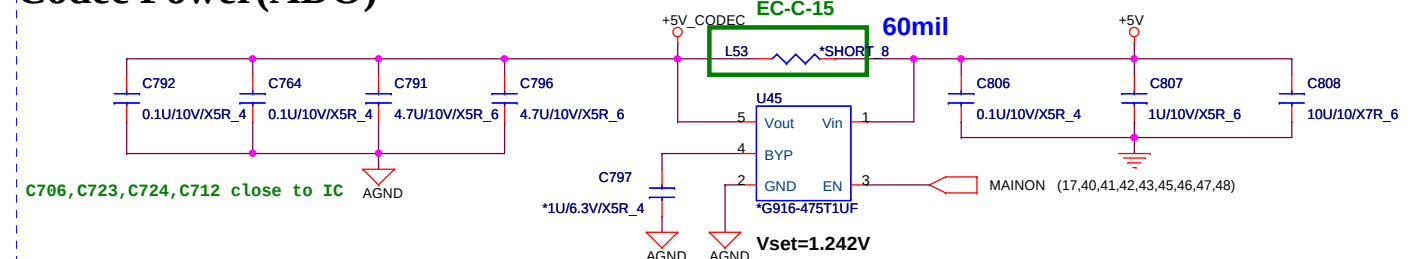
## Back light



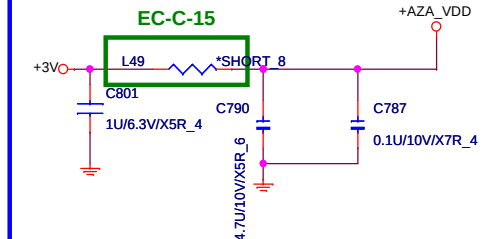
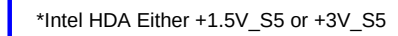




## Close to CODEC



## HDA Power(ADO)

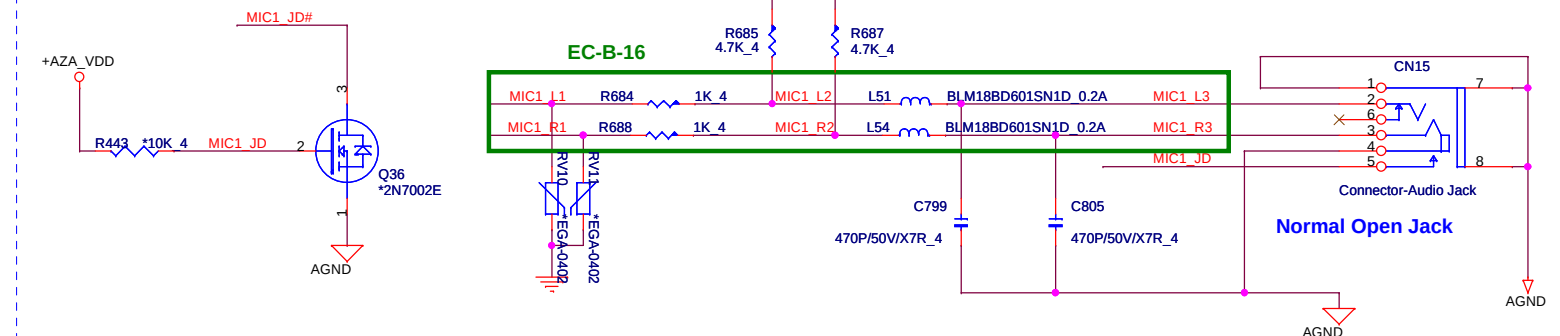


**(AMP)** **EC-B-08**

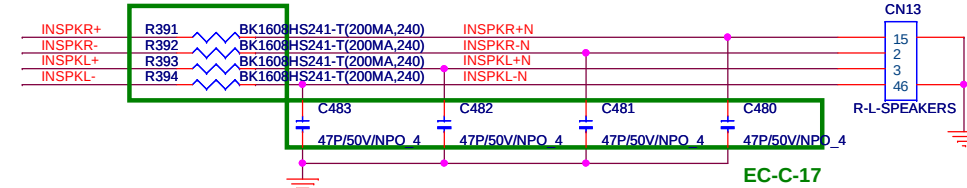
NORMAL OPEN : mount R47

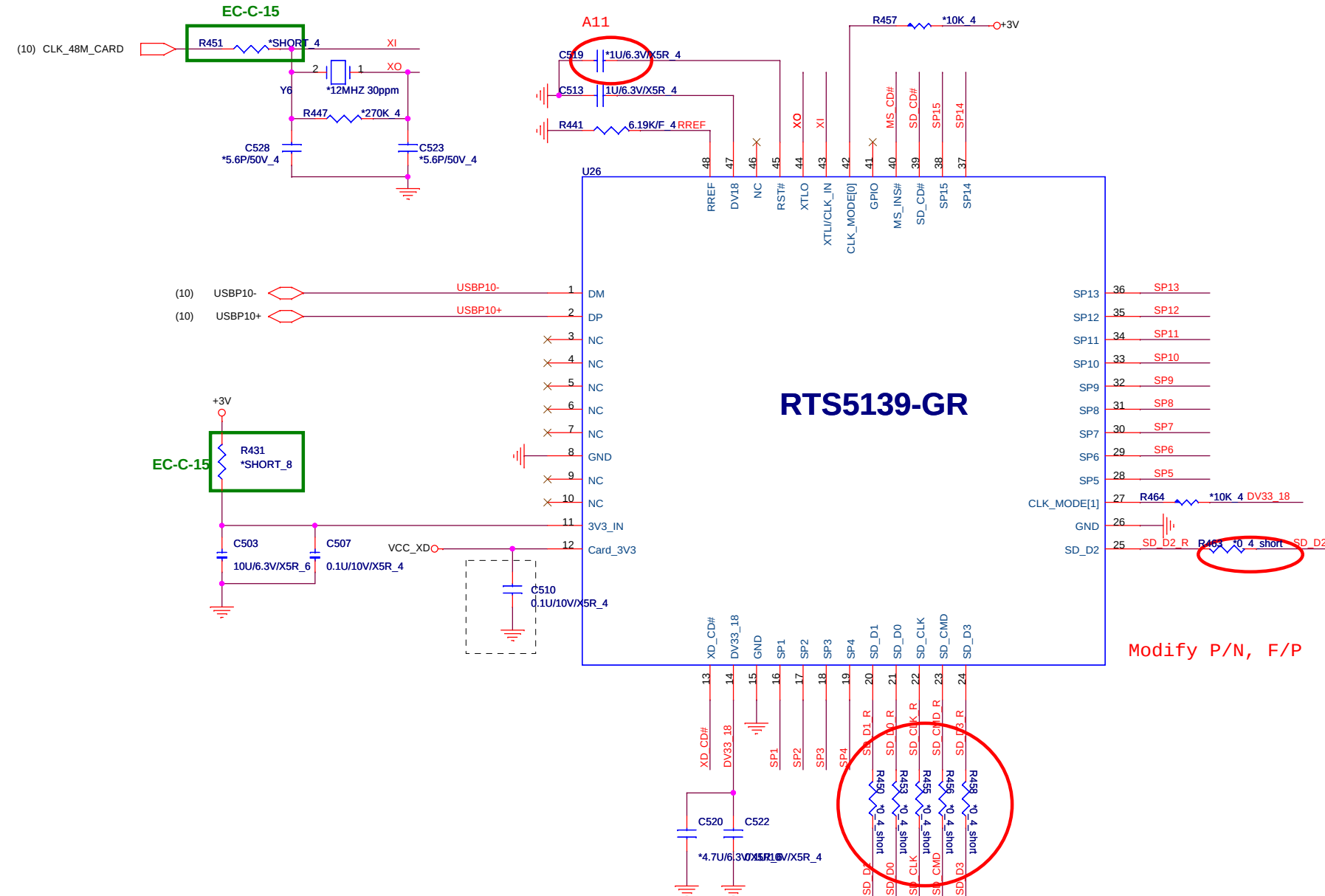
MIC1\_JD# R47 0 4 MIC1\_JD

Normal Close : mount R443,Q36



EC-QV-15



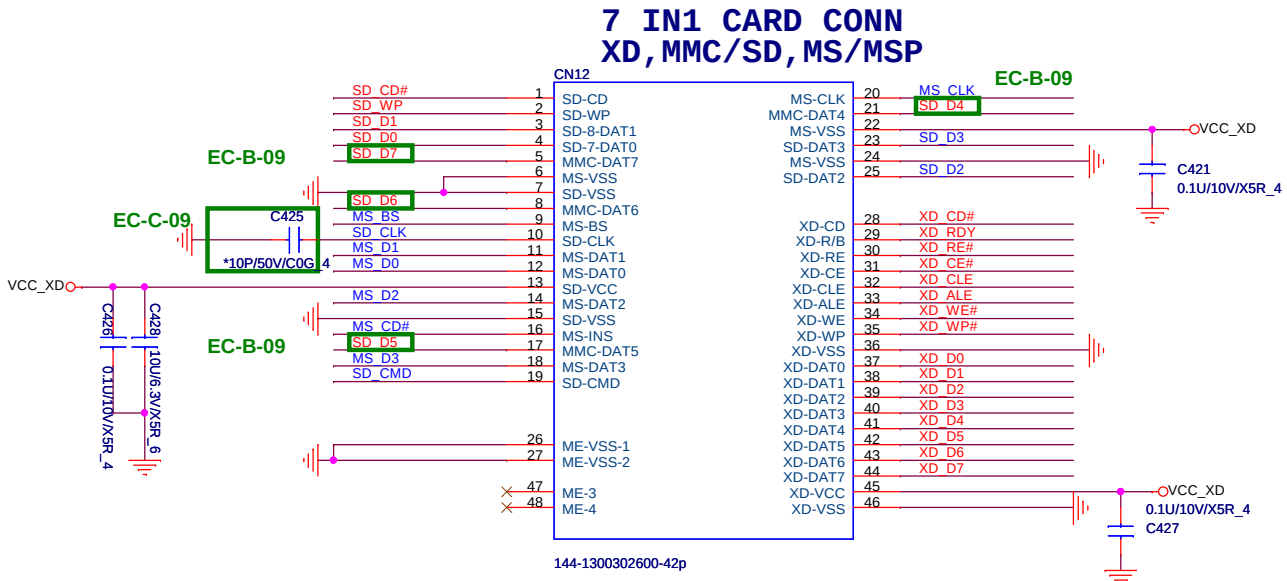


| Clock Mode strap | R9287 | R9307 |
|------------------|-------|-------|
| 48MHz            | X     | X     |
| 24MHz            | X     | O     |
| 12MHz            | O     | X     |
| 12MHz (Crystal)  | O     | O     |

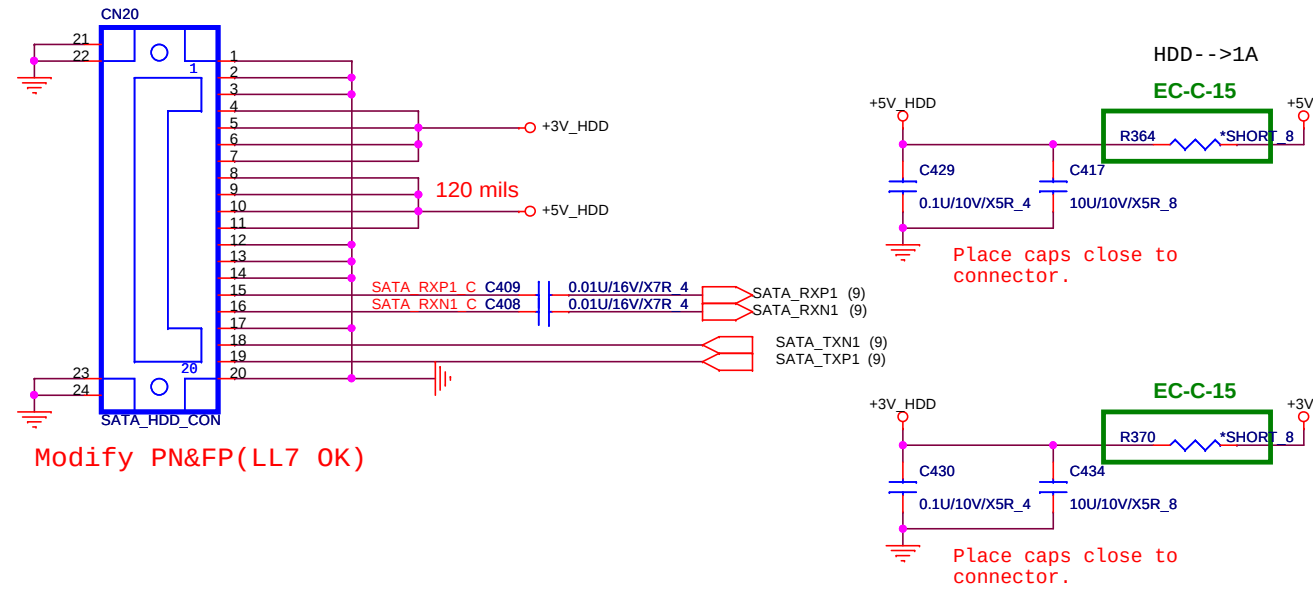
Note:

| SD/MMC    | MS    | XD     |
|-----------|-------|--------|
| SP1 SD D7 |       | XD RDY |
| SP2 SD D6 |       | XD RE# |
| SP3 SD D5 |       | XD CE# |
| SP4 SD D4 |       | XD WE# |
| SP5       | MS BS | XD CLE |
| SP6       |       | XD ALE |
| SP7       | MS D1 | XD WP# |
| SP8       |       | XD D0  |
| SP9       | MS D0 | XD D1  |
| SP10      | MS D2 | XD D2  |
| SP11      |       | XD D3  |
| SP12      | MS D3 | XD D4  |
| SP13      |       | XD D5  |
| SP14      | MS CL | XD D6  |
| SP15      | SD WP | XD D7  |

For RTS5139  
SD,MS 4bit only

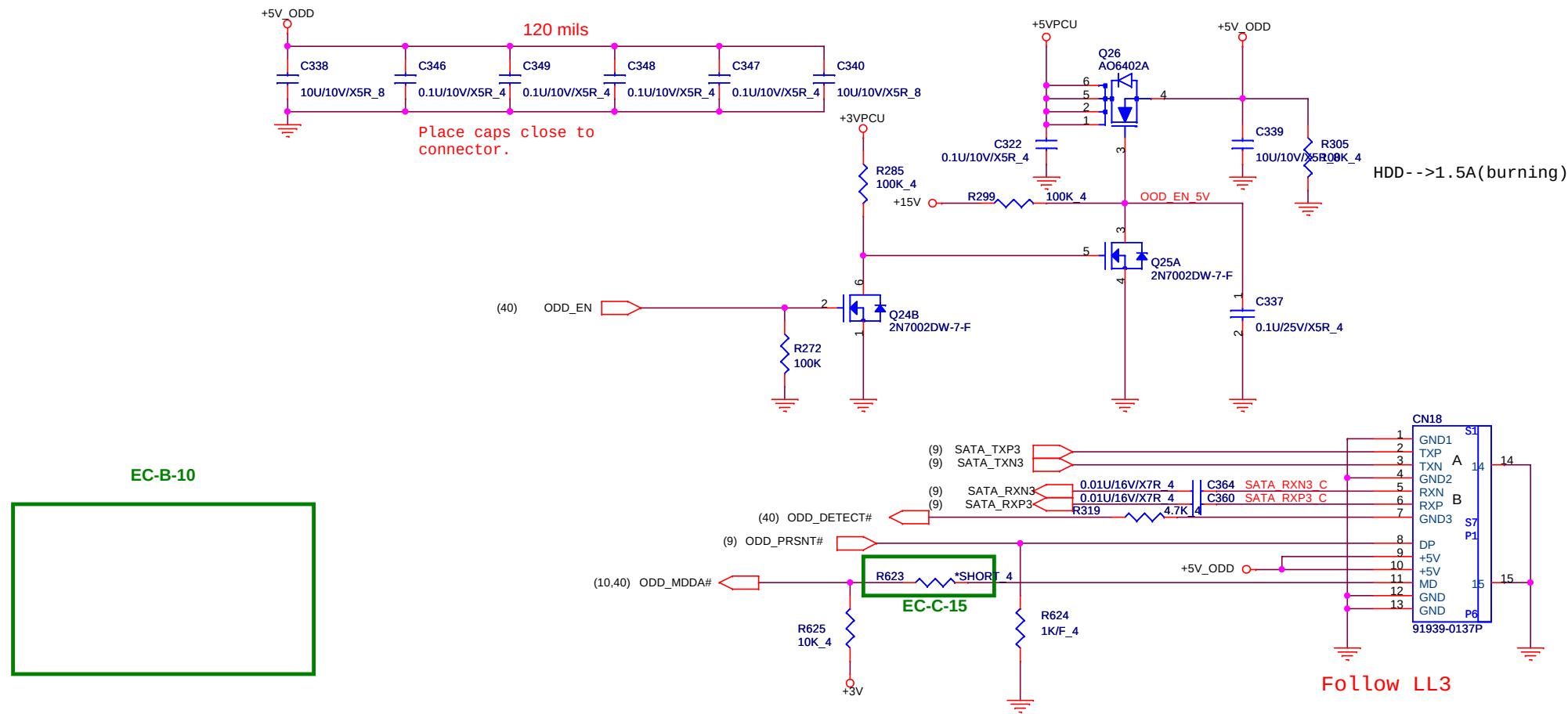


SATA HDD Connector.

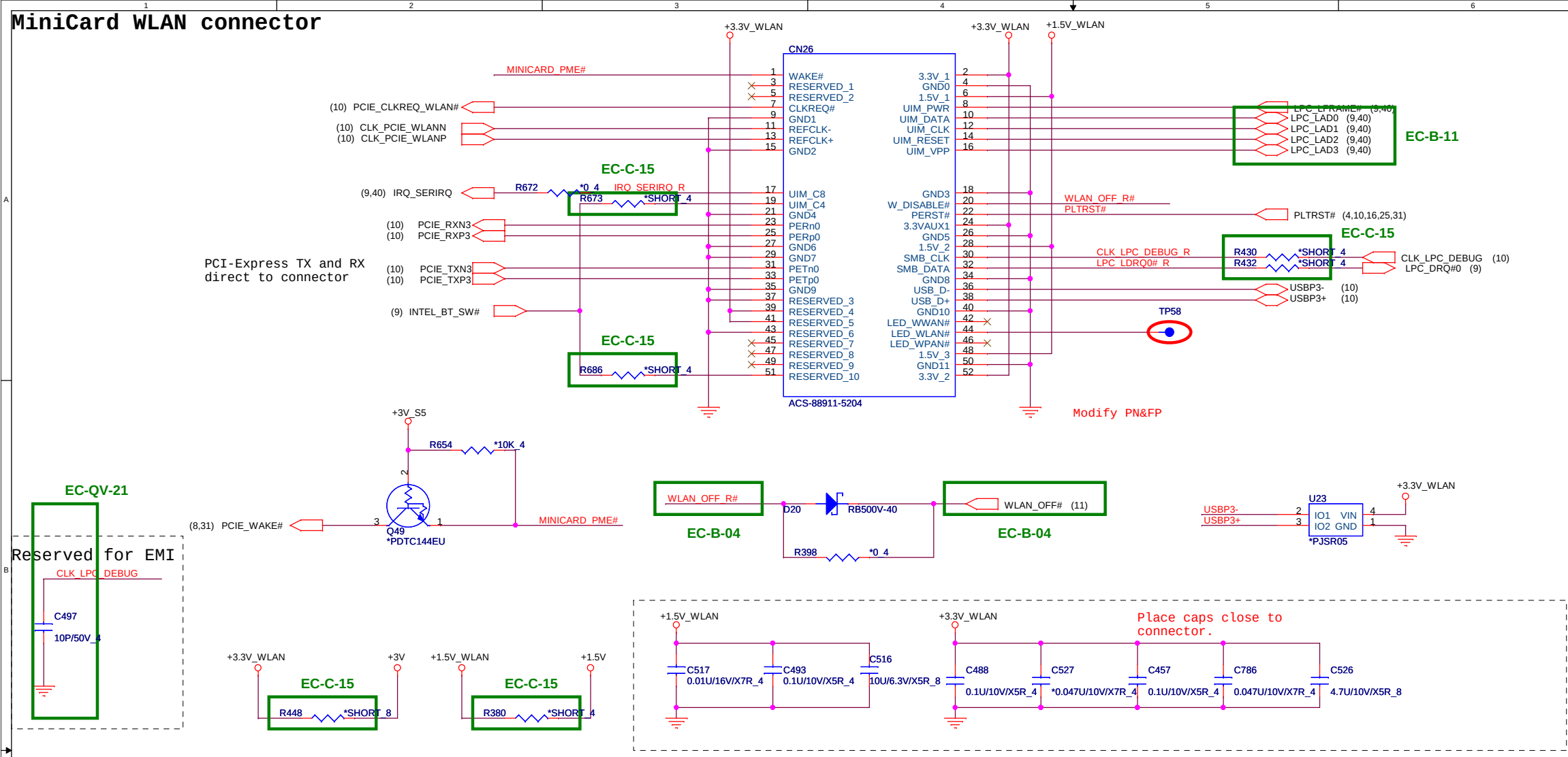


| EN | B0 | B1 | FUNCTION             |
|----|----|----|----------------------|
| 0  | X  | X  | Standby              |
| 1  | 0  | 0  | Standard SATA Output |
| 1  | 1  | 0  | Ch 0 Boost Output    |
| 1  | 0  | 1  | Ch 1 Boost Output    |
| 1  | 1  | 1  | Ch 0,1 Boost Output  |

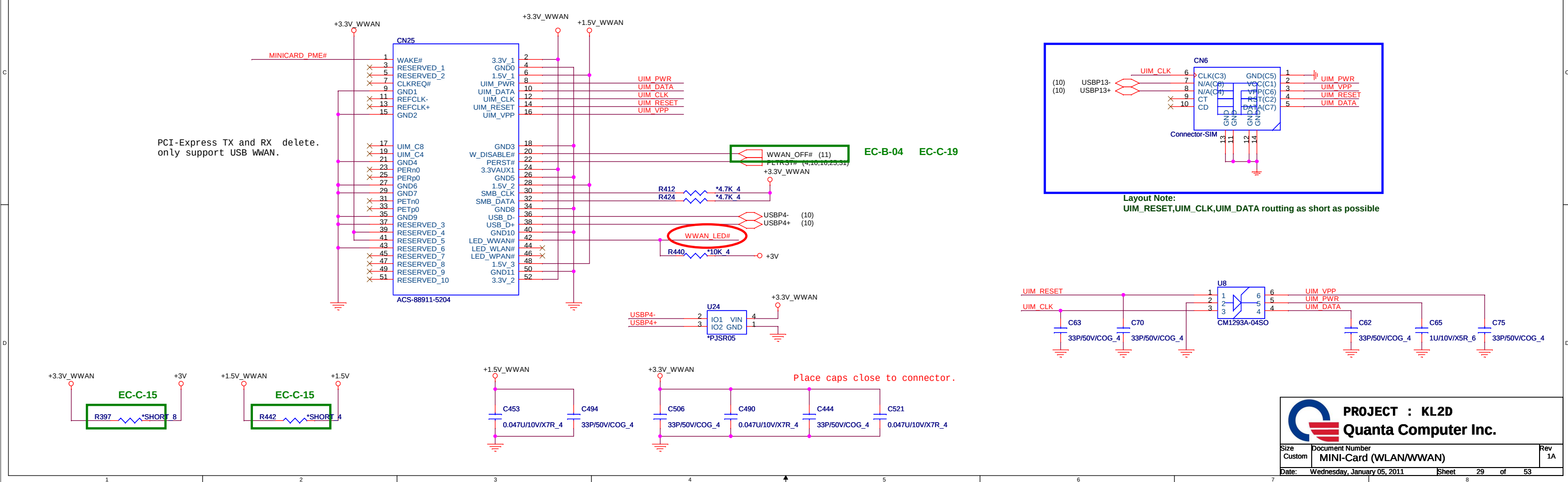
SATA ODD Connector.



MiniCard WLAN connector



MiniCard WWAN (Reserved for KL9)

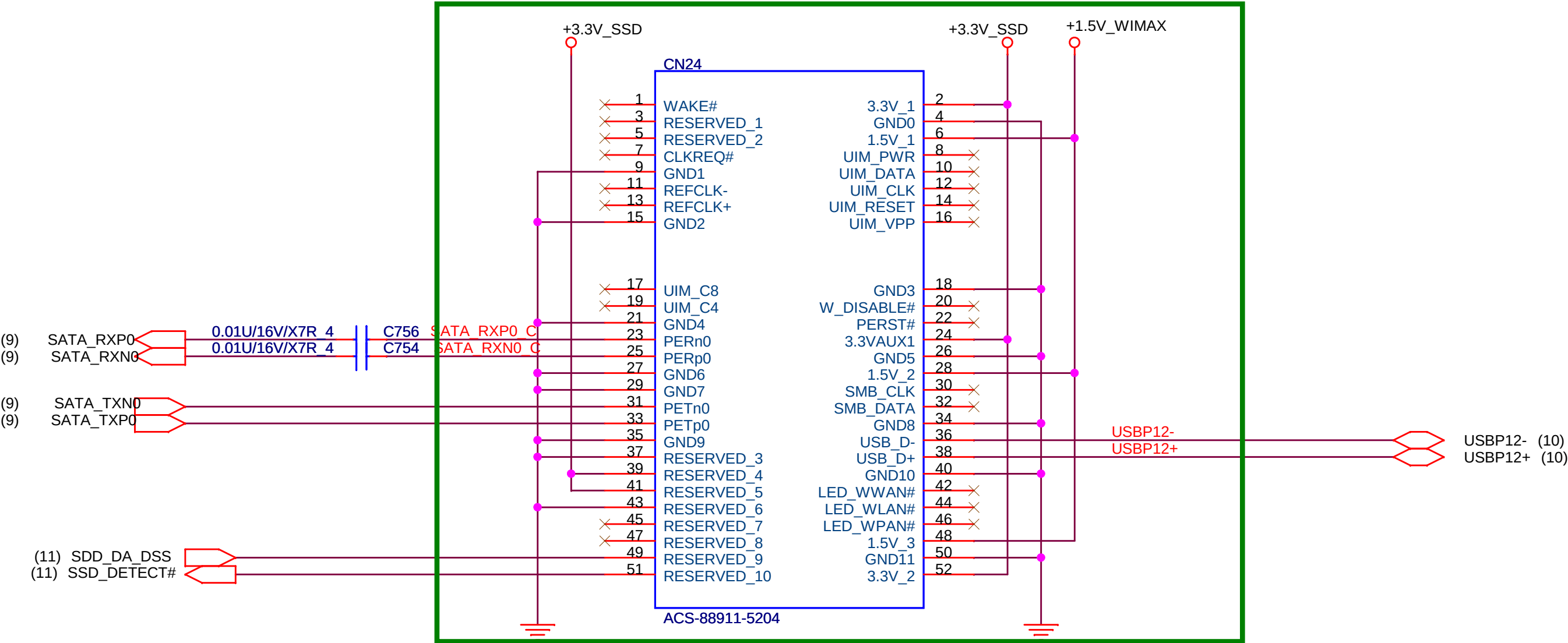




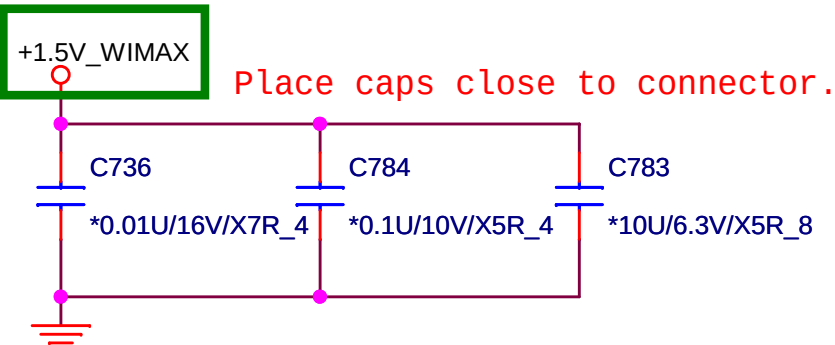
Mini PCI-E Card 3 SSD

EC-B-13

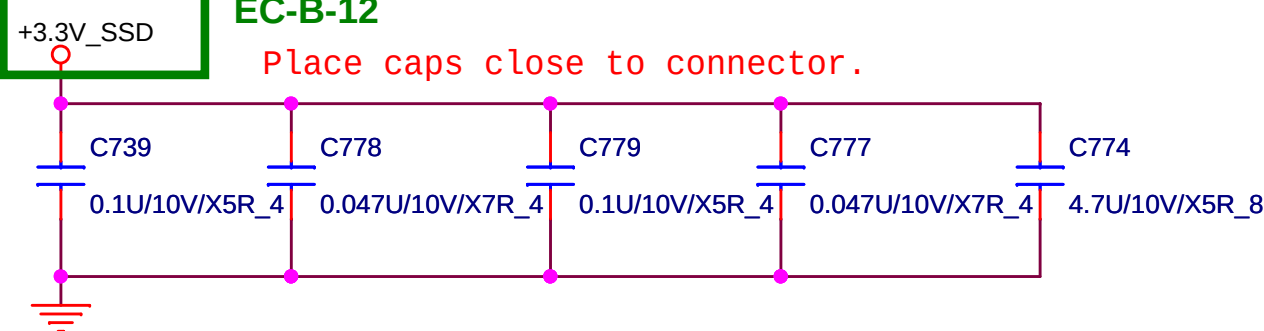
32



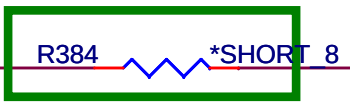
EC-B-12



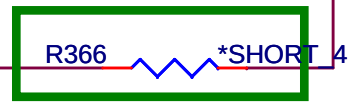
EC-B-12



EC-C-15

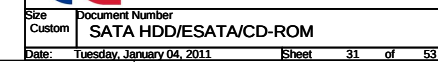


EC-C-15

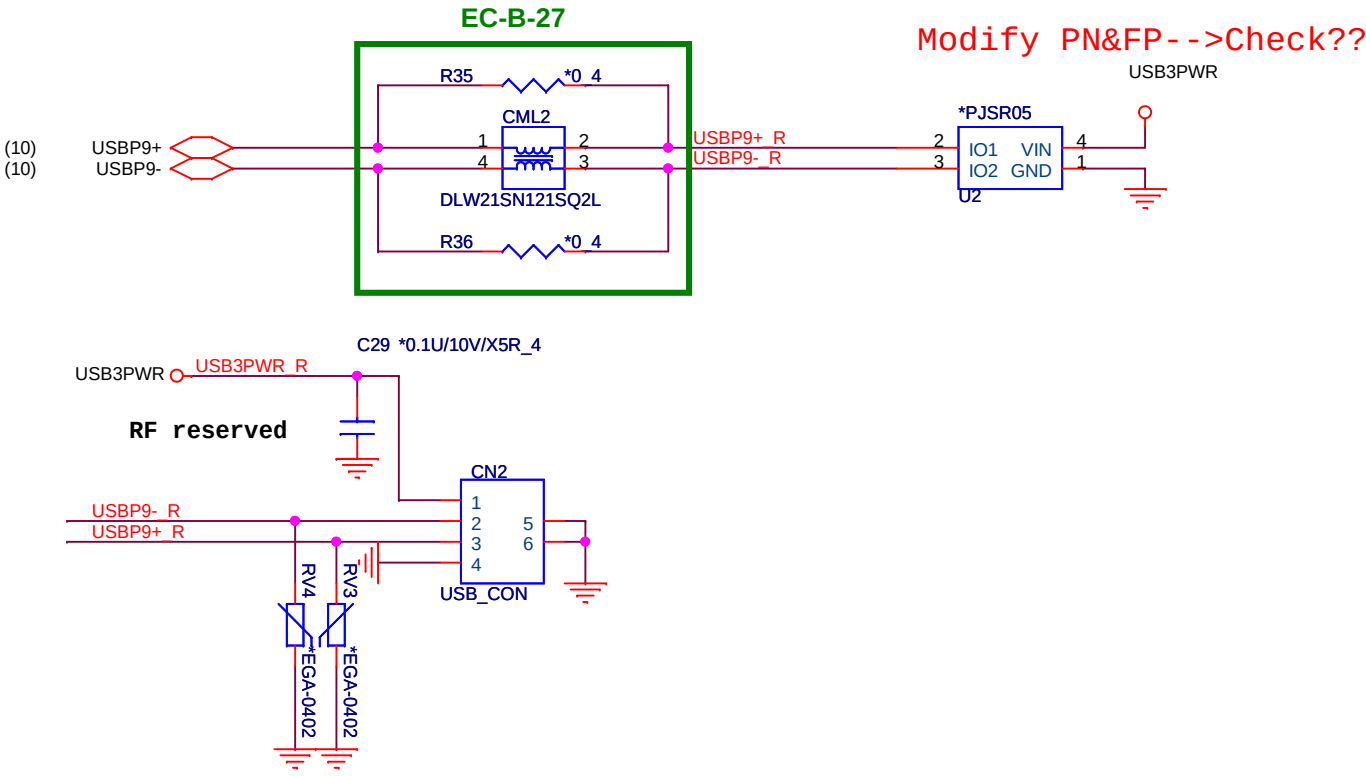
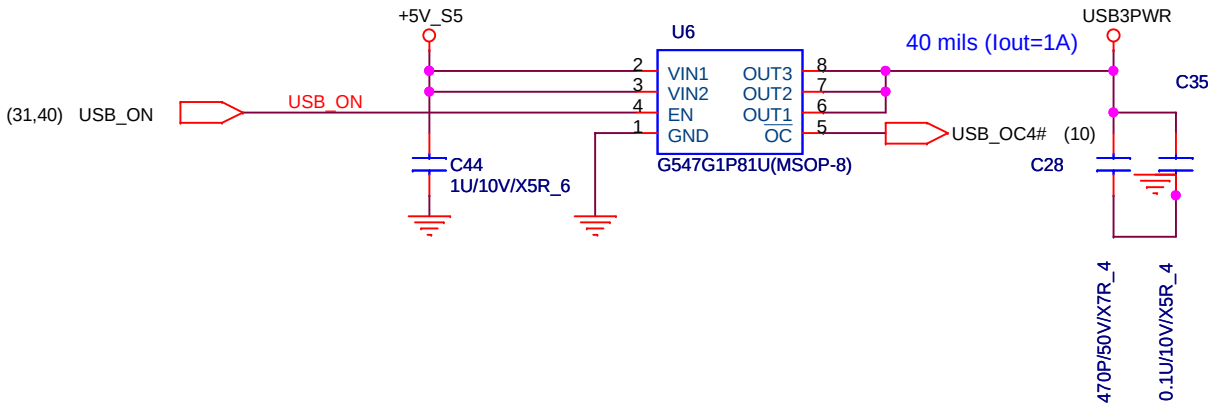


PROJECT : LL3A  
Quanta Computer Inc.

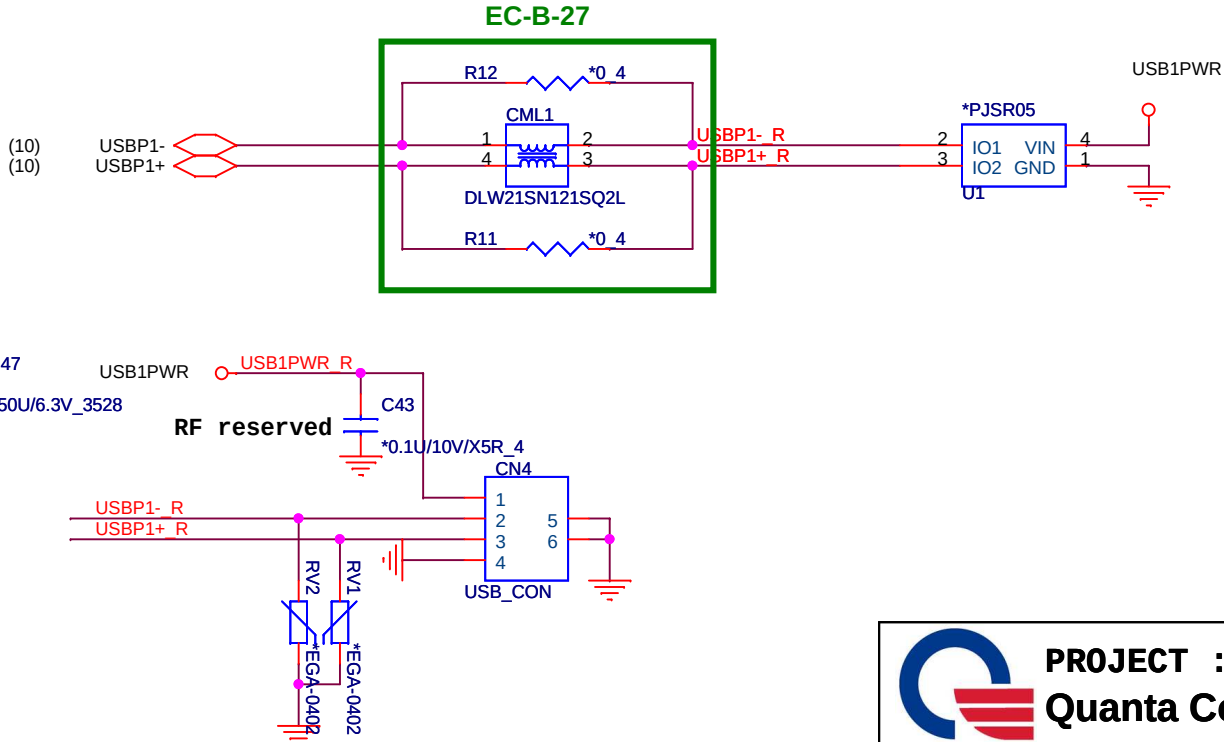
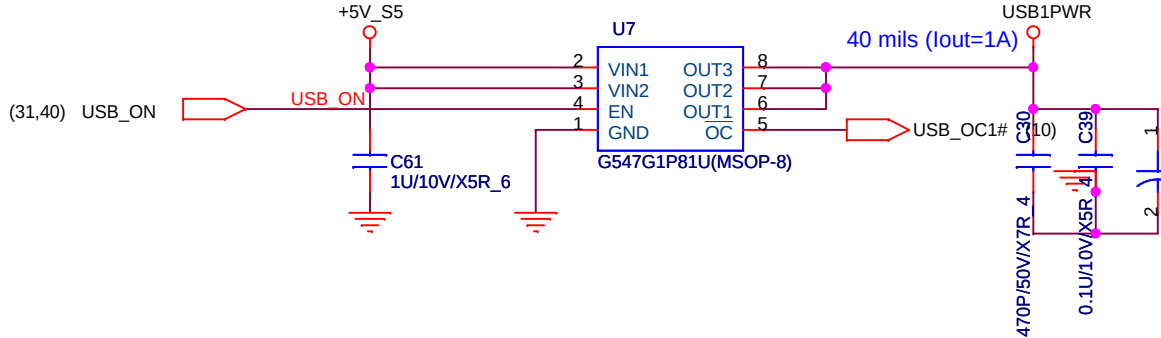
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|------------------------------------|------------------------------------|-----------|
| Size<br>Custom                     | Document Number<br>MINI Card (SSD) | Rev<br>1E |
| Date:<br>Tuesday, January 04, 2011 | Sheet<br>30 of 53                  |           |



USB2.0\*2  
External (R side) USB\*1

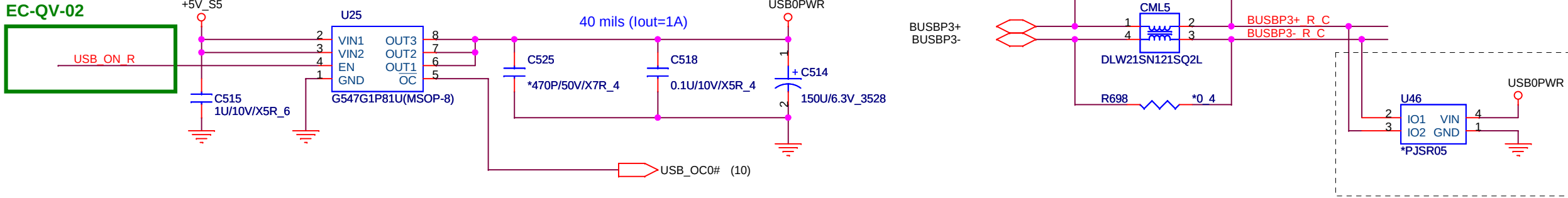


External (L side) USB\*1

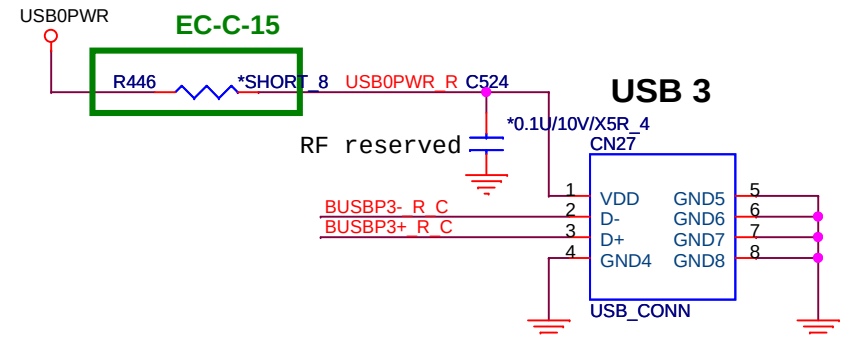
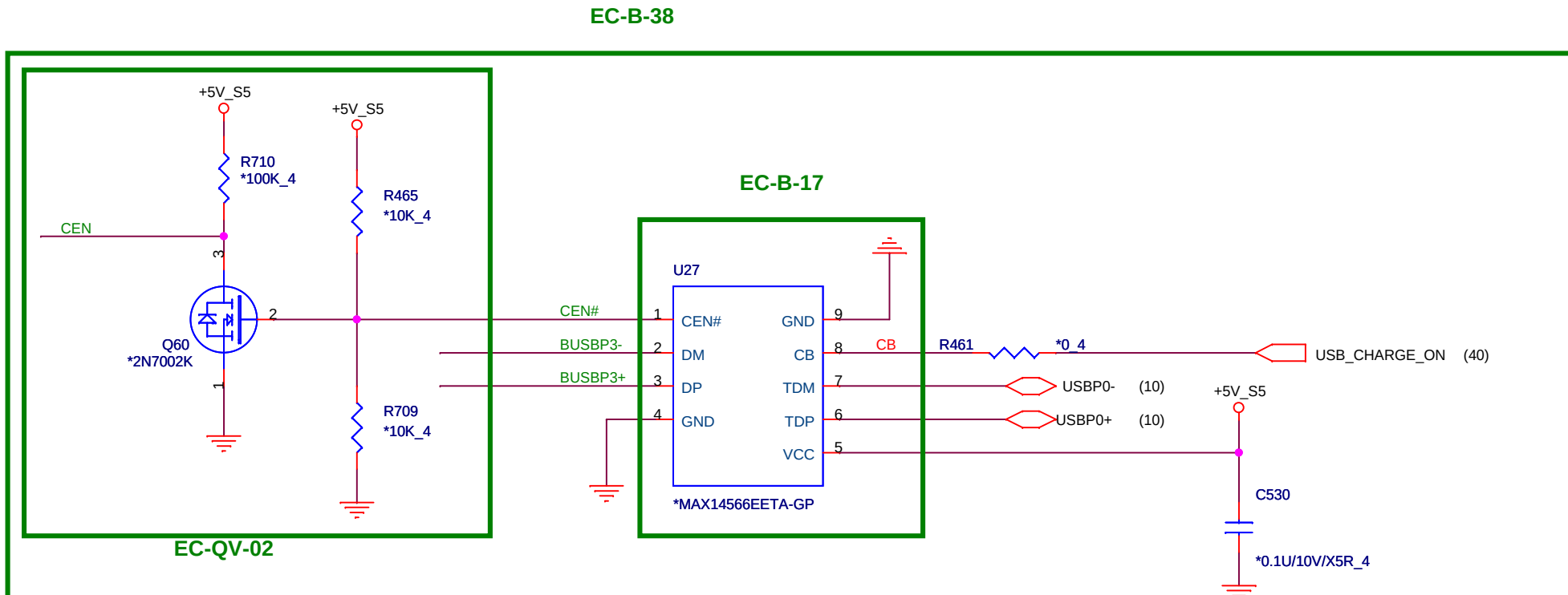


Modify FP&PN

USB



USB charger



EC-QV-02

CEN R708 \*0.4 USB\_ON R

CEN# R636 \*0.4

(40) USB\_ON R522 0.4

| USB Charge Function |      |      |      |      |         |
|---------------------|------|------|------|------|---------|
|                     | R689 | R690 | R708 | R522 | State   |
| Enable              | X    | X    | V    | X    |         |
| Disable             | V    | V    | X    | V    | Default |

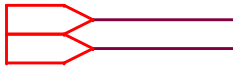
Stuff for bypass USB charger

USBP0- R689 0.4 BUSBP3-

USBP0+ R690 0.4 BUSBP3+

G-SENSOR (3-Axial)

(8,9,10,11,12,14,15,17,22,23,24,26,27,28,29,30,31,36,37,38,40,41,42,46,47,48,49) +3V  
(24,28,31,38,41,43,44,45) +15V



36

EC-B-18

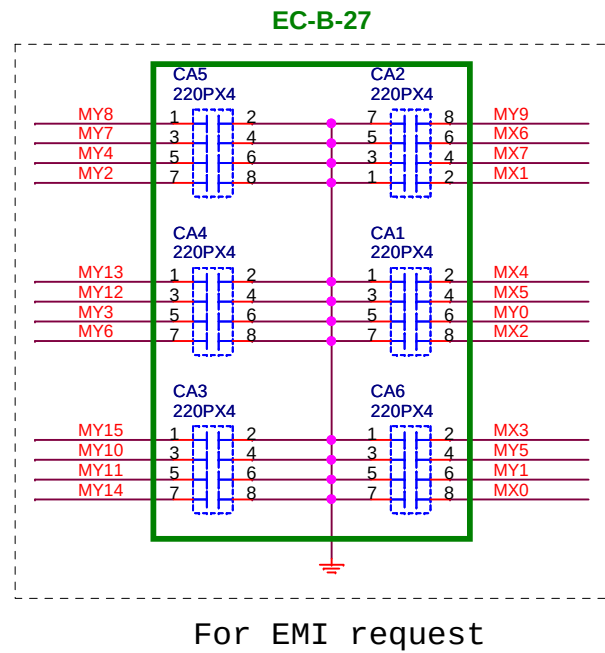
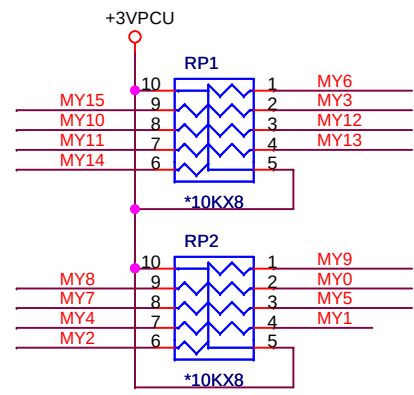
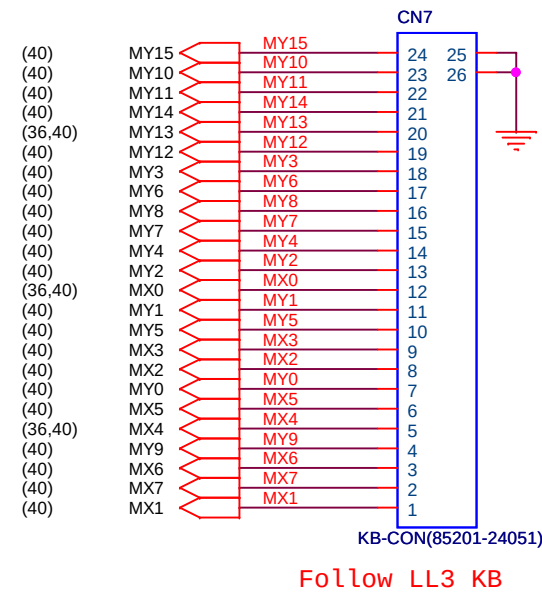


PROJECT : LL3A  
Quanta Computer Inc.

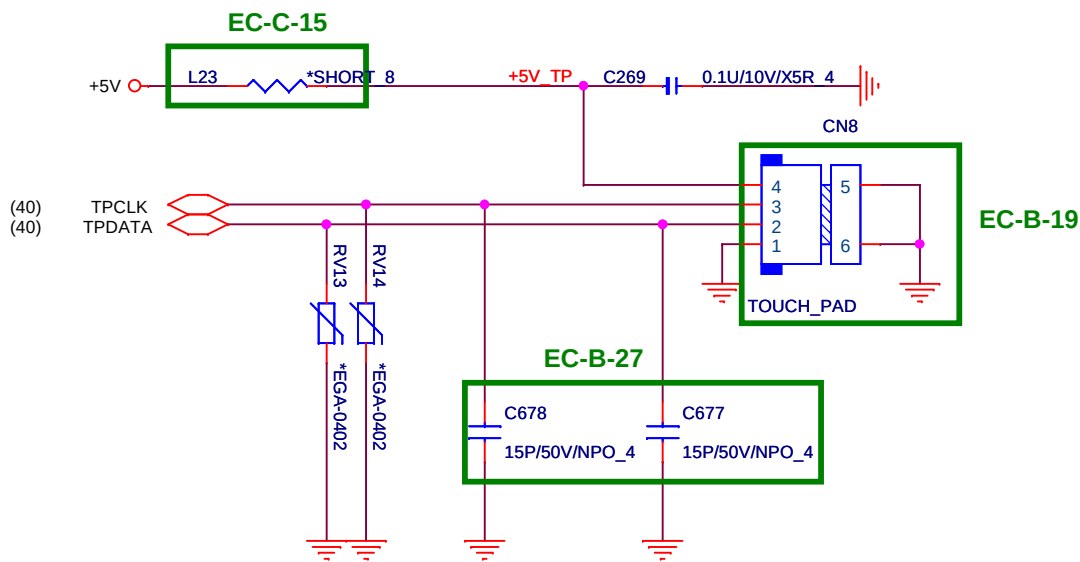
|                                    |                                    |           |
|------------------------------------|------------------------------------|-----------|
| Size<br>Custom                     | Document Number<br><b>G-SENSOR</b> | Rev<br>1E |
| Date:<br>Tuesday, January 04, 2011 | Sheet<br>34 of 53                  |           |

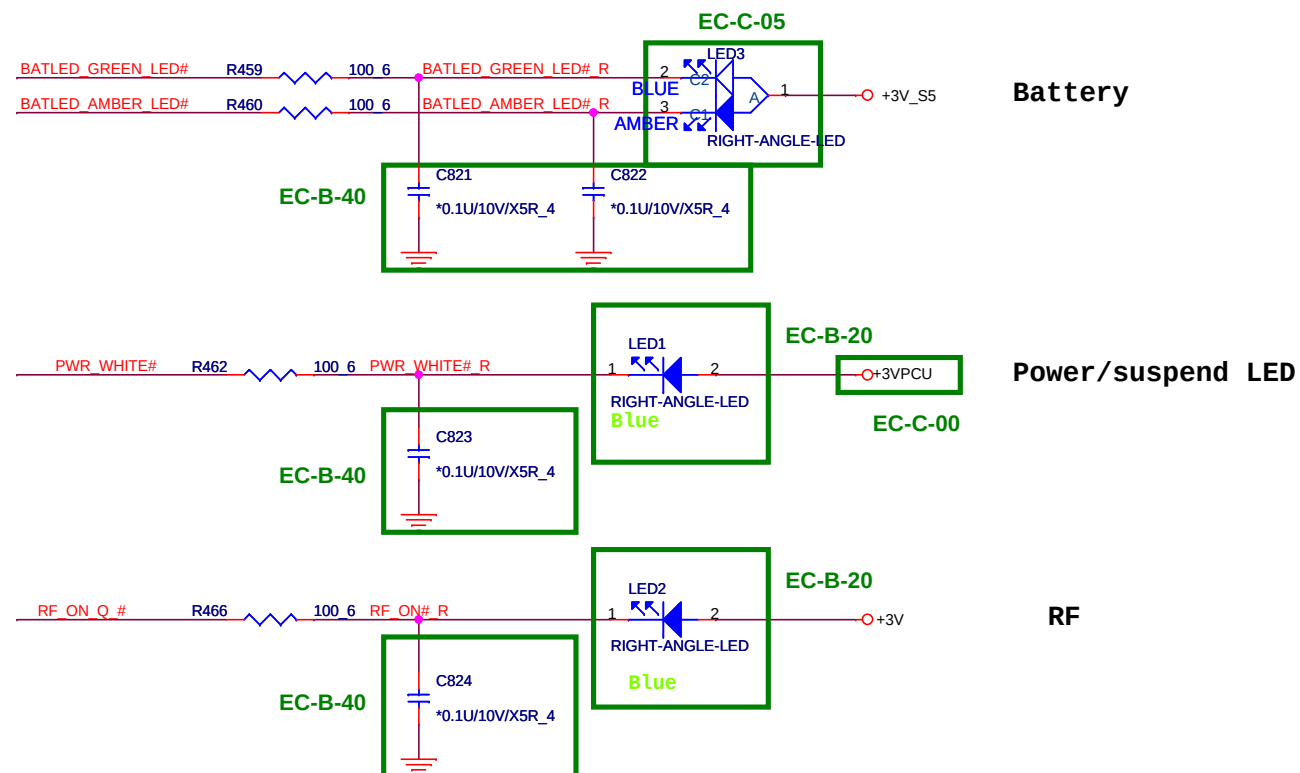
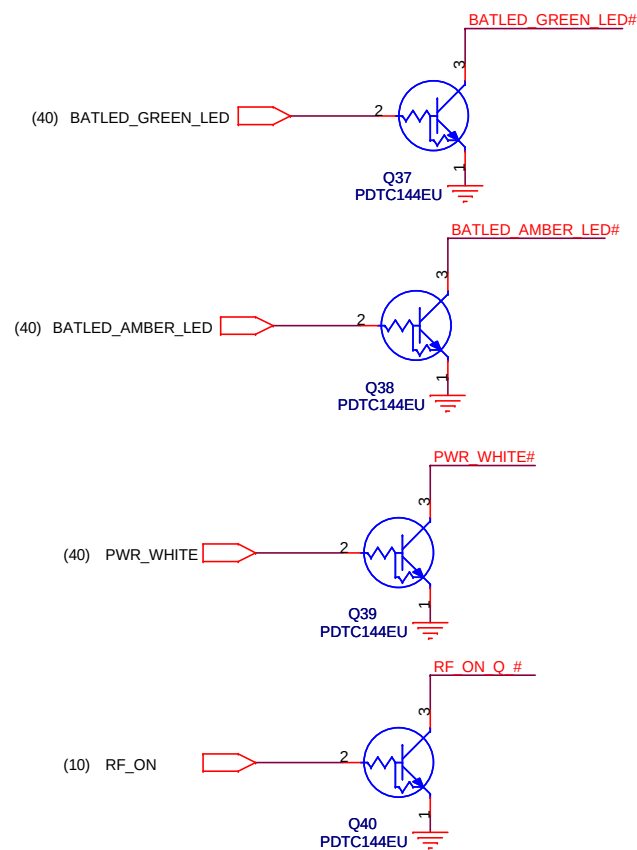


KEYBOARD



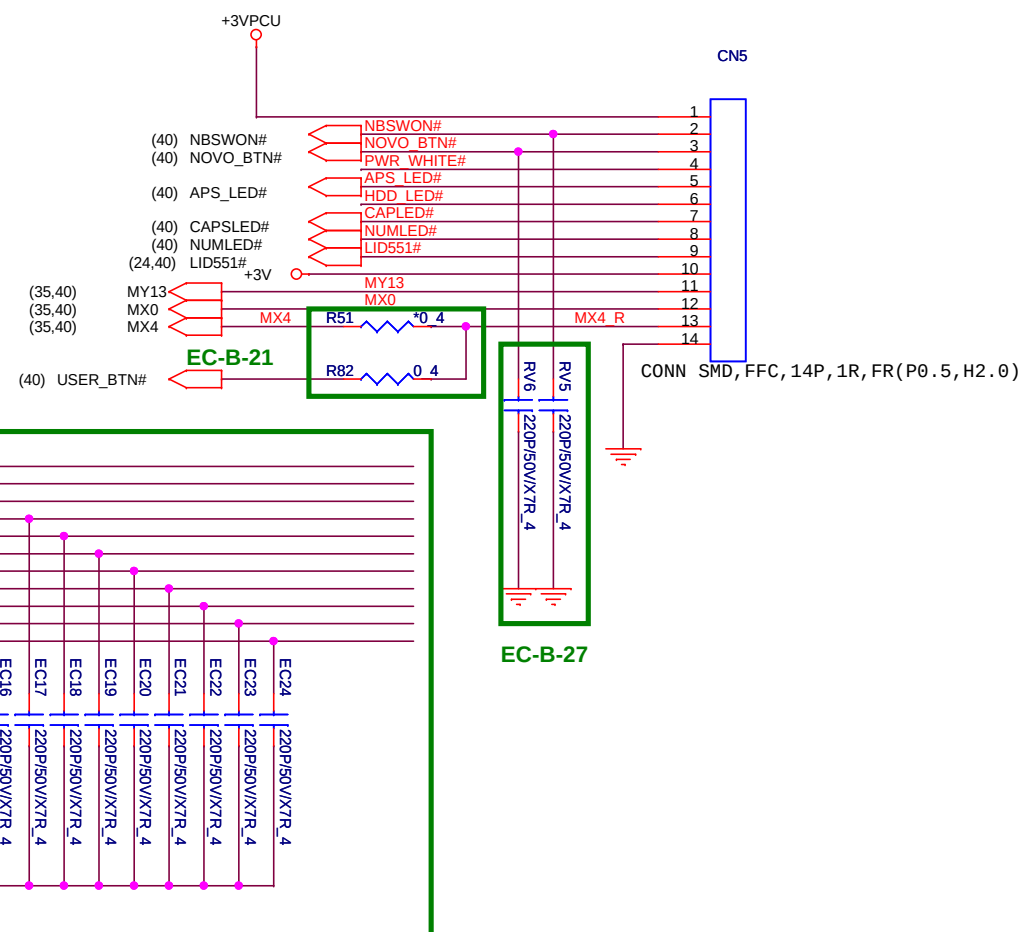
Touch pad



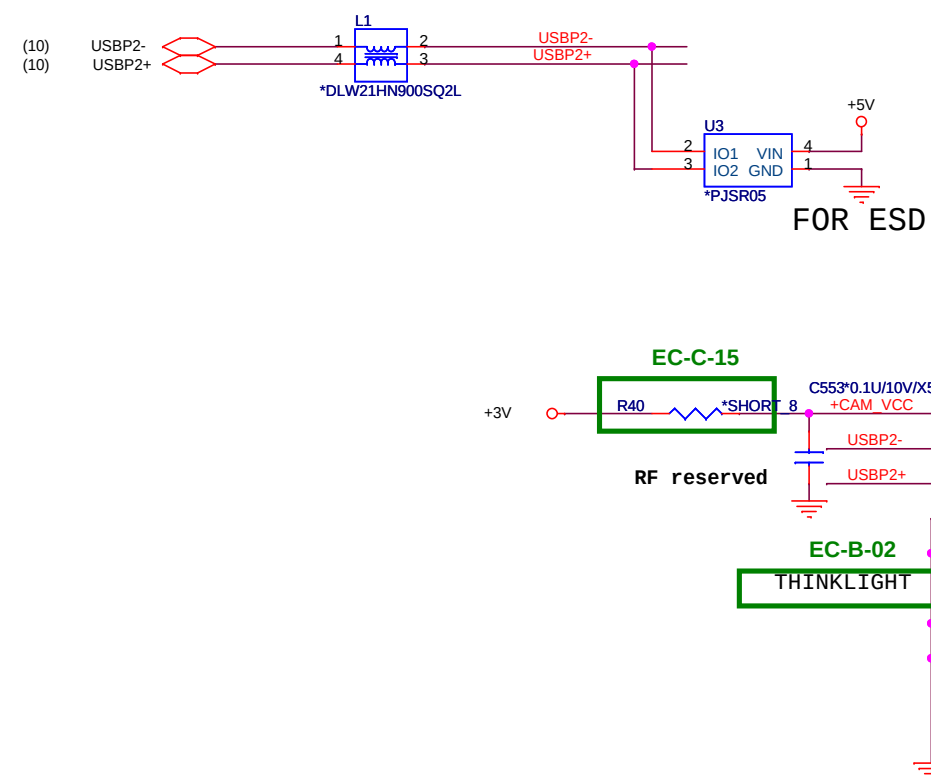


## B to B connector

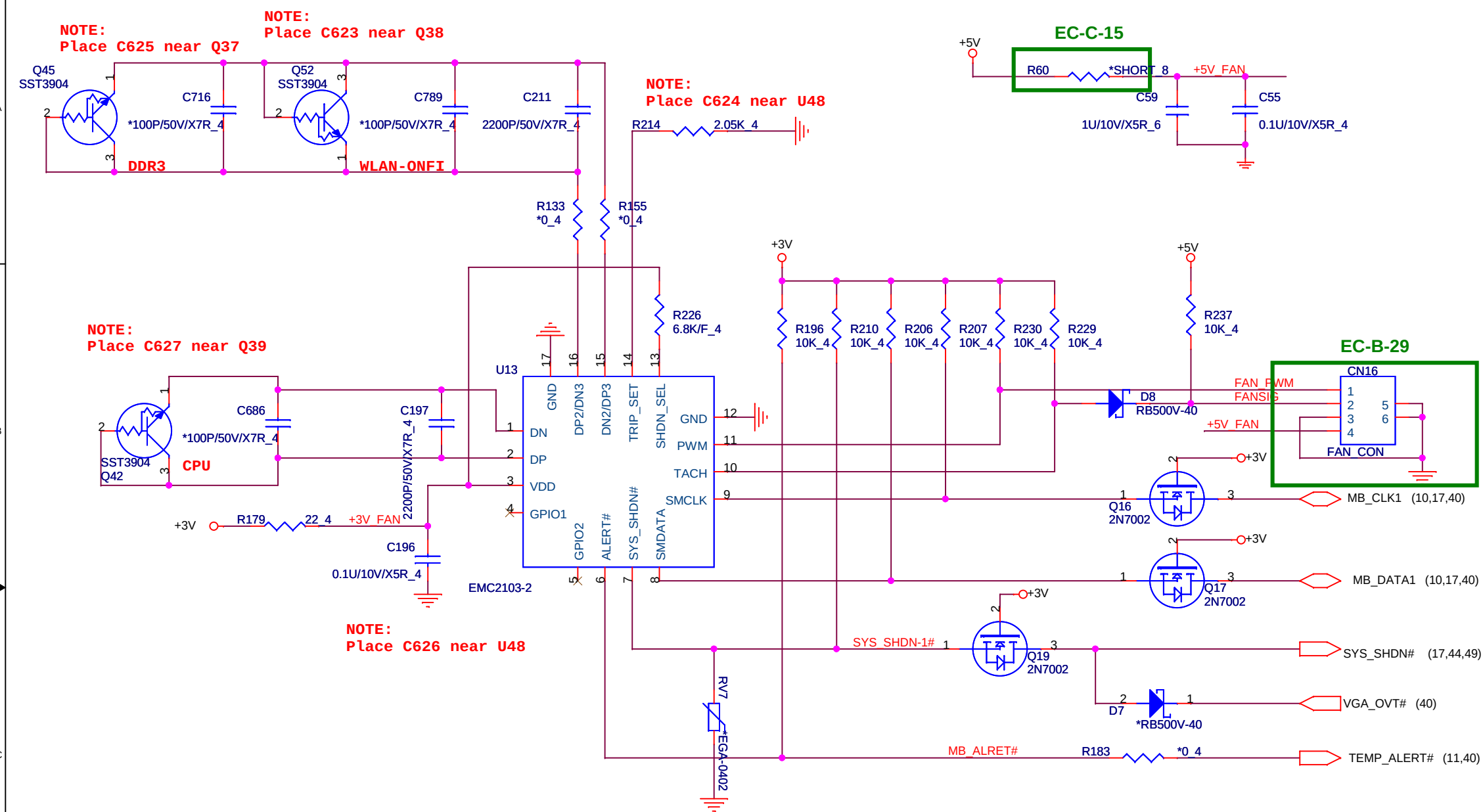
HDD (9) SATA\_ACT# R42 100 6 HDD\_LED#



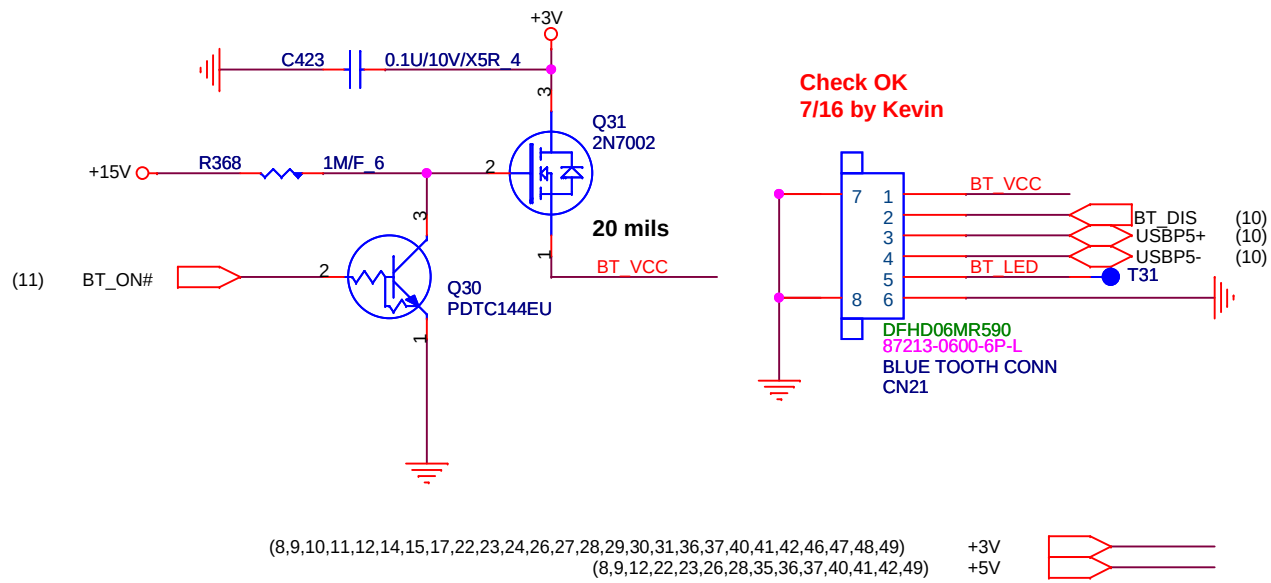
## CAMERA &amp; Keyboard light



FAN CONTROL

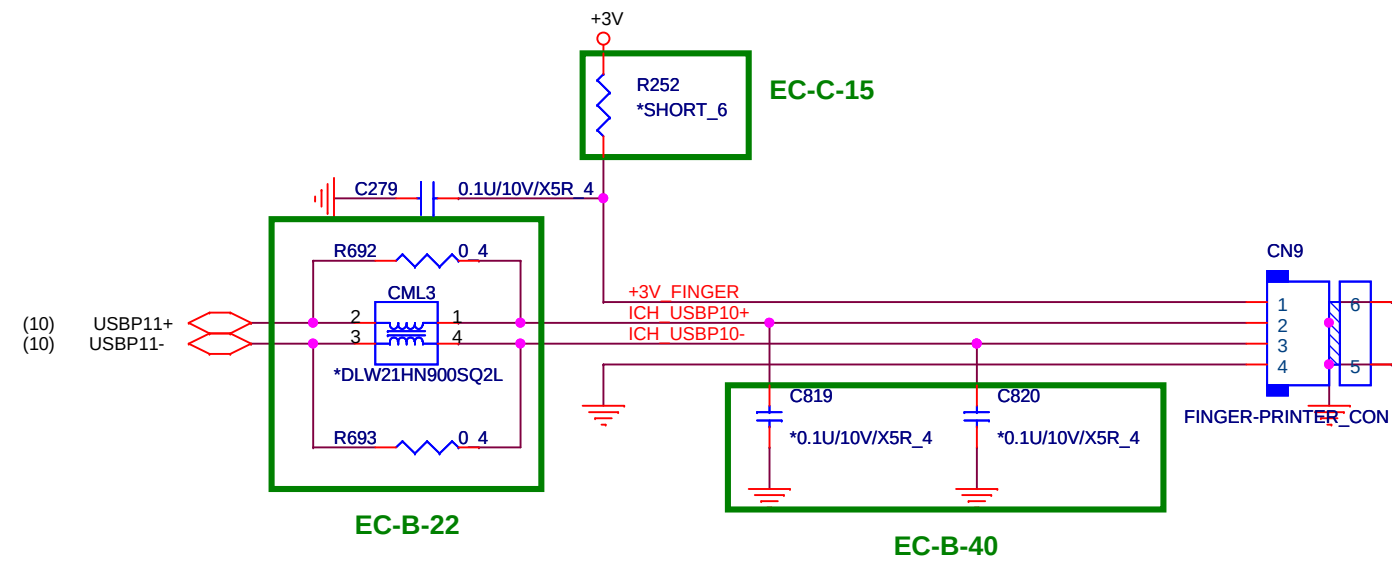


## BLUETOOTH



## FINGER PRINTER

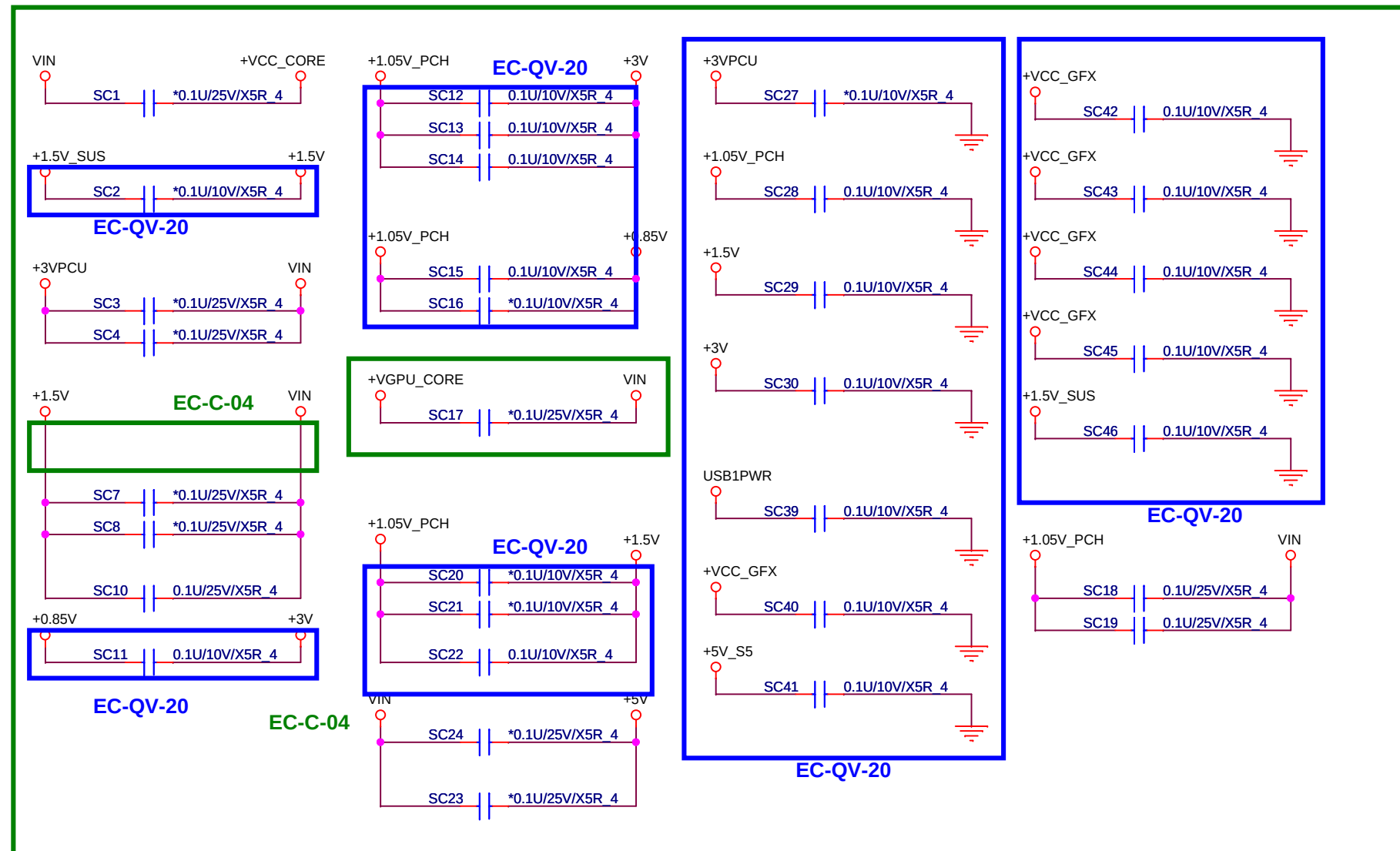
34



## TPM

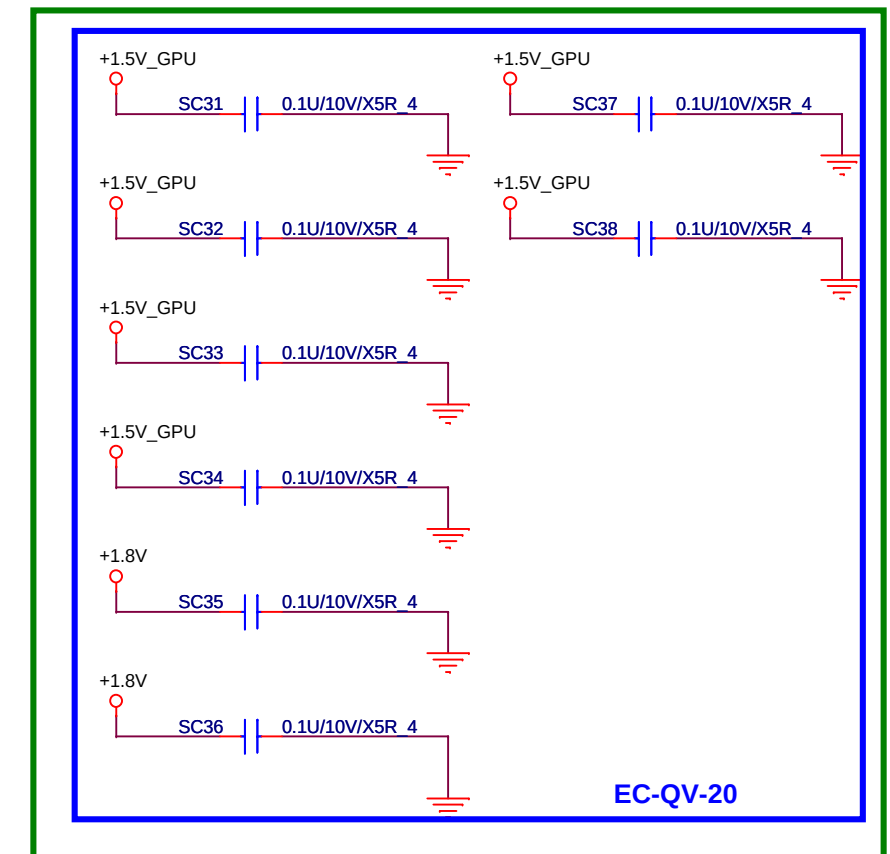
## ESD suggestions

**EC-B-28**



## EMI suggestions

**EC-C-01**

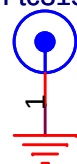
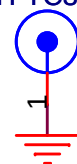


**PROJECT :LL3A**  
**Quanta Computer Inc.**

|           |                                     |                |
|-----------|-------------------------------------|----------------|
| Size<br>B | Document Number<br><b>BT/FP/TPM</b> | Rev<br>1E      |
| Date:     | Wednesday, January 05, 2011         | Sheet 38 of 53 |

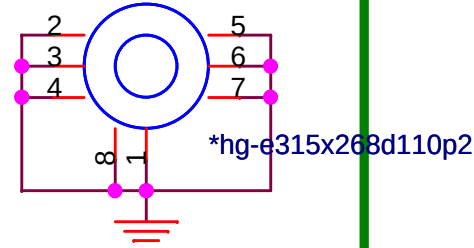
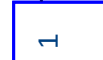
## WLAN/WWAN/Mini-SSD Nuts

EC-B-37

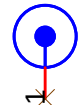
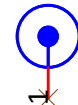
HOLE3  
\*h-tc315bc118d118p2HOLE5  
\*H-TC315BC110D110P2

EC-C-02

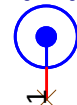
HOLE9

PAD1  
\*spad-spe236x766np

EC-QV-16

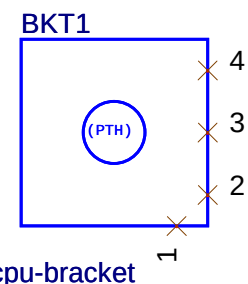
HOLE8  
\*h-o315x334d315x334nHOLE7  
\*h-o276x866d276x866n

FAN Hole

HOLE4  
\*h-c1260d1260na1300

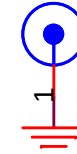
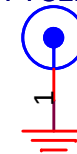
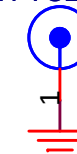
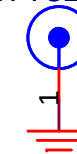
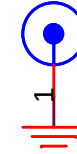
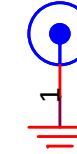
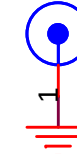
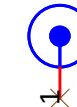
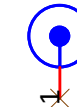
EC-B-40

CPU BKT



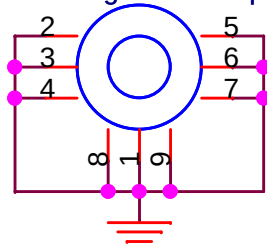
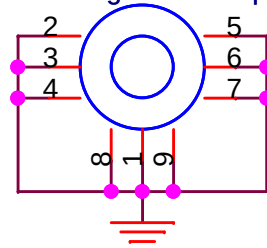
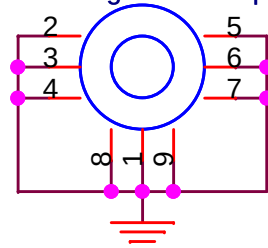
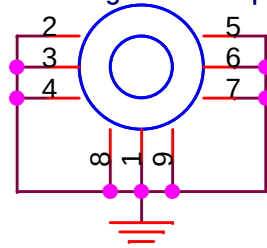
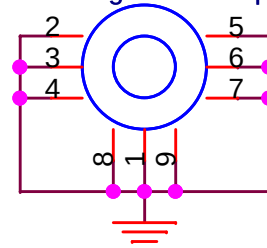
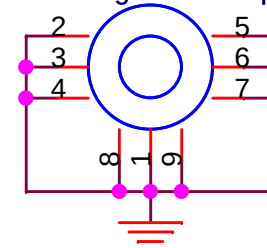
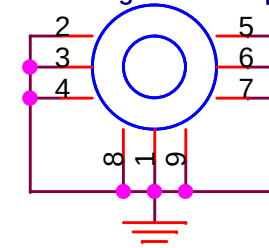
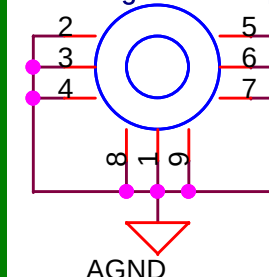
EC-B-37

VGA Nuts

HOLE21  
h-tc142bc217d102p2HOLE22  
H-TC217BC142D102P2HOLE23  
H-TC217BC142D102P2HOLE24  
H-TC217BC142D102P2HOLE26  
H-TC217BC75I75D35P2HOLE27  
H-TC217BC75I75D35P2HOLE25  
H-TC217BC75I75D35P2HOLE15  
\*h-c118d118nHOLE18  
\*h-c118d118n

EC-C-02

EC-B-40 EC-C-02

HOLE1  
\*hg-c315d110p2HOLE2  
\*hg-c315d110p2HOLE6  
\*hg-c315d110p2HOLE14  
\*hg-c315d110p2HOLE16  
\*hg-c315d110p2HOLE17  
\*hg-c315d110p2HOLE20  
\*hg-c315d110p2HOLE19  
\*hg-c315d110p2

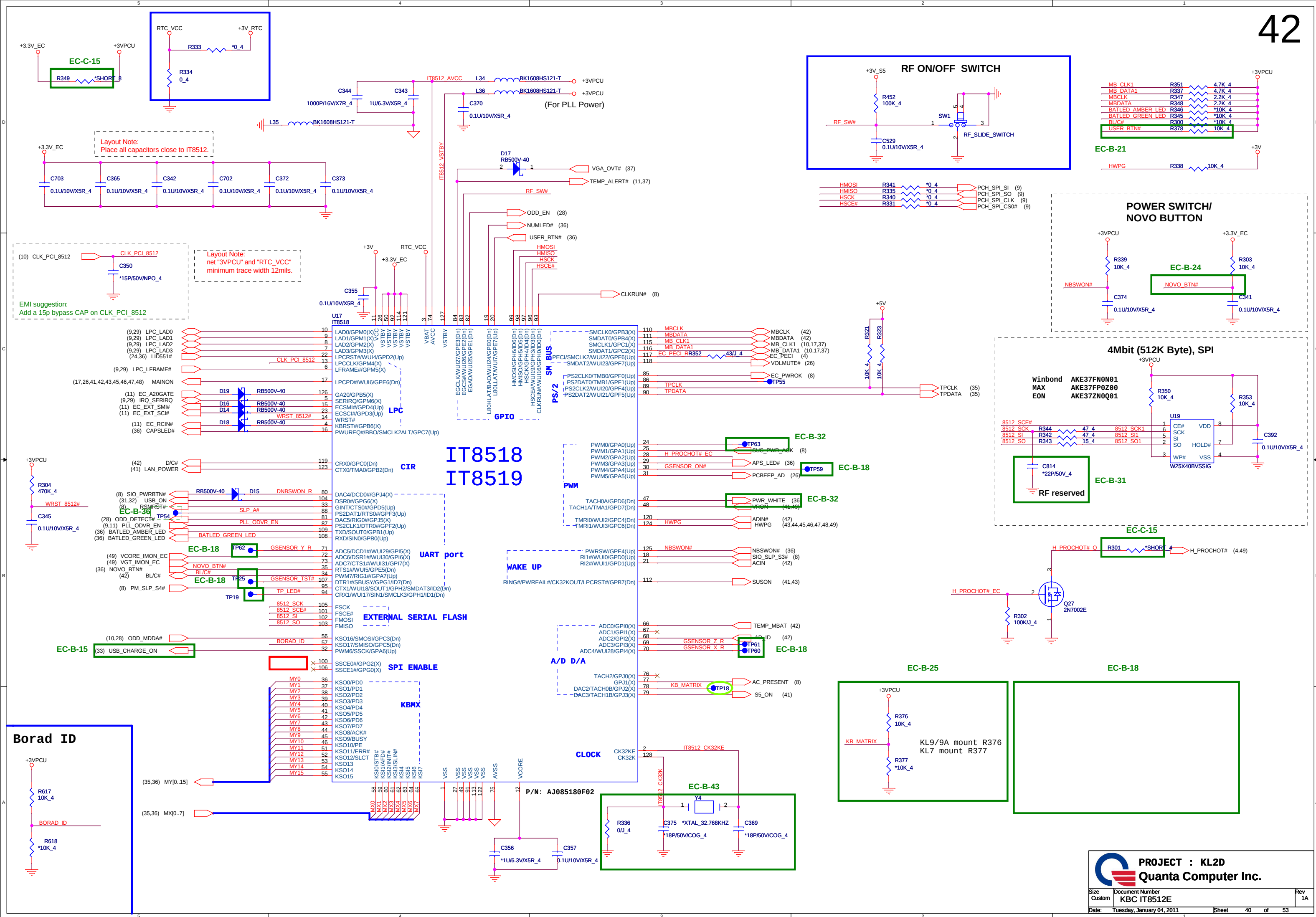
**PROJECT : KL2D**  
**Quanta Computer Inc.**

Size  
ADocument Number  
HOLD & SKEWRev  
1A

Date: Tuesday, January 04, 2011

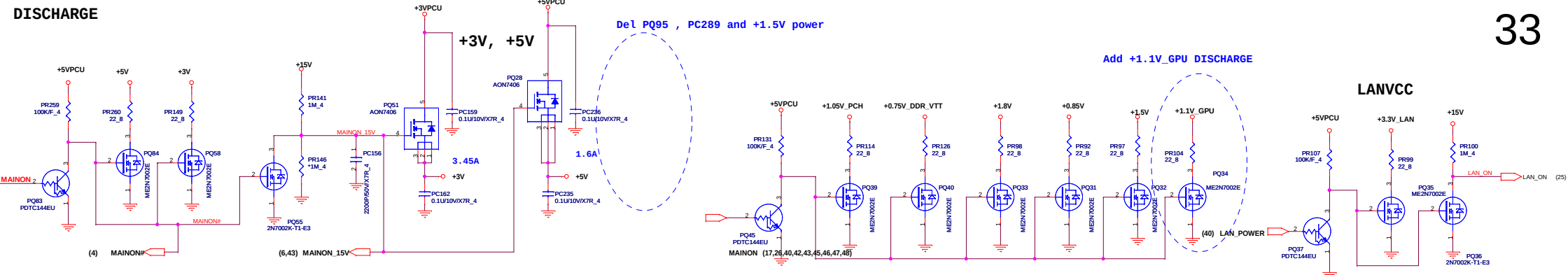
Sheet 39 of 53



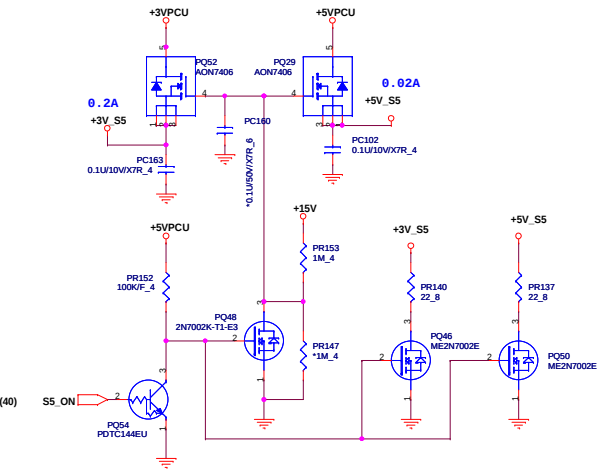


DISCHARGE

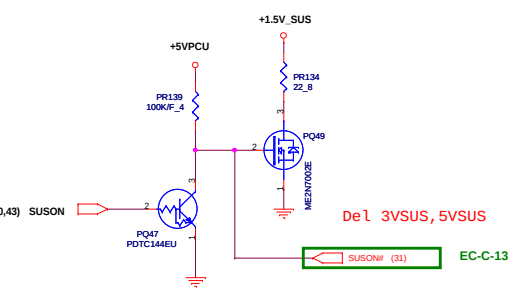
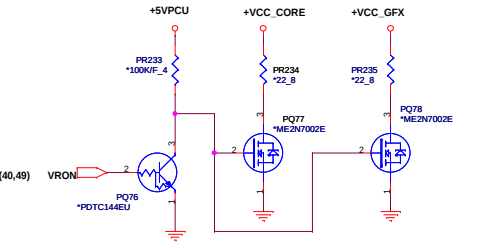
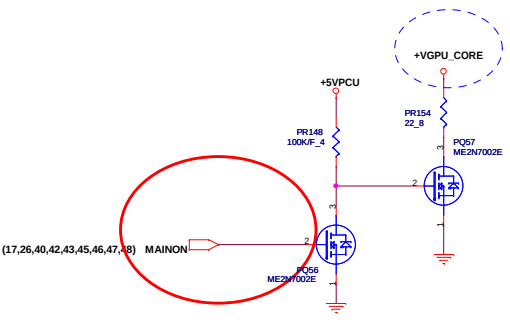
33



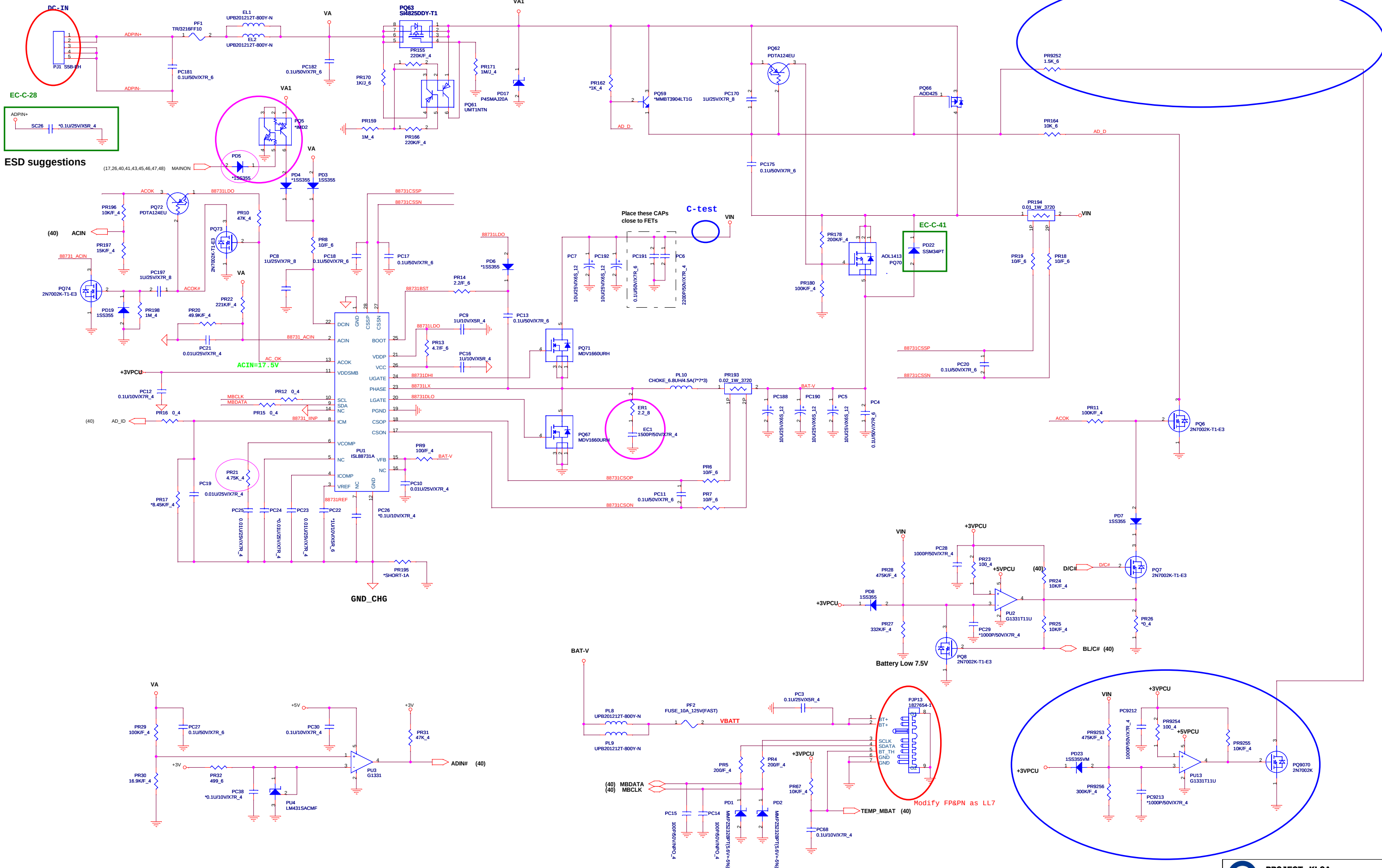
3V\_S5, 5V\_S5



Change net +V6PU\_CORE



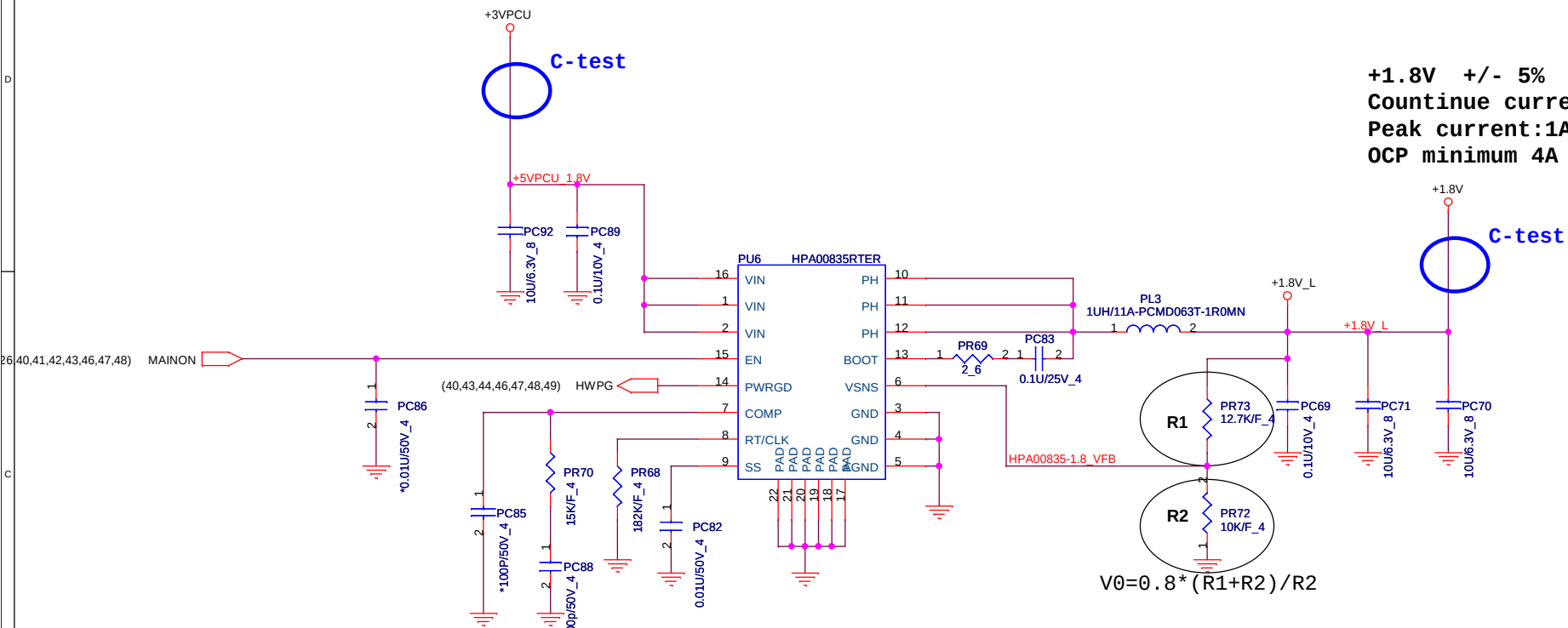
Modify FP&amp;PN as LL7



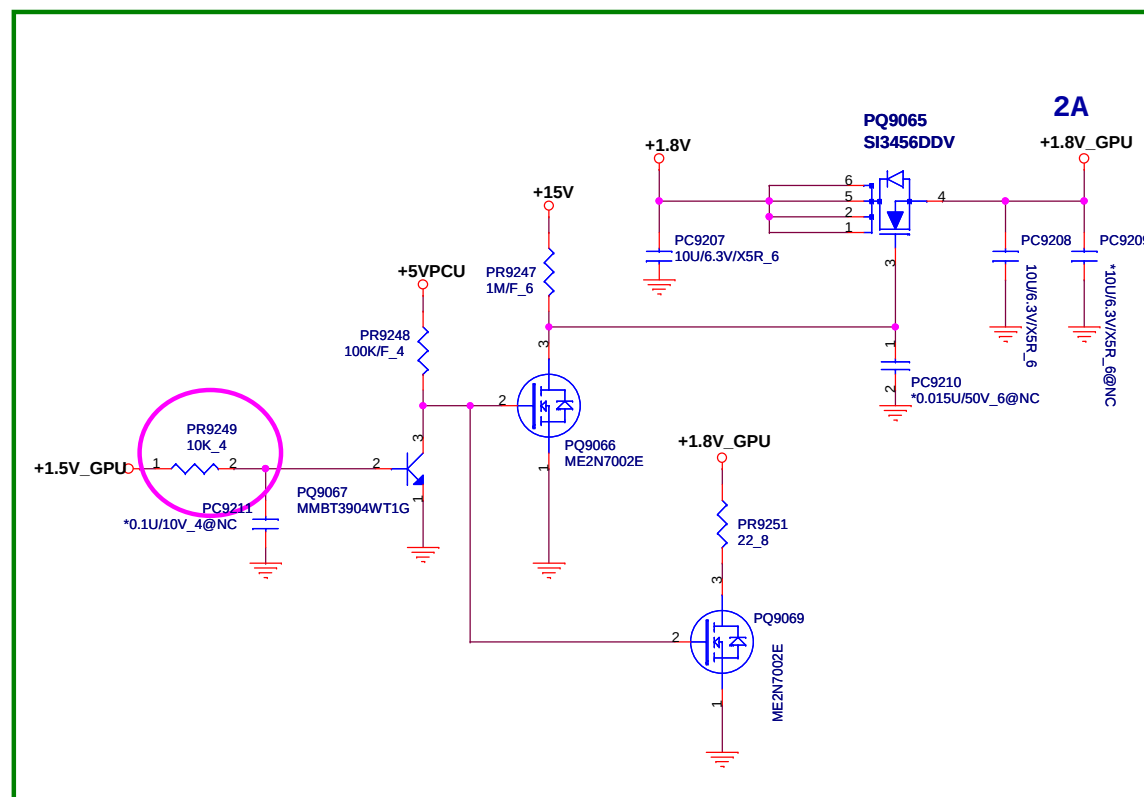




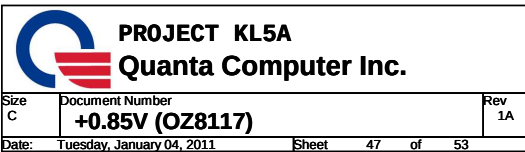




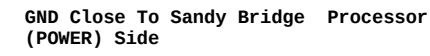
EC-C-26







- ▶ NVDDD <= VDD33+0.5 V
- ▶ FBVDDQ <= VDD33+0.5 V



|                                                                                                                                                                                                                |                                              |                |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------|-----------|
|  <div style="display: inline-block; vertical-align: middle;"> <b>PROJECT KL5A</b><br/> <b>Quanta Computer Inc.</b> </div> |                                              |                |           |
| Size<br>C                                                                                                                                                                                                      | Document Number<br><b>+1.05_PCH (OZ8117)</b> |                | Rev<br>1A |
| Date:                                                                                                                                                                                                          | Tuesday, January 04, 2011                    | Sheet 48 of 53 |           |





| EC #    | Page                          | Description                                                                                   | Part Affected                                                                                                                        |
|---------|-------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| EC-B-00 | 4                             | Change U18 output type from coms to open-drain.                                               | U18                                                                                                                                  |
| EC-B-01 | 10                            | Remove TPM funtion. (LPC interface)                                                           | U14                                                                                                                                  |
| EC-B-02 | 10,38                         | Remove keyboard light funtion.                                                                | U14,CN3                                                                                                                              |
| EC-B-03 | 11                            | Switch pins between BOARD_ID0 and BOARD_ID1.                                                  | U14                                                                                                                                  |
| EC-B-04 | 11,31                         | Change net name from WIMAX_OFF# TO WLAN_OFF#, also add pull high 10K with GPIO16              | U14,R530                                                                                                                             |
| EC-B-05 | 11                            | Add board id table.                                                                           |                                                                                                                                      |
| EC-B-06 | 25                            | Change ESD protection components.                                                             | U47,U48                                                                                                                              |
| EC-B-07 | 26                            | Modify the DIS backlight pin connecting to LVDS connector.                                    | CN1                                                                                                                                  |
| EC-B-08 | 28                            | Add R37 and R47 for normal-open audio jack.                                                   | R37,R47                                                                                                                              |
| EC-B-09 | 29                            | Add SD_4~SD_7 to support SD 3.0                                                               | CN12                                                                                                                                 |
| EC-B-10 | 30                            | Remove 2nd battery switch.It is only for KL8/8A.                                              |                                                                                                                                      |
| EC-B-11 | 31                            | Correct the LPC connection for debug card.                                                    | CN26                                                                                                                                 |
| EC-C-12 | 32                            | change capacitors connection from +3V to +3.3V_SSD,+1.5V to +1.5V_WIMAX                       |                                                                                                                                      |
| EC-B-13 | 32                            | Remove unused nets in SSD connector.                                                          | CN24                                                                                                                                 |
| EC-B-14 | 33                            | Switch the PCIE_TXN5 , PCIE_TXP5 for the right connection.                                    | U21                                                                                                                                  |
| EC-B-15 | 35                            | Change control signal of U25 enable pin from USB_ON to USB_CHARGE_ON for usb charge function. | U25                                                                                                                                  |
| EC-B-16 | 28                            | Modify left & right sound reverse issue                                                       |                                                                                                                                      |
| EC-B-17 | 35                            | Correct USB charge IC footprint.                                                              | U27                                                                                                                                  |
| EC-B-18 | 36,42                         | Remove G-sensor circuits.                                                                     | U17                                                                                                                                  |
| EC-B-19 | 37                            | Correct the pin connection of CN8.                                                            | CN8                                                                                                                                  |
| EC-B-20 | 38                            | Correct LED footprint.                                                                        | LED1,LED2                                                                                                                            |
| EC-B-21 | 38                            | Add R51,R82 for user button defined as a power button function.                               | R51,R82,R378                                                                                                                         |
| EC-B-22 | 40                            | Add R692,R693 and disable EMI solution "CML3".                                                | R692,R693,CML3                                                                                                                       |
| EC-B-23 | 40                            | Remove TPM circuits.                                                                          |                                                                                                                                      |
| EC-B-24 | 42                            | Correct connection and net name of NOVO_BTN#                                                  |                                                                                                                                      |
| EC-B-25 | 42                            | Add KB_MATRIX signal for different keyboard matrix selection between KL7 and KL9.             | R376,R377                                                                                                                            |
| EC-B-26 | 46,48                         | Add a +1.5V_GPU and +1.8V_GPU circuits for GPU power sequence tuning.                         | PQ9064,PC9202,PC9203,P9204,PC9205,PC9206,PR9246                                                                                      |
| EC-B-27 | 10,24,27<br>33,34,14<br>37,38 | Add EMI solution                                                                              | EC11,EC12,R658,R659,R660,R661,EC13,C711,CML1,R11,R12,CML2,R35,R36,CML4,R649,R651,C677,C678<br>CA1,CA2,CA3,CA4,CA5,CA6,EC14~EC26,C466 |
| EC-B-28 | 40                            | Add ESD solution                                                                              | SC1~SC30                                                                                                                             |
| EC-B-29 | 39                            | Follow PDC standard parts pin definition                                                      | CN16                                                                                                                                 |

|                                                                                       |                                                      |                                       |                  |
|---------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------|------------------|
|  | <b>PROJECT : KL2D</b><br><b>Quanta Computer Inc.</b> |                                       |                  |
|                                                                                       | Size<br>Custom                                       | Document Number<br><b>EC RECORD B</b> | Rev<br><b>1A</b> |
| Date:                                                                                 | Thursday, November 04, 2010                          | Sheet 51 of 53                        |                  |

| EC #    | Page | Description                                                                                                       | Part Affected                                                                                                                                                                                                                                                                                                                                            |
|---------|------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EC-C-00 | 36   | Change power LED pull high power plane to +3VPCU to fix system enter S3 can't flicker issue                       | LED1                                                                                                                                                                                                                                                                                                                                                     |
| EC-C-01 | 38   | Add EMI solution                                                                                                  | SC31,SC32,SC33,SC34,SC35,SC36,SC37,SC38                                                                                                                                                                                                                                                                                                                  |
| EC-C-02 | 39   | Modify Hole footprint                                                                                             | Hole1,Hole2,Hole6,Hole14,Hole16,Hole17,Hole20,Hole19,Hole9,Hole8                                                                                                                                                                                                                                                                                         |
| EC-C-03 | 12   | Reserve CAP for +VCCA FDI_VRM                                                                                     | C825                                                                                                                                                                                                                                                                                                                                                     |
| EC-C-04 | 38   | Delete ESD solution SC9 due to +1.5V power plane was deleted & change SC17 power plane bridge, add some component | SC9,SC17,SC25,SC39,SC40,SC41,SC42,SC43,SC44,SC45,SC14,SC18,SC19,SC28,SC29,SC30,SC10,SC11<br>SC12,SC13,SC15,SC22,SC46                                                                                                                                                                                                                                     |
| EC-C-05 | 36   | Modify LED footprint to fix SMT issue                                                                             | LED3                                                                                                                                                                                                                                                                                                                                                     |
| EC-C-06 | 25   | Modify bead footprint to fix SMT issue                                                                            | L37                                                                                                                                                                                                                                                                                                                                                      |
| EC-C-07 | 21   | Modify VRAM footprint to fix SMT issue                                                                            | U4,U5,U28,U29                                                                                                                                                                                                                                                                                                                                            |
| EC-C-08 | 4    | Un-stuff AND gate component                                                                                       | U18,C366,R314,R313                                                                                                                                                                                                                                                                                                                                       |
| EC-C-09 | 27   | Change SD_CLK CAP from 33p to 10p                                                                                 | C425                                                                                                                                                                                                                                                                                                                                                     |
| EC-C-10 | 22   | Change RES value to fix HDMI test fail item                                                                       | R399,R400,R401,R402,R403,R404,R405,R406                                                                                                                                                                                                                                                                                                                  |
| EC-C-11 | 26   | Reserve diode for EAPD pin                                                                                        | D23                                                                                                                                                                                                                                                                                                                                                      |
| EC-C-12 | 9    | Un-stuff JTAG RES                                                                                                 | R181,R182,R528,R534,R533,R538,R168                                                                                                                                                                                                                                                                                                                       |
| EC-C-13 | 31   | Add USB3.0 schematic                                                                                              | CN23,R369,R371,CML4,R637,R657,R704,R705,Q53,Q54,C706,C826,C827,C828,R636,R155,R707,C829,Q58,C830                                                                                                                                                                                                                                                         |
| EC-C-14 | 31   | Modify TP footprint to 3050                                                                                       | TP20,TP21,TP22,TP23                                                                                                                                                                                                                                                                                                                                      |
| EC-C-15 |      | Modify 0 ohm RES to short pad                                                                                     | R592,R608,R326,R274,R287,R292,R502,R504,R505,R512,R513,R521,R527,R91,R245,R258,R587,R301<br>R648,R451,R380,R442,R366,R425,R623,R673,R686,R77,R430,R432,R101,R212,R221,R232,R235,R239<br>R563,R586,R588,R600,R66,R71,R94,R391,R392,R393,R394,R252,R242,R248,R250,R603,R65,R233<br>R4,R50,R652,L49,L53,R431,R364,R370,R397,R448,R384,R446,L23,R40,R60,R349 |
| EC-C-16 | 12   | Change 0.002 ohm RES to 0 ohm                                                                                     | R247,R261,R254                                                                                                                                                                                                                                                                                                                                           |
| EC-C-17 | 26   | Stuff ESD solution                                                                                                | C480,C481,C482,C483                                                                                                                                                                                                                                                                                                                                      |
| EC-C-18 | 10   | Remove RF_ON pull down RES                                                                                        | R522                                                                                                                                                                                                                                                                                                                                                     |
| EC-C-19 | 29   | Add net WWAN_OFF# to disable WWAN function                                                                        |                                                                                                                                                                                                                                                                                                                                                          |
| EC-C-20 | 12   | Change 0 ohm RES to bead                                                                                          | R222                                                                                                                                                                                                                                                                                                                                                     |
| EC-C-21 | 4    | For INTEL design guide definition and material shortage, will change to 25.5ohm                                   | R315                                                                                                                                                                                                                                                                                                                                                     |
| EC-C-22 | 6    | Stuff SVID DATA pull high RES                                                                                     | R286                                                                                                                                                                                                                                                                                                                                                     |
|         |      |                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                          |

| EC #     | Page | Description                                                                                         | Part Affected                                                                                                                                        |
|----------|------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| EC-QV-00 | 42   | Add power circuit for charge issue.                                                                 | PR9252,PR9253,PR9254,PR9255,PR9256,PC9212,PC9213,PD23,PU13,PQ9070                                                                                    |
| EC-QV-01 | 37   | Add 0 Ohm for thermal sensor of DDR & WWAN                                                          | R133,R155                                                                                                                                            |
| EC-QV-02 | 33   | Modify the USB charge circuit(Add option and reverse circuit, change the USB switch EN to USB_ON_R) | R522,R636,R708,R709,R710,Q60                                                                                                                         |
| EC-QV-03 | 11   | Change the SV_DET pull high to +3V_S5 from +3V                                                      |                                                                                                                                                      |
| EC-QV-04 | 25   | Change footprint of L37 for SMT request                                                             | L37                                                                                                                                                  |
| EC-QV-05 | 43   | Change footprint of PL4, PL6, PL7, PL15, PL16                                                       | PL4, PL6, PL7, PL15, PL16                                                                                                                            |
| EC-QV-06 | 49   | Add capacitor in +VCC_GFX                                                                           | PC260                                                                                                                                                |
| EC-QV-07 | 4    | Add un-stuff capacitor on H_PWRGOOD_R                                                               | C900                                                                                                                                                 |
| EC-QV-08 | 4    | Change to component stuff                                                                           | R325,Q29                                                                                                                                             |
| EC-QV-09 | 8    | Add un-stuff capacitor on SYS_PWROK                                                                 | C901                                                                                                                                                 |
| EC-QV-10 | 8    | Change R493 to un-stuff                                                                             | R493                                                                                                                                                 |
| EC-QV-11 | 9    | Change footprint to shortpad                                                                        | R409,R153                                                                                                                                            |
| EC-QV-12 | 9    | Add 1M ohm on ACZ_SYNC_CODEC                                                                        | R711                                                                                                                                                 |
| EC-QV-13 | 9    | Change C119,C118 to 18p from 6p for RTC issue                                                       | C118,C119                                                                                                                                            |
| EC-QV-14 | 30   | Change component to un-stuff.                                                                       | C783,C784,C736                                                                                                                                       |
| EC-QV-15 | 26   | Change the footprint for EMI solution stuff BEAD                                                    | R391,R392,R393,R394                                                                                                                                  |
| EC-QV-16 | 39   | Change HOLE8 to NTPH                                                                                |                                                                                                                                                      |
| EC-QV-17 | 31   | Modify USB3.0 circuit for power saving.                                                             | Q35,R414,R415,D28,R712,Q62,R716,C950,Q63,R717,C951,C952,R713,R714,Q61,C953,R715,R421                                                                 |
| EC-QV-18 | 22   | Change the footprint to 4 GND PIN for HDMI                                                          | CN22                                                                                                                                                 |
| EC-QV-19 | 49   | Change PC47 to stuff                                                                                | PC47                                                                                                                                                 |
| EC-QV-20 | 38   | Change the P/N for cost saving                                                                      | SC2,SC11,SC12,SC13,SC14,SC15,SC16,SC20,SC21,SC22,SC27,SC28,SC29,SC30,SC39,SC40,SC41,SC42,SC43,SC44,SC45,SC46,SC31,SC32,SC33,SC34,SC35,SC36,SC37,SC38 |
| EC-QV-21 | 29   | Change C497 to stuff                                                                                | C497                                                                                                                                                 |
|          |      |                                                                                                     |                                                                                                                                                      |